# Demystifying Artificial Intelligence Sorcery

## (Part 2: Machine Learning)[a]

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[a]Available @ https://github.com/a-mhamdi/jlai/

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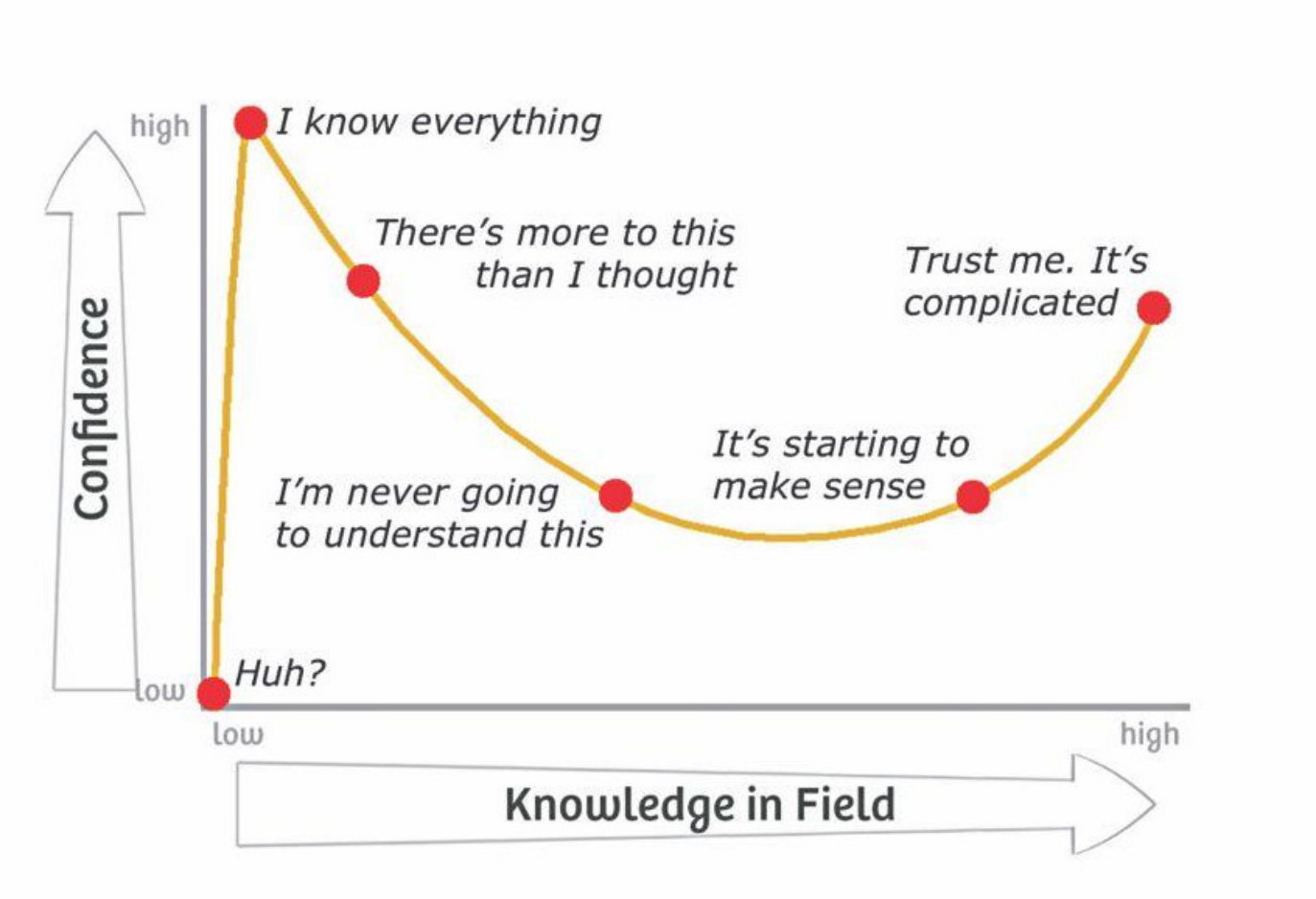
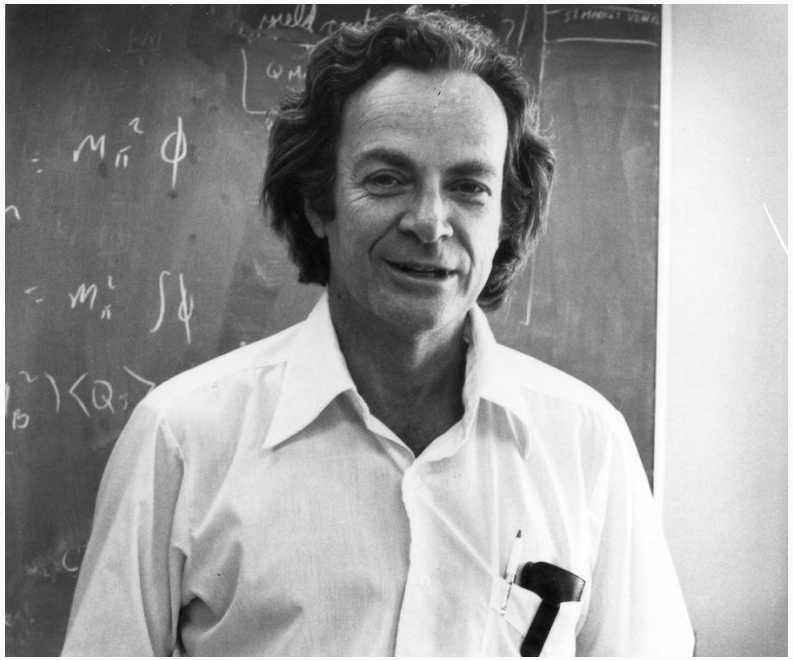
# Dunning-Kruger Effect




Kruger, J. and Dunning, D. (1999) *Unskilled and unaware of it: How difficulties in recognizing one's own incompetence lead to inflated self-assessments.* **J Pers Soc Psychol.** 77(6) pp. 1121–1134.
doi 10.1037/0022-3514.77.6.1121

"Knowledge isn't free. You have to pay attention."

***Richard P. Feynman***

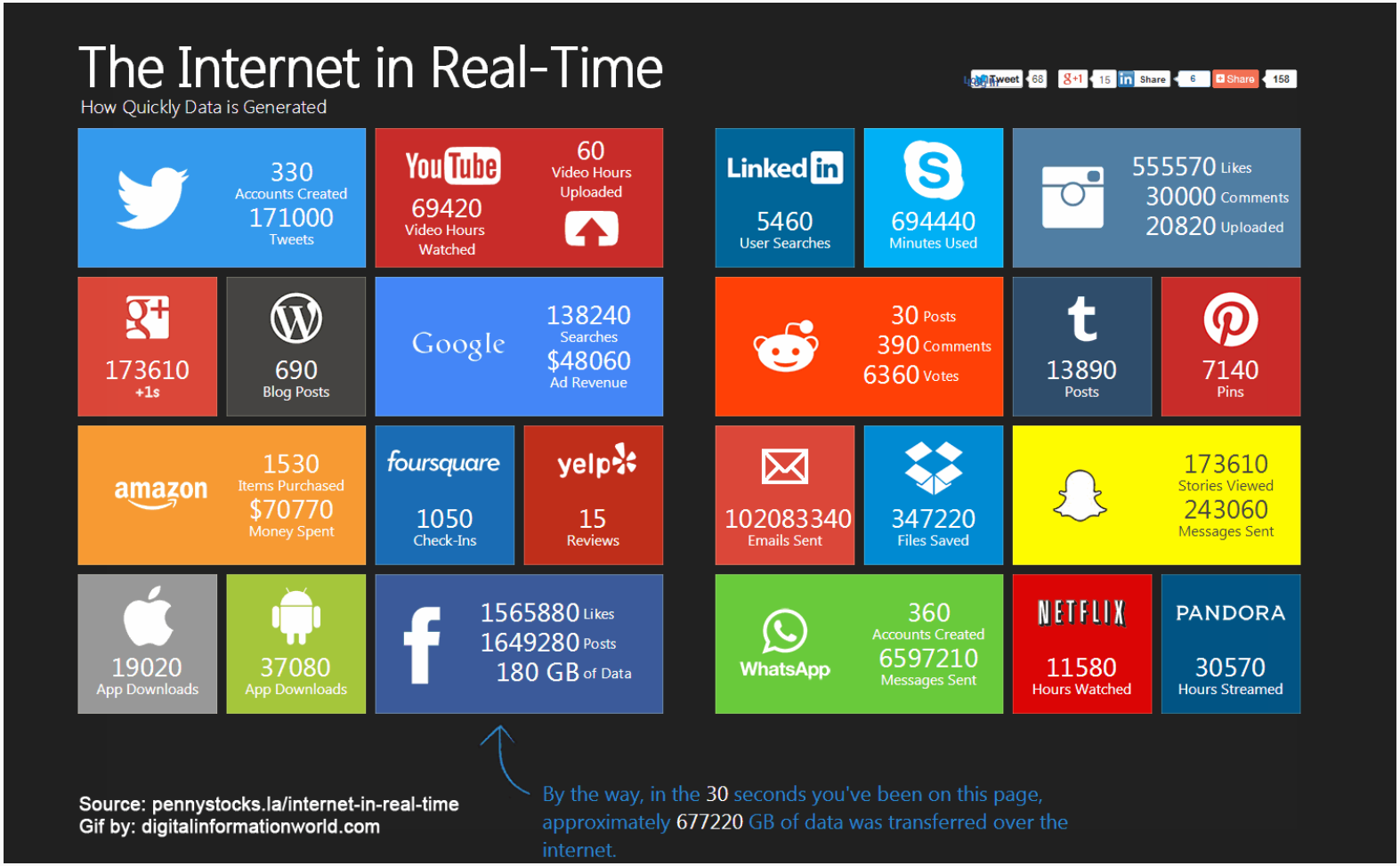


# Roadmap

1. An overview
2. Supervised Learning
3. Unsupervised Learning
4. Complementary Lab. Project
5. ML Landscape through Quizzes

# An overview

# Global Data Traffic




Update on the internet in real time is available here.

# Literature Review (1/2)

"The field of machine learning is concerned with the question of how to construct computer programs that automatically improve with experience."

Mitchell, T. (1997) *Machine Learning*. **McGraw-Hill International Editions. McGraw-Hill.**

# Literature Review (2/2)

"Machine learning (ML) is a scientific discipline that concerns developing learning capabilities in computer systems. Machine learning is one of central areas of Artificial Intelligence (AI). It is an interdisciplinary area that combines results from statistics, logic, robotics, computer science, computational intelligence, pattern recognition, data mining, cognitive science, and more."

Wojtusiak, J. (2012) Machine learning. **In *Encyclopedia of the Sciences of Learning*, pages 2082–2083. Springer US.**

# Debrief

**Arthur Samuel (1959)**
Machine Learning: Field of study that gives computers the ability to learn without being explicitly programmed.

**Tom Mitchell (1998)**
Well-posed Learning Problem: A computer is said to learn from experience $\mathcal{E}$ with respect to some task $\mathcal{T}$ and some performance measure $\mathcal{P}$, if its performance on $\mathcal{T}$, as measured by $\mathcal{P}$, improves with experience $\mathcal{E}$.

**Task #1**
Suppose your email program watches which emails you do or do not mark as spam, and based on that learns how to better filter spam. What is the task $\mathcal{T}$ in this setting?

1. Classifying emails as spam or not spam;
2. Watching you label emails as spam or not spam;
3. The number (or fraction) of emails correctly classified as spam/not spam;
4. None of the above-this not a machine learning problem.

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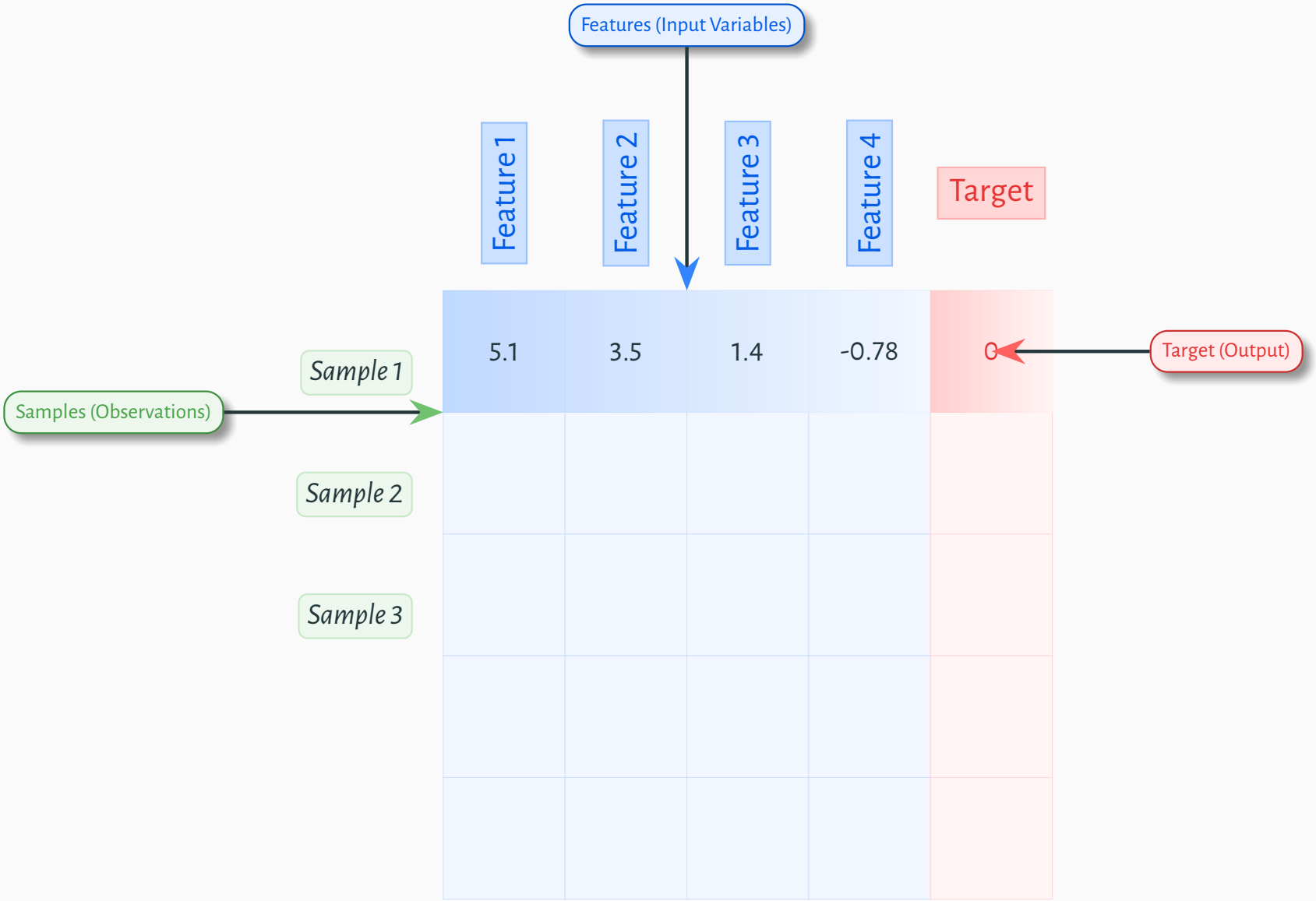
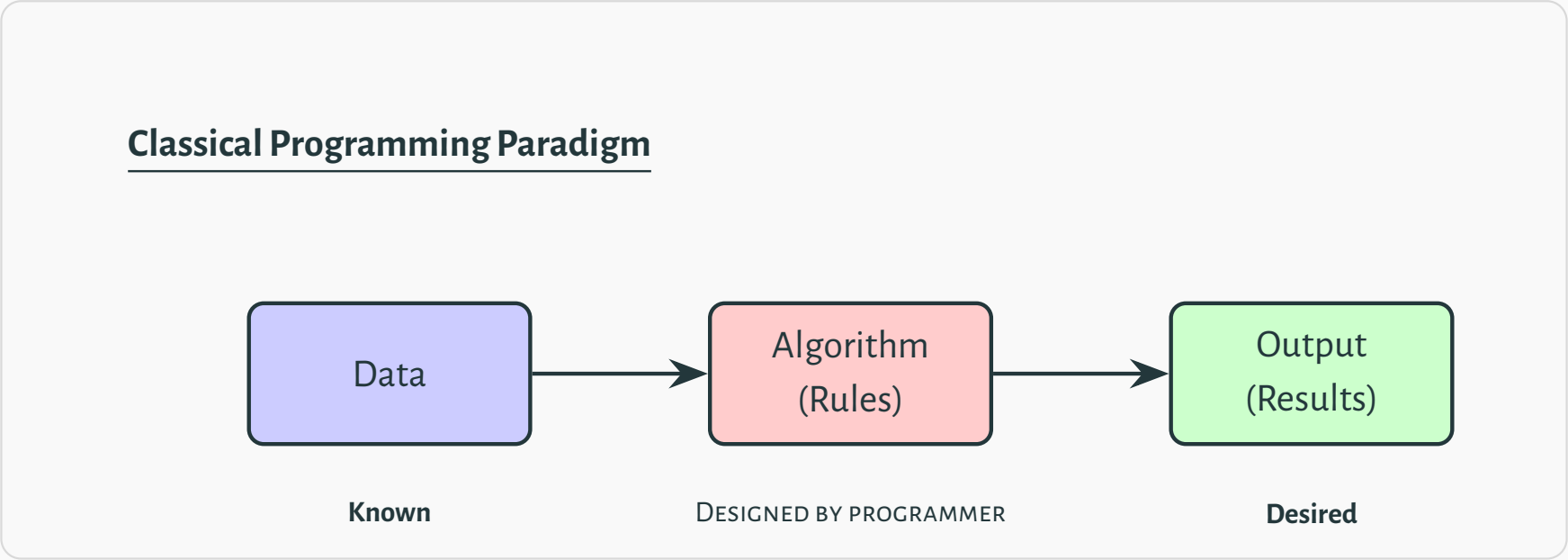
# Terminology (1/4)

- Dataset is a structured collection of data organized for analysis or processing. Good datasets often contain diverse, representative examples with clear labels for the information they contain.
  *(e.g., census records, medical images, customer purchases.)*
- Feature is an input variable or an attribute used to make predictions.
- Target is the output variable the model is trained to predict.
- Model is an algorithm that learns patterns from data to make predictions or decisions. Models transform inputs into outputs based on parameters that are optimized during training.
  *(e.g., neural networks, decision trees, regression.)*

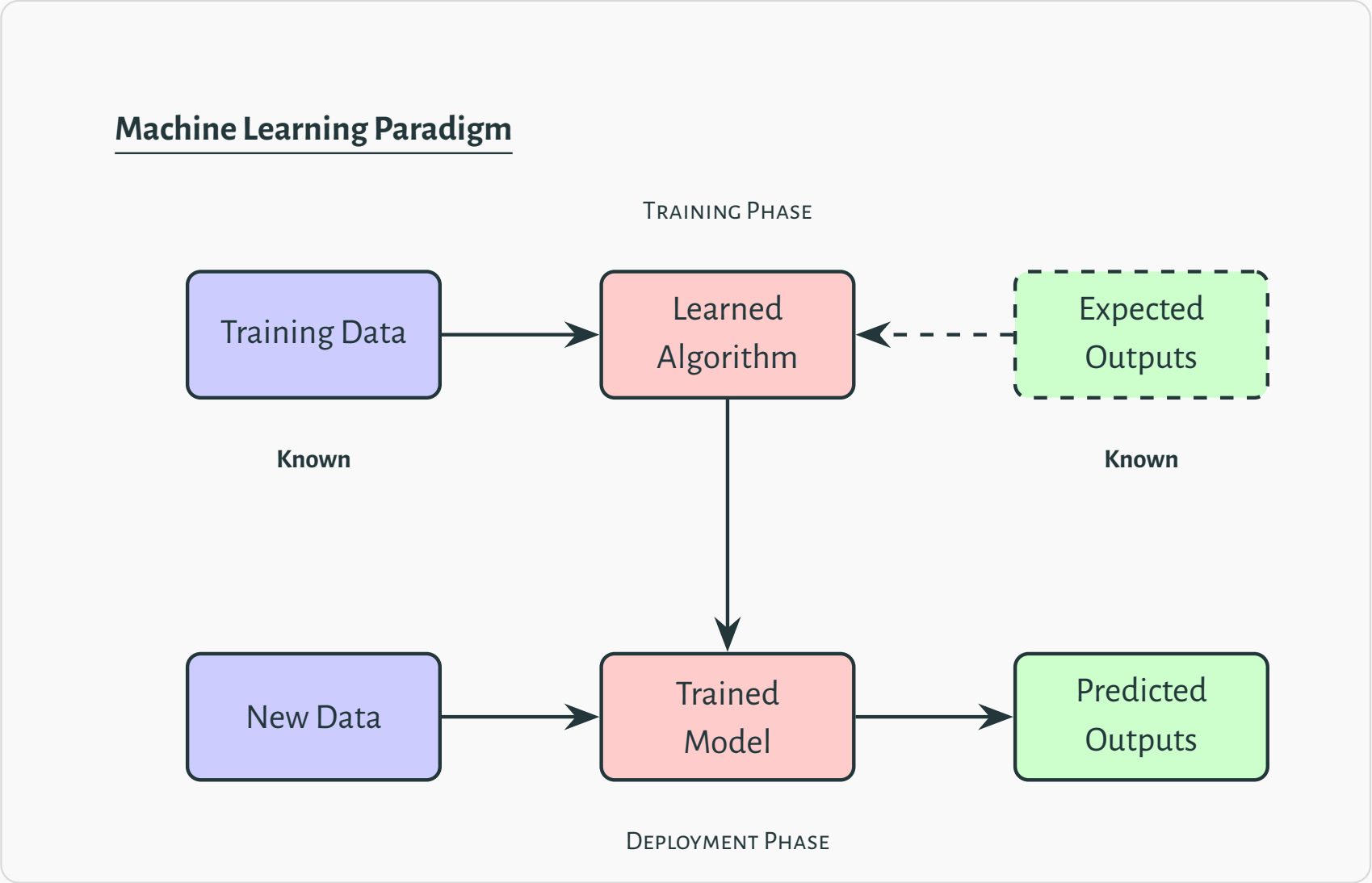
# Terminology (2/4)



# Terminology (3/4)



# Terminology (4/4)



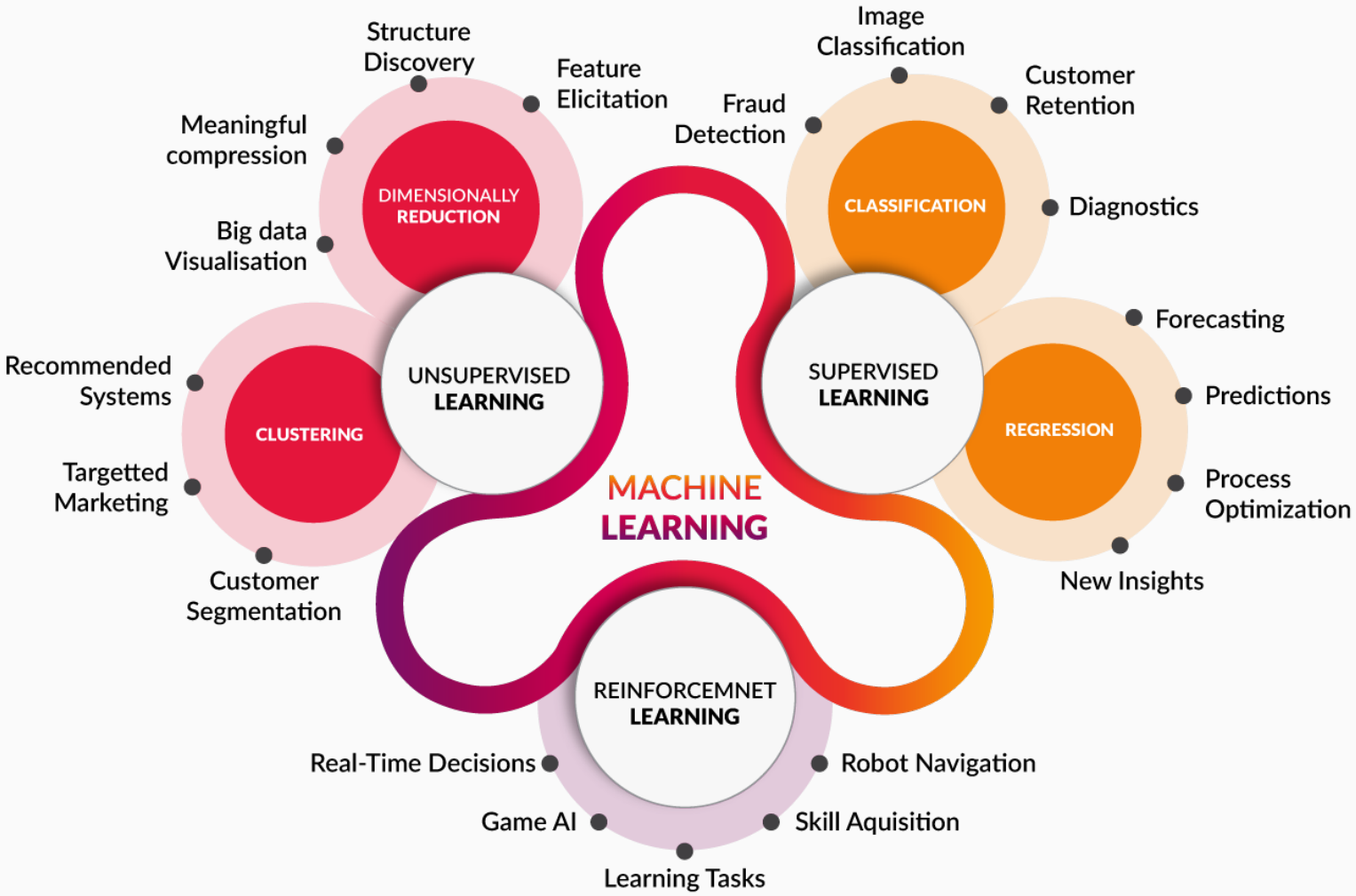
# Learning Paradigms

- Supervised Learning is a machine learning approach where algorithms learn from labeled training data to predict outputs for new inputs. The algorithm is "supervised" by being shown correct answers during training, allowing it to learn the relationship between features and target variables.
  *(e.g., spam filtering, image classification, price prediction.)*
- Unsupervised Learning is a machine learning approach where algorithms find patterns in data without explicit labels or guidance. These algorithms identify inherent structures in data, such as groupings, anomalies, or distribution characteristics.
  *(e.g., clustering, anomaly detection, dimensionality reduction.)*

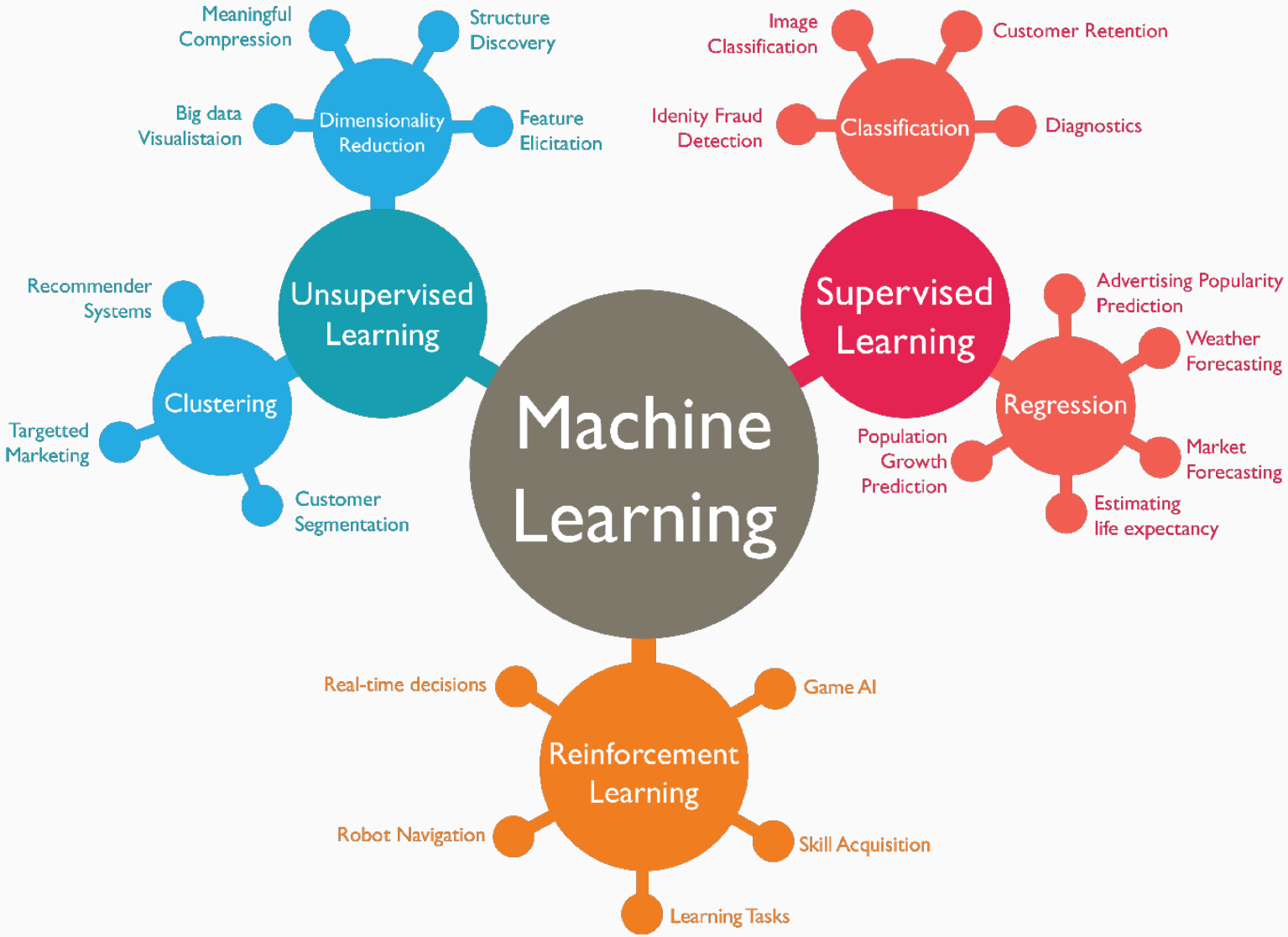
# Learning Paradigms

## Common Tasks

- Regression is a supervised learning technique that predicts continuous numerical values based on input features. It models relationships between dependent and independent variables by finding the best-fitting mathematical function.
  *(e.g., price forecasting, temperature prediction, sales modeling.)*
- Classification is a supervised learning task where algorithms learn to assign input data to predefined categories or classes. The model identifies decision boundaries that separate different classes based on feature patterns.
  *(e.g., spam detection, sentiment analysis, disease diagnosis.)*
- Clustering is an unsupervised learning technique that groups similar data points together based on their inherent characteristics. It identifies natural groupings without predefined labels by optimizing for high intra-cluster similarity and low inter-cluster similarity.
  *(e.g., customer segmentation, document grouping, image compression.)*

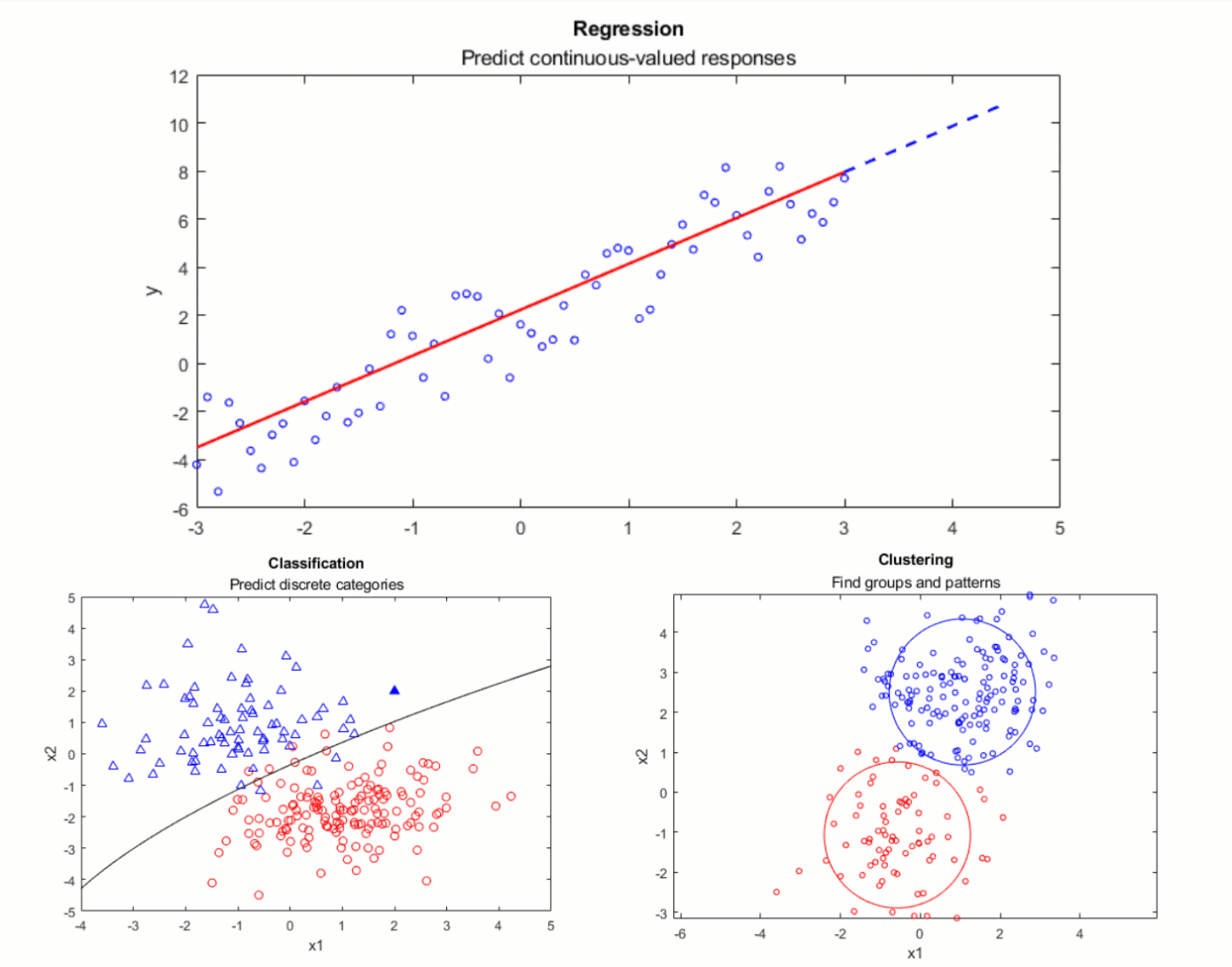



https://www.cognub.com/index.php/cognitive-platform/




https://vitalflux.com/great-mind-maps-for-learning-machine-learning/

# Regression | Classification | Clustering




https://github.com/MathWorks-Teaching-Resources/Machine-Learning-for-Regression

# REMINDER

# Programming Language

julialang.org/

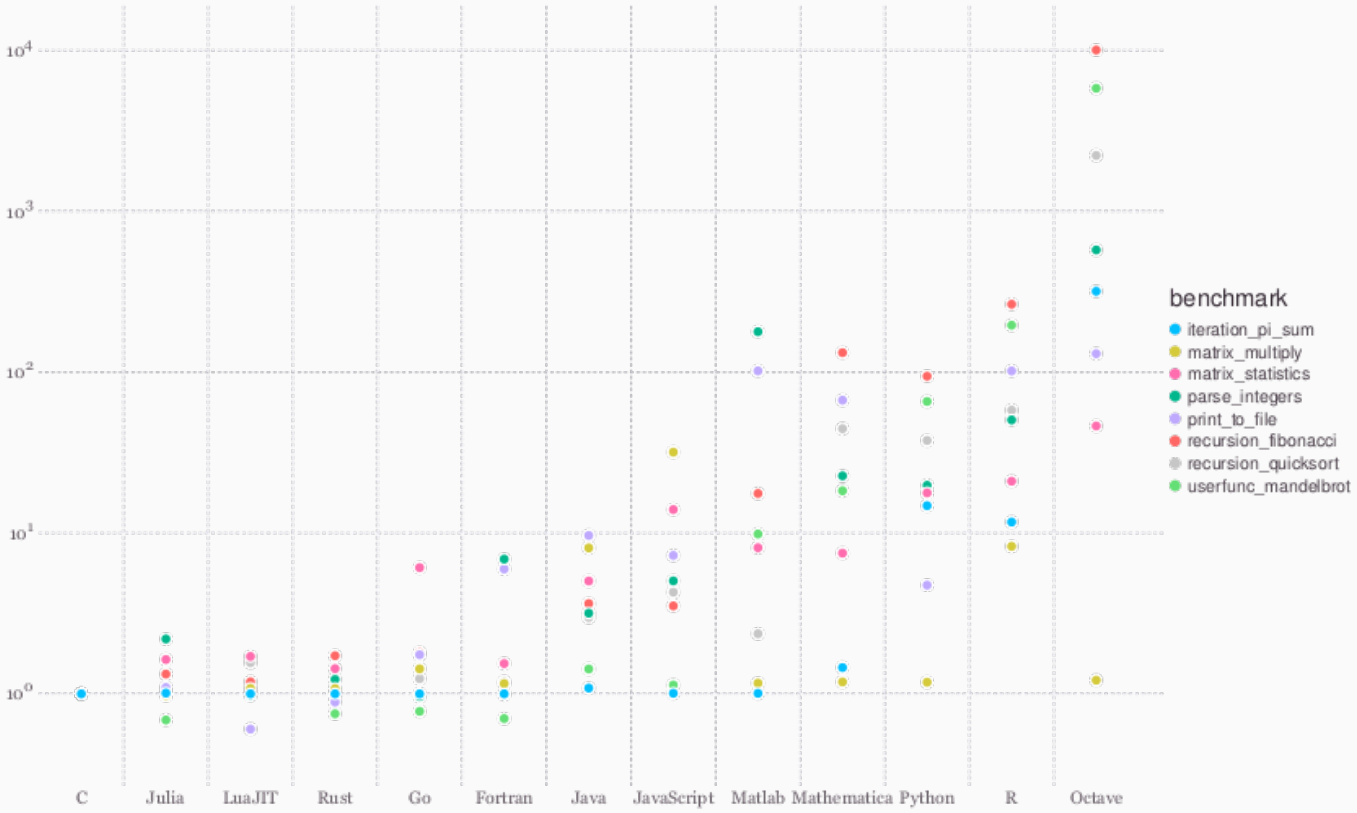
# Development Environments

- ▲ $ docker compose up
- ▼ $ docker compose down

# Julia in a Nutshell

- **Fast:** native code for multiple platforms via `LLVM`;
- **Dynamic:** good support for interactive use *(like a a scripting language)*;
- **Reproducible:** environment recreation across platforms, with pre-built binaries;
- **Composable:** multiple dispatch as a paradigm *(oop & functional programming)*;
- **General:** asynchronous I/O, metaprogramming, debugging, logging; profiling, pkg, ...
- **Open Source:** GitHub repository at https://github.com/JuliaLang/julia.

# Julia Micro-Benchmarks (1/2)




https://julialang.org/benchmarks

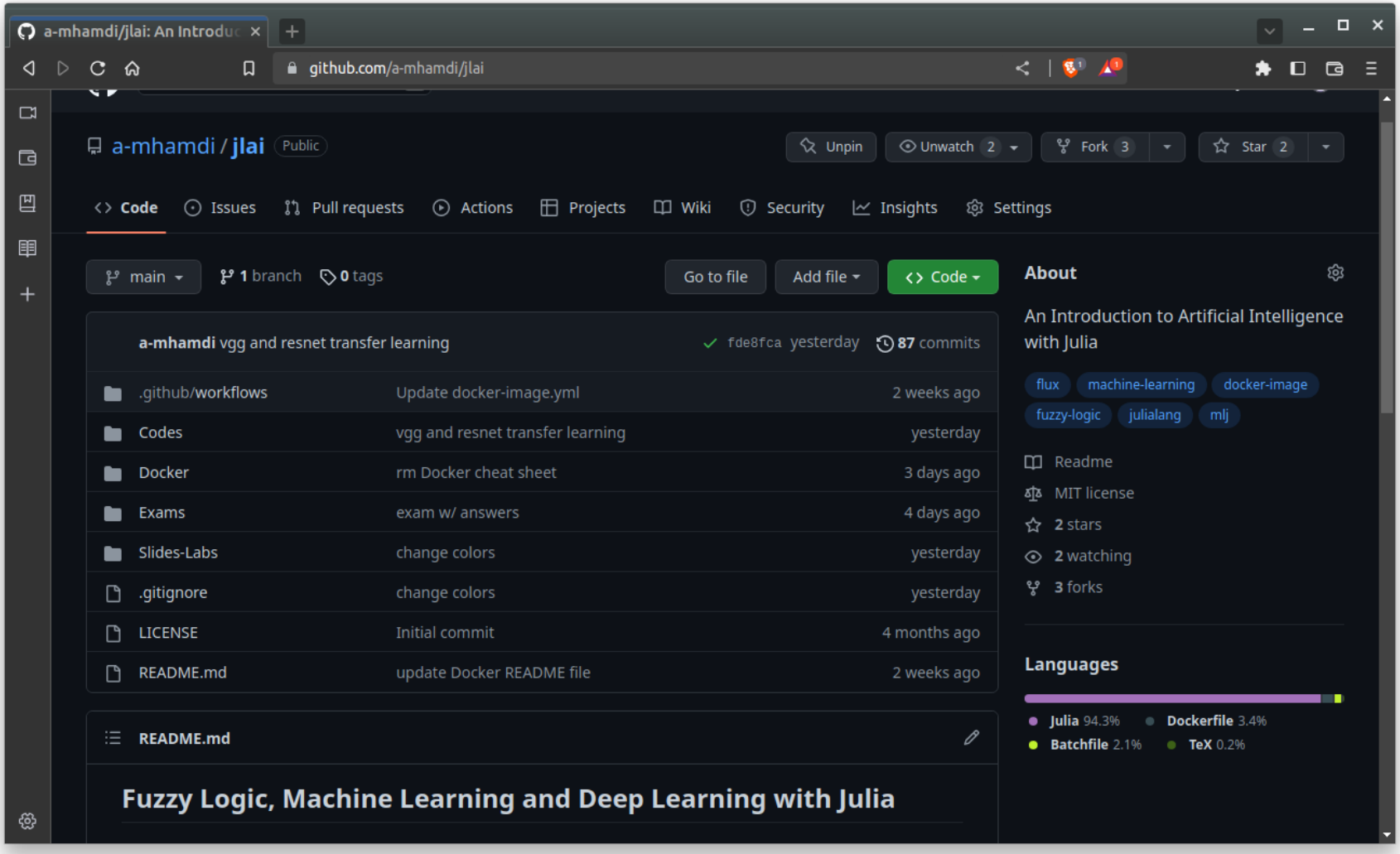
# Julia Micro-Benchmarks (2/2)

**Geometric Means[1] of Micro-Benchmarks by Language**

| | | |
|---|---|---|
| 1 | C | 1.0 |
| 2 | Julia | 1.17006 |
| 3 | LuaJIT | 1.02931 |
| 4 | Rust | 1.0999 |
| 5 | Go | 1.49917 |
| 6 | Fortran | 1.67022 |
| 7 | Java | 3.46773 |
| 8 | JavaScript | 4.79602 |
| 9 | Matlab | 9.57235 |
| 10 | Mathematica | 14.6387 |
| 11 | Python | 16.9262 |
| 12 | R | 48.5796 |
| 13 | Octave | 338.704 |

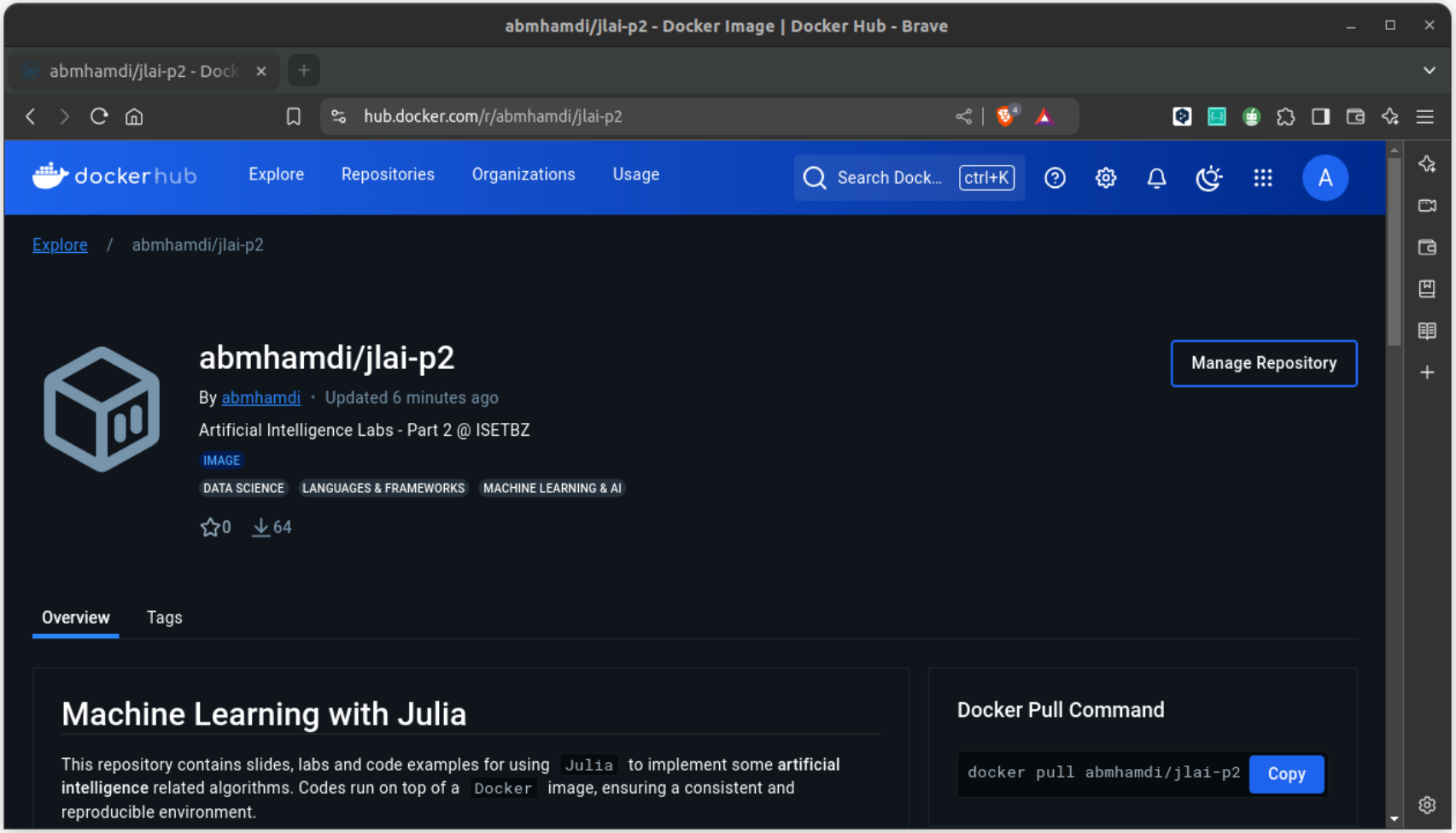
[1] Measure of central tendency expressed as $(x_1 \times x_2 \times \cdots \times x_n)^{1/n}$

# Source Control Management (SCM)



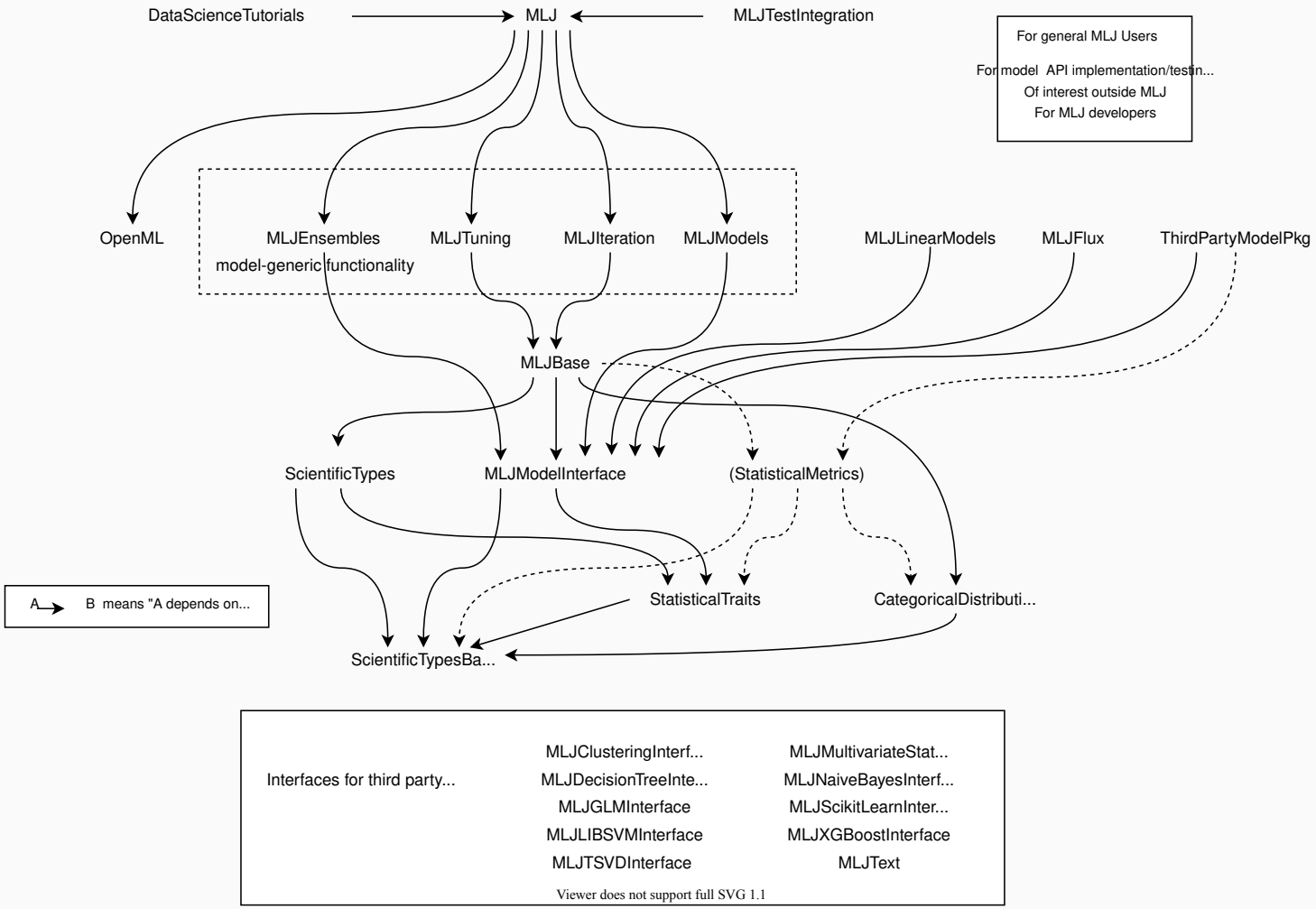

https://github.com/a-mhamdi/jlai

# Docker Image




https://hub.docker.com/r/abmhamdi/jlai-p2

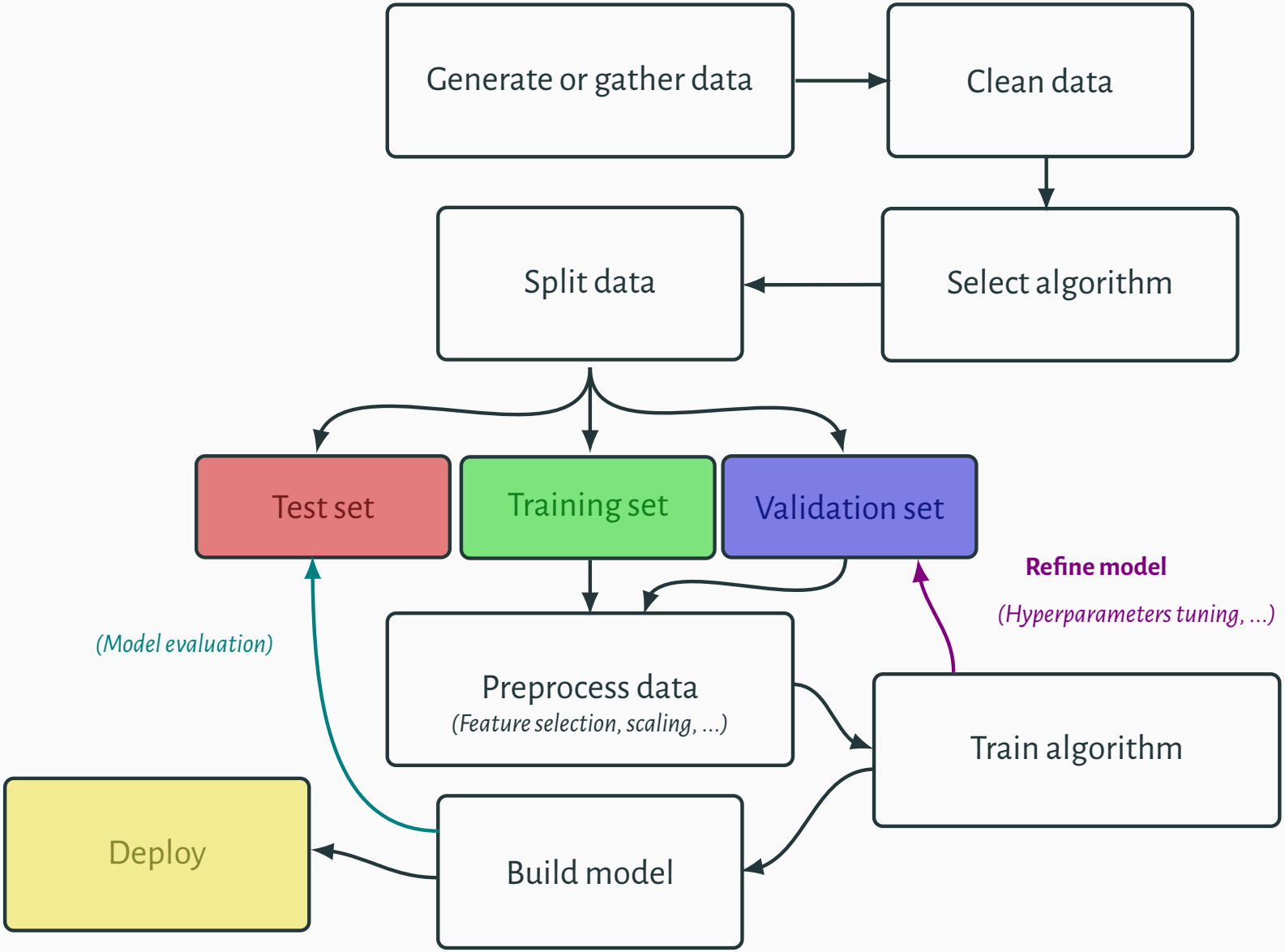
# A Machine Learning Framework for Julia




https://docs.juliahub.com/MLJ/

# Supervised Learning

# Workflow in Machine Learning



# Data Preprocessing

## How?

**Cleaning** Identifying and correcting or removing inaccuracies and inconsistencies in the data.

**Transformation** Converting data from one format or structure to another.

**Normalization** Scaling the data so that it fits within a specific range. This is often done to make the data more amenable to certain operations or algorithms.

# Data Preprocessing

## Why?

- Raw data is often messy and may need to be cleaned and formatted before it can be used for machine learning.
  *(This may involve removing missing or invalid data, handling outliers, and encoding categorical variables.)*
- Normalizing the data can help to scale the features so that they are on the same scale.
  *(This can be important for algorithms that use distance measures, as features on different scales can dominate the distance measure.)*
- Preprocessing techniques such as feature selection and feature extraction can help to reduce the dimensionality of the data.
  *(This may improve the performance of the model and reduce the risk of overfitting.)*
- Preprocessing techniques such as feature selection can help to identify the most important features in the data.
  *(This can make the model more interpretable and easier to understand.)*

# Data Preprocessing

## Feature Scaling

**Normalization** *(Min-Max Scaling)*

$$X \triangleq \frac{X - \text{minimum}(X)}{\text{maximum}(X) - \text{minimum}(X)}$$

▲ Algorithms that need bounded values

**Standardization** *(Z-score Normalization)*

$$X \triangleq \frac{X - \mu}{\sigma}$$

▲ Assuming normally distributed data

**Robust Scaling**

$$X \triangleq \frac{X - \text{median}(X)}{\text{IQR}}$$

▲ Datasets with significant outliers

**Max Abs Scaling**

$$X \triangleq \frac{X}{\left|\text{maximum}(X)\right|}$$

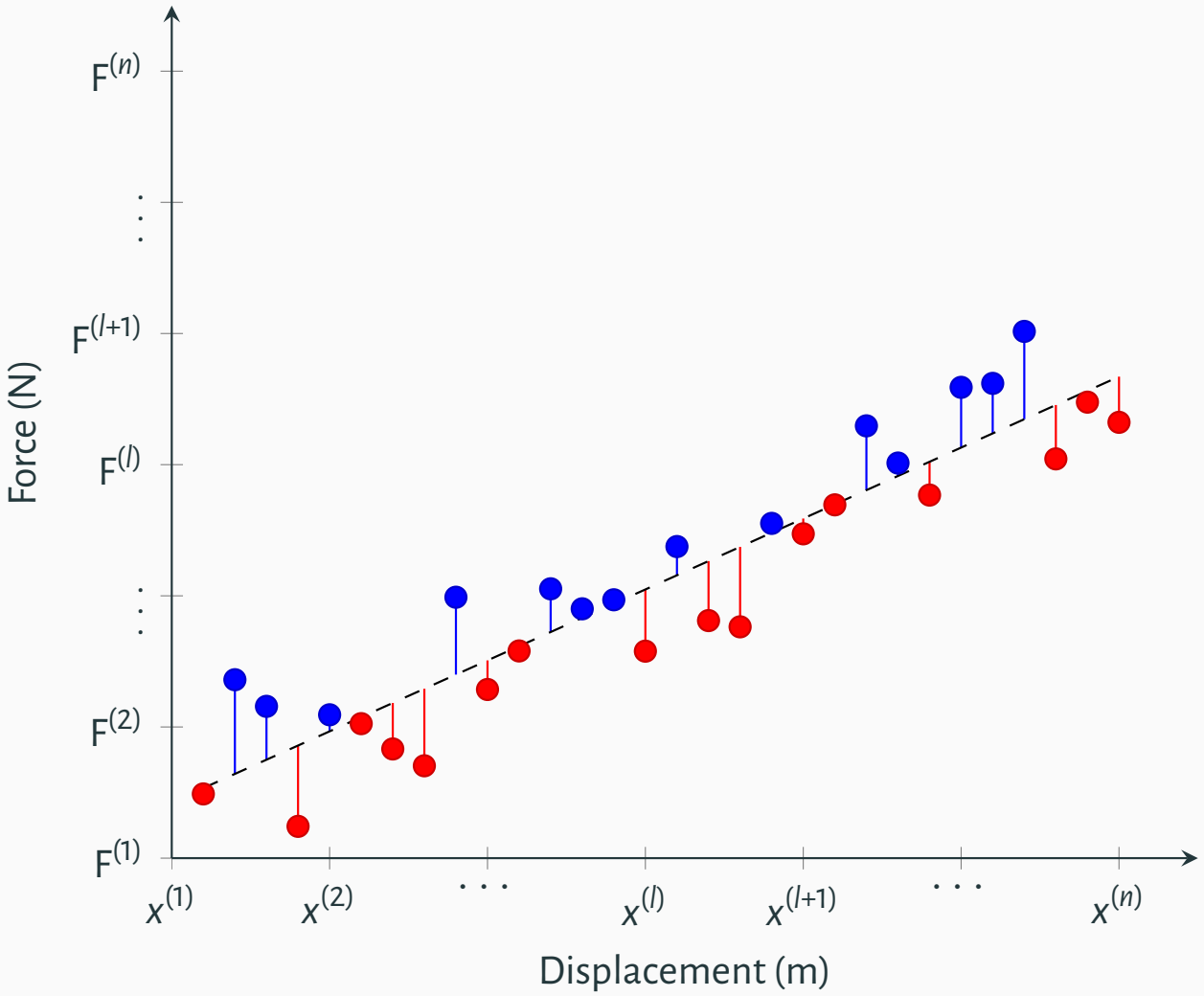
▲ Sparse data, preserving original signs

# Data Preprocessing Template

The code is available @ github.com/a-mhamdi/jlai ⟶ *Codes* ⟶ *Julia* ⟶ *Part-2*

⟶ *Pluto* ⟶ *ml-workflows.jl*

⟶ *Jupyter* ⟶ *ml-workflows.ipynb*


Force (N)
$F^{(n)}$
$F^{(l+1)}$
$F^{(l)}$
$F^{(2)}$
$F^{(1)}$
$x^{(1)}$
$x^{(2)}$
$x^{(l)}$
$x^{(l+1)}$
$x^{(n)}$
Displacement (m)

Consider the example of a spring. Our main goal is to determine the stiffness $k$ of this spring, given some experimental data. The mathematical model *(Hooke's law)*:

$$F = kx \tag{1}$$

Restoring force is proportional to displacement.

**Table 1:** Measurements of couple $\left(x^{(p)},\ F^{(p)}\right)$

| | | | | | |
|---|---|---|---|---|---|
| $x^{(p)}$ | $x^{(1)}$ | $\cdots$ | $x^{(l)}$ | $\cdots$ | $x^{(n)}$ |
| $F^{(p)}$ | $F^{(1)}$ | $\cdots$ | $F^{(l)}$ | $\cdots$ | $F^{(n)}$ |

$$\begin{aligned} F^{(p)} &= \hat{F}^{(p)} + \varepsilon^{(p)} \\ &= kx^{(p)} + \varepsilon^{(p)}, \end{aligned} \tag{2}$$

where $F^{(p)}$ denotes the unknown real value of the force applied to the spring. In order to estimate the stiffness value $k$, we can consider the quadratic criterion:

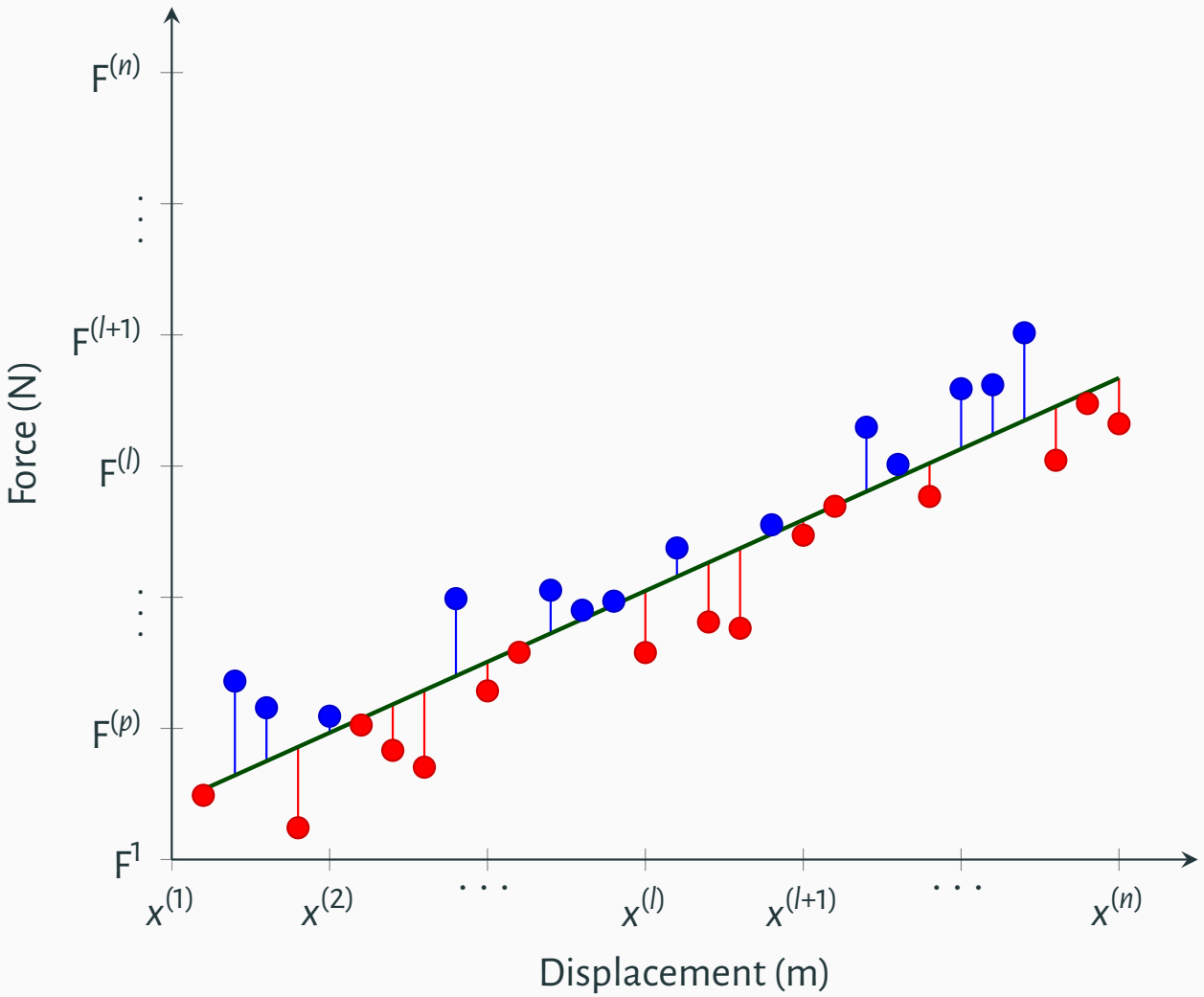
$$\begin{aligned} \mathcal{J} &= \sum_{p=1}^{n} \left(\varepsilon^{(p)}\right)^2 \\ &= \sum_{p=1}^{n} \left(F^{(p)} - kx^{(p)}\right)^2 \end{aligned}$$

$$\frac{\partial \mathcal{J}}{\partial k} = 0 \tag{3}$$

$$2\sum_{p=1}^{n}\left(F^{(p)} - kx^{(p)}\right)\sum_{p=1}^{n}\frac{\partial\left(F^{(p)} - kx^{(p)}\right)}{\partial k} = 0$$

$$\sum_{p=1}^{n}\left(F^{(p)} - kx^{(p)}\right)\sum_{p=1}^{n}x^{(p)} = 0$$

$$\sum_{p=1}^{n}F^{(p)}\,x^{(p)} = k\sum_{p=1}^{n}\left(x^{(p)}\right)^2 \iff \boxed{\hat{k} = \frac{\sum_{p=1}^{n}F^{(p)}\,x^{(p)}}{\sum_{p=1}^{n}\left(x^{(p)}\right)^2}}$$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$F^{(l)}$
$F^{(p)}$
$F^{1}$
$x^{(1)}$
$x^{(2)}$
$x^{(l)}$
$x^{(l+1)}$
$x^{(n)}$
Displacement (m)

# Practical Implementation

The code is available @ github.com/a-mhamdi/jlai ⟶ *Codes* ⟶ *Julia* ⟶ *Part-2*

⟶ *Pluto* ⟶ *simple-regression.jl*

⟶ *Jupyter* ⟶ *simple-regression.ipynb*

This example consists on determining the unknown couple $(y_0,\ v_0)$ of a mobile solid. We assume that the trajectory is linear. The mathematical model that relates the position $y$ to time $t$ is given by this equation:

$$y = y_0 + v_0 t \tag{4}$$

**Table 2:** Measurements of position $y$

| | | | | | |
|---|---|---|---|---|---|
| $t^{(k)}$ | $t^{(1)}$ | $\cdots$ | $t^{(l)}$ | $\cdots$ | $t^{(n)}$ |
| $y^{(k)}$ | $y^{(1)}$ | $\cdots$ | $y^{(l)}$ | $\cdots$ | $y^{(n)}$ |

$$\begin{aligned} y^{(k)} &= \hat{y}^{(k)} + \varepsilon^{(k)} \\ &= y_0 + v_0 t^{(k)} + \varepsilon^{(k)}, \end{aligned} \tag{5}$$

where $y^{(k)}$ denotes the unknown real value of the position $y$ at time point $t^{(k)}$.

In order to estimate the values taken by the couple $\begin{bmatrix} y_0, & v_0 \end{bmatrix}^T$, we consider the quadratic criterion again, as follows:

$$\begin{aligned} \mathcal{J} &= \sum_{k=1}^{n} \left(\varepsilon^{(k)}\right)^2 \\ &= \boldsymbol{\varepsilon}^T \times \boldsymbol{\varepsilon} \end{aligned}$$

The vector $\boldsymbol{\varepsilon}$ is set by $\varepsilon^{(k)}$, $\forall k \geq 1$:

$$\varepsilon = \begin{bmatrix} \varepsilon^{(1)} & \cdots & \varepsilon^{(n)} \end{bmatrix}^T$$

$$\frac{\partial \mathcal{J}}{\partial \begin{bmatrix} y_0 \\ \\ v_0 \end{bmatrix}} = 0 \tag{6}$$

# Practical Implementation

The code is available @ github.com/a-mhamdi/jlai ⟶ *Codes* ⟶ *Julia* ⟶ *Part-2*

⟶ *Pluto* ⟶ *multivariable-regression.jl*

⟶ *Jupyter* ⟶ *multivariable-regression.ipynb*

Consider the following multivariable equation:

$$y = \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_m x_m \tag{7}$$

For a particular single measurement, eq. (7) can be updated as

$$y^{(k)} = \theta_1 x_1^{(k)} + \theta_2 x_2^{(k)} + \cdots + \theta_m x_m^{(k)} + \varepsilon^{(k)}$$

We denote hereafter by $\boldsymbol{\theta}$ the vector $\begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_m \end{bmatrix}$. The function $y^{(k)}$ becomes:

$$y^{(k)} = \underbrace{\left[x_1^{(k)}, x_2^{(k)}, \cdots, x_m^{(k)}\right]}_{\mathbf{x}^{(k)\,T}} \boldsymbol{\theta} + \varepsilon^{(k)}$$

We assume that we have $n$ measurements for $y$. Then we can transform the previous equation into

$$\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon},$$

where $\mathbf{y}^T = \left[y^{(1)}, y^{(2)}, \cdots, y^{(n)}\right], \mathbf{X} = \begin{bmatrix} \mathbf{x}^{(1)\,T} \\ \mathbf{x}^{(2)\,T} \\ \vdots \\ \mathbf{x}^{(n)\,T} \end{bmatrix}$, and $\boldsymbol{\varepsilon}^T = \left[\varepsilon^{(1)}, \varepsilon^{(2)}, \cdots, \varepsilon^{(n)}\right]$.

We can consider the mean squared error or quadratic criterion in order to compute the approximated value of $\boldsymbol{\theta}$:

$$\begin{aligned} \mathcal{J} &= \sum_{k=1}^{n} \left(\varepsilon^{(k)}\right)^2 \\ &= \boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon} \end{aligned}$$

The best well estimated value of $\hat{\boldsymbol{\theta}}$ corresponds to the absolute minimum of $\mathcal{J}$. This leads to calculate the gradient of $\mathcal{J}$ with respect to $\boldsymbol{\theta}$:

$$\frac{\partial \mathcal{J}}{\partial \boldsymbol{\theta}} = \frac{\partial \left(\boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon}\right)}{\partial \boldsymbol{\theta}}$$

$$\frac{\partial \left(\boldsymbol{\varepsilon}^T \boldsymbol{\varepsilon}\right)}{\partial \boldsymbol{\theta}} = 2 \left(\frac{\partial \boldsymbol{\varepsilon}}{\partial \boldsymbol{\theta}}\right)^T \boldsymbol{\varepsilon}$$

Recall that $\boldsymbol{\varepsilon} = \mathbf{y} - \mathbf{X}\boldsymbol{\theta}$, the term $\dfrac{\partial \boldsymbol{\varepsilon}}{\partial \boldsymbol{\theta}}$ hence becomes:

$$\frac{\partial \boldsymbol{\varepsilon}}{\partial \boldsymbol{\theta}} = -\mathbf{X}$$

$$\begin{aligned} \frac{\partial J}{\partial \boldsymbol{\theta}} &= 2(-\mathbf{X})^{\mathsf{T}} \left(\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\right) \\ &= 0 \end{aligned}$$

The vector $\hat{\boldsymbol{\theta}}$ is given by $\boxed{\hat{\boldsymbol{\theta}} = \left(\mathbf{X}^T\mathbf{X}\right)^{-1}\mathbf{X}^T\mathbf{y}}$

**$\mathbf{X}^T\mathbf{X}$ is not invertible (singular/degenerate)**

▼ Redundant Features

Some features are linearly dependent, i.e, $\exists$ some $x_p \propto$ some $x_l$, e.g., $x_p$ in feet and $x_l$ in m.

▼ Too many features

Fewer observations compared to the number of features, i.e, $m \geq n$.

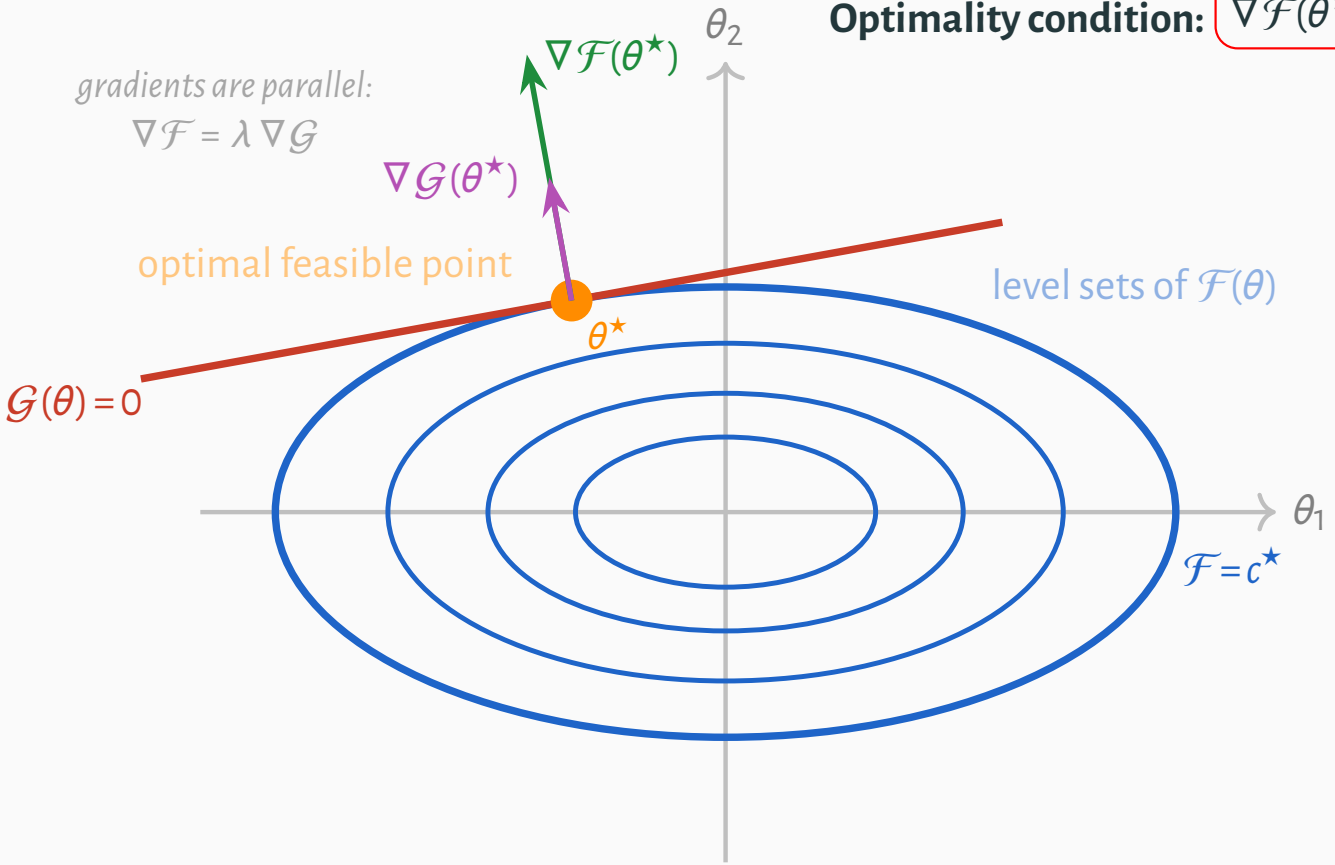
- Delete some features
- Add extra observations
- Use regularization: $\underbrace{\boxed{\lambda \sum_{i=2}^{m} |\theta_i|}}_{\text{Lasso}} \underbrace{\boxed{\frac{1}{2}\lambda \sum_{i=2}^{m} \theta_i^2}}_{\text{Ridge}} \underbrace{\boxed{r\lambda \sum_{i=2}^{m} |\theta_i| + \frac{(1-r)}{2}\lambda \sum_{i=2}^{m} \theta_i^2}}_{\text{Elastic net}}$

$$\hat{\boldsymbol{\theta}} = \left( \mathbf{X}^{\mathsf{T}}\mathbf{X} + \lambda \begin{bmatrix} 0 & \cdots & \cdots & \cdots & 0 \\ \vdots & 1 & \ddots & & \vdots \\ \vdots & \ddots & 1 & \ddots & \vdots \\ \vdots & & \ddots & 1 & 0 \\ 0 & \cdots & \cdots & 0 & 1 \end{bmatrix}_{(m,\,m)} \right)^{-1} \mathbf{X}^{\mathsf{T}}\mathbf{y}$$

# Lagrange Multipliers

**Problem:** $\max_{\theta} \; \mathcal{F}(\theta) \quad \text{subject to} \quad \mathcal{G}(\theta) = 0$

**Optimality condition:** $\boxed{\nabla \mathcal{F}(\theta^\star) \;=\; \lambda \, \nabla \mathcal{G}(\theta^\star)}$




**Lagrangian:** $\mathcal{L}(\theta, \lambda) = \mathcal{F}(\theta) - \lambda \, \mathcal{G}(\theta), \quad \nabla_\theta \mathcal{L} = \mathbf{0}, \quad \mathcal{G}(\theta^\star) = 0$

# Regularization: Lasso vs. Ridge

**Why Regularize?**

Minimize the **penalized loss**

$$\mathcal{L}(\boldsymbol{\theta}) = \underbrace{\sum_{k=1}^{n} \left(y^{(k)} - \hat{y}^{(k)}\right)^2}_{\text{data fit}} + \lambda\, \mathcal{G}(\boldsymbol{\theta})$$

Regularization controls model complexity and prevents overfitting.

**Ridge — $L_2$ ($p = 2$)**

$$\mathcal{G}(\boldsymbol{\theta}) = \|\boldsymbol{\theta}\|_2^2 = \sum_i \theta_i^2$$

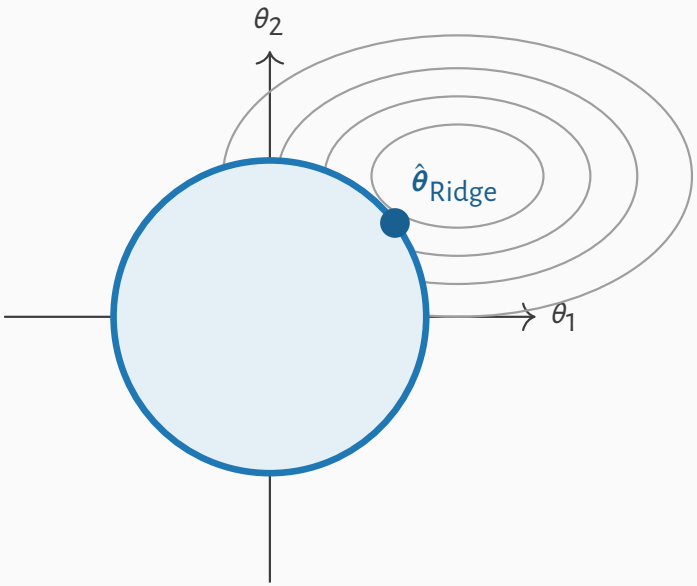
- Isosurface: **circle / sphere**
- Shrinks all weights *smoothly*
- No exact zeros $\Longrightarrow$ keeps all features

**Lasso — $L_1$ ($p = 1$)**

$$\mathcal{G}(\boldsymbol{\theta}) = \|\boldsymbol{\theta}\|_1 = \sum_i |\theta_i|$$

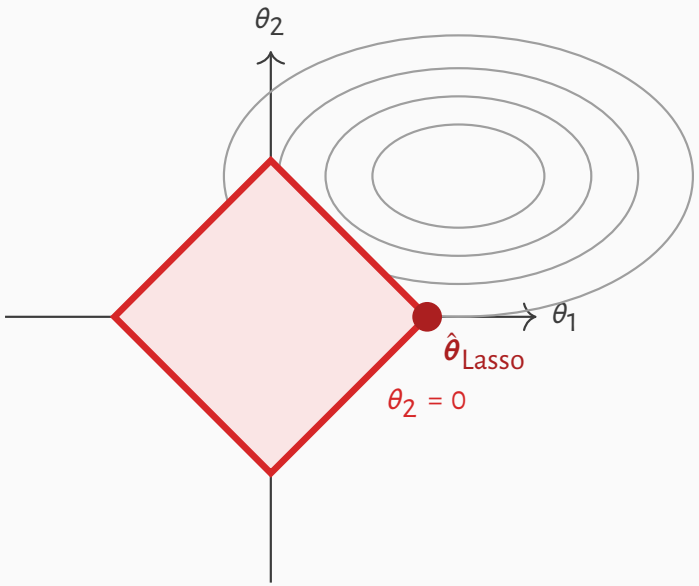
- Isosurface: **diamond / cross-polytope**
- Solution hits *corners* $\Longrightarrow$ sparse weights
- Exact zeros $\Longrightarrow$ built-in **feature selection**

# Geometry Intuition




Smooth tangency $\Longrightarrow$ no exact zeros

**Ridge ($L_2$)**




Corner tangency $\Longrightarrow$ exact zeros

**Lasso ($L_1$)**

The loss contours expand from the OLS minimum $\hat{\boldsymbol{\theta}}_{\text{OLS}}$ until they first hit the constraint region.

- The diamond's corners lie on the axes $\Longrightarrow$ tangency happens at a corner, setting one parameter to **exactly 0** (sparsity).
- The circle has no corners $\Longrightarrow$ tangency is smooth, all weights are merely **shrunk**.

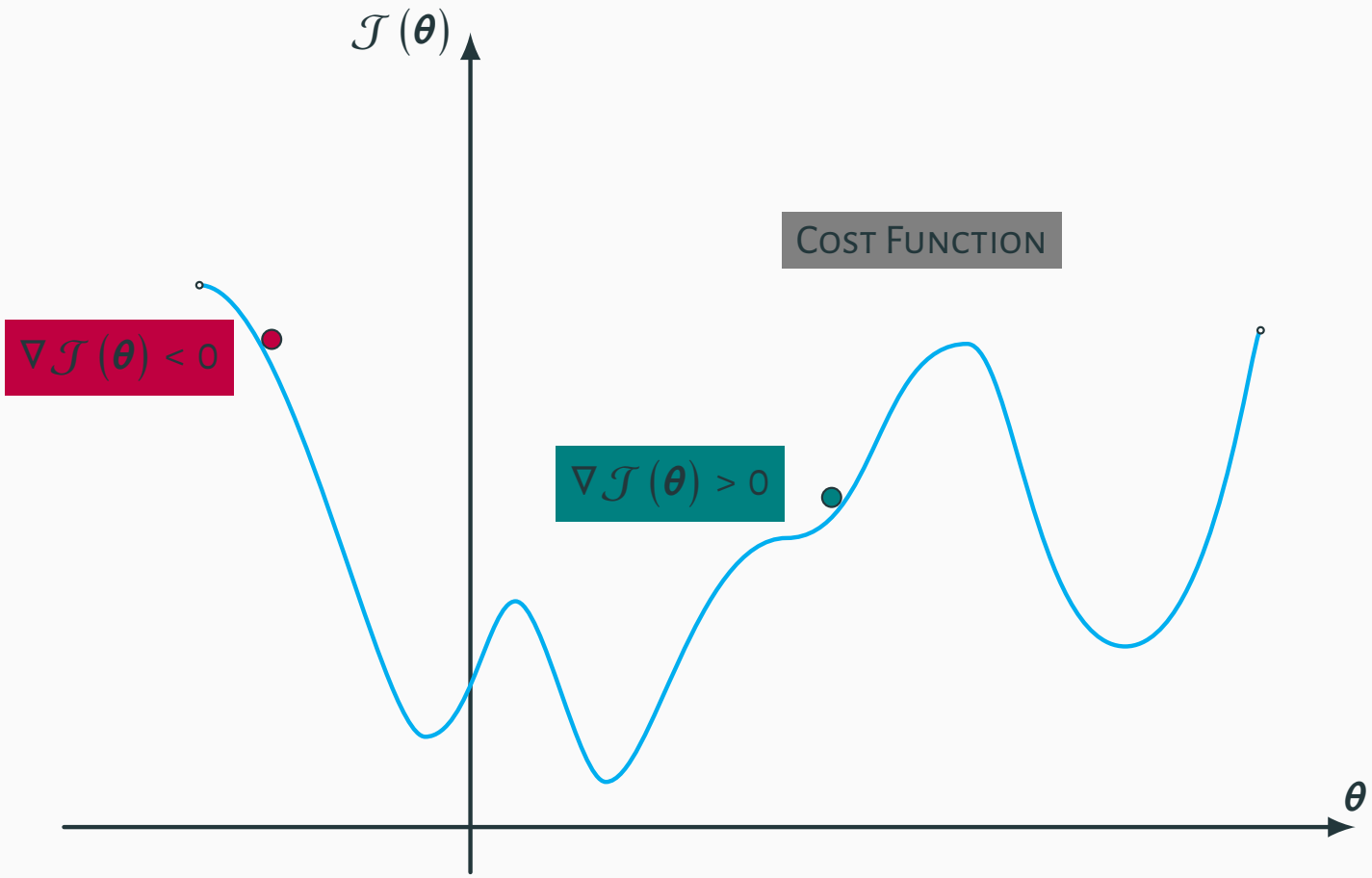
# Practical Implementation

The code is available @ github.com/a-mhamdi/jlai ⟶ *Codes* ⟶ *Julia* ⟶ *Part-2*

⟶ *Pluto* ⟶ *polynomial-regression.jl*

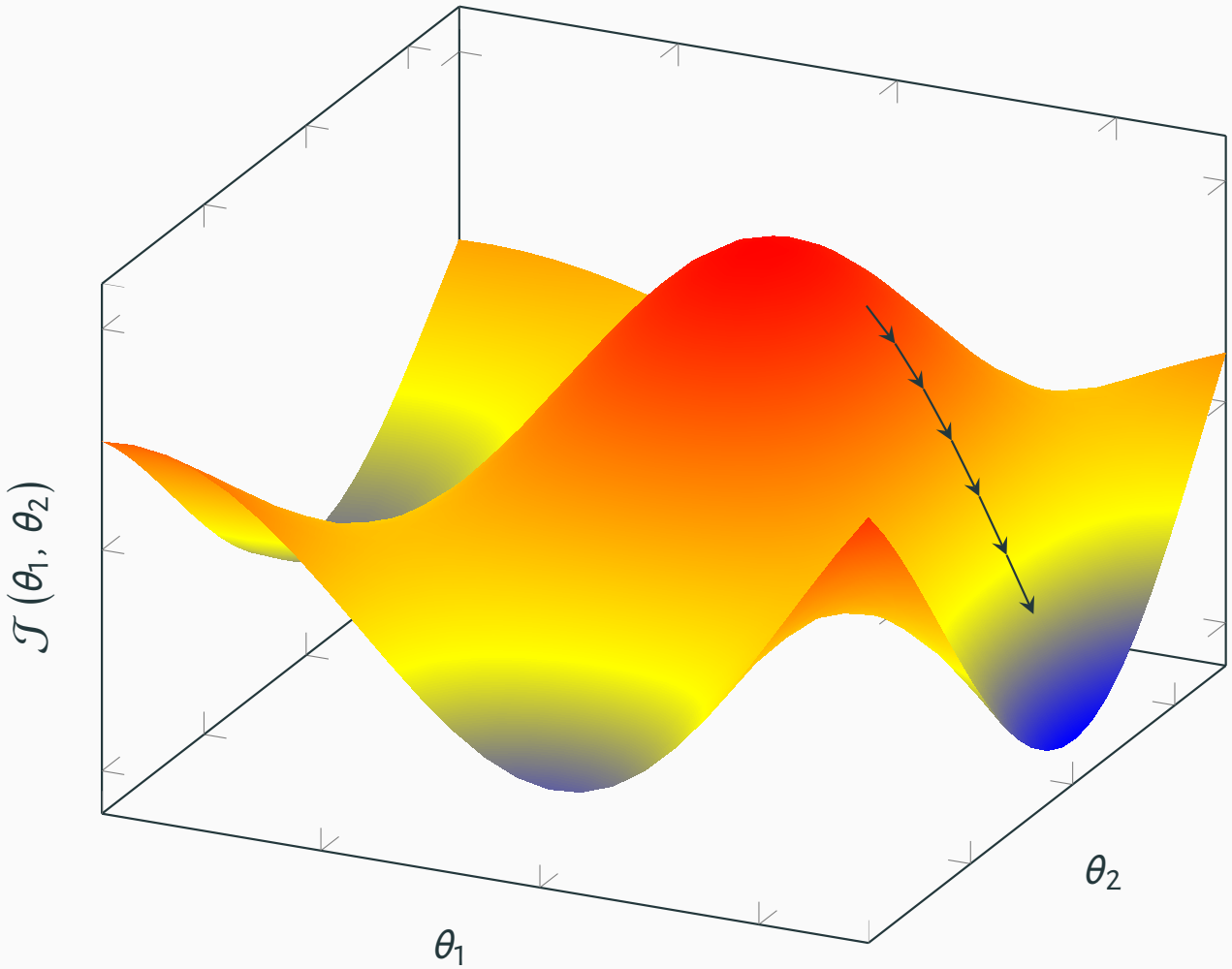
⟶ *Jupyter* ⟶ *polynomial-regression.ipynb*

# Gradient Descent




$$\underbrace{\boldsymbol{\theta}}_{\text{New value}} \triangleq \underbrace{\boldsymbol{\theta}}_{\text{Old value}} - \overbrace{\alpha}^{\text{Weighting factor}} \underbrace{\nabla \mathcal{J}(\boldsymbol{\theta})}_{\text{Gradient term is steepest ascent}}$$

# Gradient Descent




① Start with some random values of $\theta_1$ and $\theta_2$

② Keep changing $\theta_1$ and $\theta_2$ to reduce $\mathcal{J}(\theta_1, \theta_2)$ until we hopefully end up at minimum

# Gradient Descent

$$\boxed{\theta_i \triangleq \theta_i - \underbrace{\alpha}_{\text{Learning Rate}} \frac{\partial \mathcal{J}}{\partial \theta_i}}$$

Recall that

$$\mathcal{J} = \frac{1}{2n} \sum_{k=1}^{n} \left( y^{(k)} - h_{\boldsymbol{\theta}} \left( \mathbf{x}^{(k)} \right) \right)^2 \implies \frac{\partial \mathcal{J}}{\partial \theta_i} = -\frac{1}{n} \sum_{k=1}^{n} \left( y^{(k)} - h_{\boldsymbol{\theta}} \left( \mathbf{x}^{(k)} \right) \right) x_i^{(k)}$$

$$\boxed{\theta_1 \triangleq \theta_1 + \alpha \frac{1}{n} \sum_{k=1}^{n} \left( y^{(k)} - h_{\boldsymbol{\theta}} \left( \mathbf{x}^{(k)} \right) \right) x_1^{(k)}} \quad \boxed{\theta_2 \triangleq \theta_2 + \alpha \frac{1}{n} \sum_{k=1}^{n} \left( y^{(k)} - h_{\boldsymbol{\theta}} \left( \mathbf{x}^{(k)} \right) \right) x_2^{(k)}}$$

$$\vdots$$

$$\boxed{\theta_m \triangleq \theta_m + \alpha \frac{1}{n} \sum_{k=1}^{n} \left( y^{(k)} - h_{\boldsymbol{\theta}} \left( \mathbf{x}^{(k)} \right) \right) x_m^{(k)}}$$

# Gradient Descent

$$\boxed{\theta_i \triangleq \theta_i - \underbrace{\alpha}_{\text{Learning Rate}} \frac{\partial \mathcal{J}}{\partial \theta_i}}$$

Recall that with $\mathbf{L}_2$ regularization term

$$\mathcal{J} = \frac{1}{2n}\sum_{k=1}^{n}\left(y^{(k)} - h_{\boldsymbol{\theta}}\left(\mathbf{x}^{(k)}\right)\right)^2 + \frac{\lambda}{2n}\sum_{i=2}^{m}\theta_i^2 \quad \Longrightarrow \quad \frac{\partial \mathcal{J}}{\partial \theta_i} = -\frac{1}{n}\sum_{k=1}^{n}\left(y^{(k)} - h_{\boldsymbol{\theta}}\left(\mathbf{x}^{(k)}\right)\right)x_i^{(k)} + \frac{\lambda}{n}\theta_i \quad \text{iff } i \neq 1$$

$$\boxed{\theta_1 \triangleq \left(1 - \cancel{\alpha\frac{\lambda}{n}}\right)\theta_1 + \alpha\frac{1}{n}\sum_{k=1}^{n}\left(y^{(k)} - h_{\boldsymbol{\theta}}\left(\mathbf{x}^{(k)}\right)\right)x_1^{(k)}}$$

$$\boxed{\theta_2 \triangleq \left(1 - \alpha\frac{\lambda}{n}\right)\theta_2 + \alpha\frac{1}{n}\sum_{k=1}^{n}\left(y^{(k)} - h_{\boldsymbol{\theta}}\left(\mathbf{x}^{(k)}\right)\right)x_2^{(k)}}$$

$$\vdots$$

$$\boxed{\theta_m \triangleq \left(1 - \alpha\frac{\lambda}{n}\right)\theta_m + \alpha\frac{1}{n}\sum_{k=1}^{n}\left(y^{(k)} - h_{\boldsymbol{\theta}}\left(\mathbf{x}^{(k)}\right)\right)x_m^{(k)}}$$

# Hands-on Example (1/3)

**Task #2**

The yield $y$ of a chemical process is a random variable whose value is considered to be a linear function of the temperature $x$. The following data of corresponding values of $x$ and $y$ is found:

| Temperature in °C ($x$) | 0 | 25 | 50 | 75 | 100 |
|---|---|---|---|---|---|
| Yield in grams ($y$) | 14 | 38 | 54 | 76 | 95 |

The linear regression model $y = \theta_1 + \theta_2 x$ is used. Determine the values of $\theta_0$, $\theta_1$.

1. Using normal equation,
2. Using gradient descent for 5 iterations, given the following initial settings:

$$\alpha = 0.01 \qquad \text{and} \qquad \boldsymbol{\theta} = \begin{bmatrix} 1 \\ \\ 0.5 \end{bmatrix}$$

# Hands-on Example (2/3)

① **Normal Equation**

$$\mathbf{y} = \begin{bmatrix} 14 \\ 38 \\ 54 \\ 76 \\ 95 \end{bmatrix} \quad \text{and} \quad \mathbf{X} = \begin{bmatrix} 1 & 0 \\ 1 & 25 \\ 1 & 50 \\ 1 & 75 \\ 1 & 100 \end{bmatrix} \implies \hat{\boldsymbol{\theta}} = \begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{bmatrix} = \begin{bmatrix} 15.4 \\ 0.8 \end{bmatrix}$$

② **Stochastic Gradient Descent**

| $k$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $y$ | 14 | 38 | 54 | 76 | 95 |
| $h_{\boldsymbol{\theta}}\left(\mathbf{x}^{(k)}\right)$ | 1 | 13.63 | 330.999 | –9894.410 | 734688.376 |
| $\hat{\boldsymbol{\theta}} = \begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{bmatrix}$ | $\begin{bmatrix} 1.13 \\ 0.5 \end{bmatrix}$ | $\begin{bmatrix} 1.374 \\ 6.592 \end{bmatrix}$ | $\begin{bmatrix} -1.396 \\ -131.907 \end{bmatrix}$ | $\begin{bmatrix} 98.308 \\ 7345.901 \end{bmatrix}$ | $\begin{bmatrix} -7247.626 \\ -727247.475 \end{bmatrix}$ |

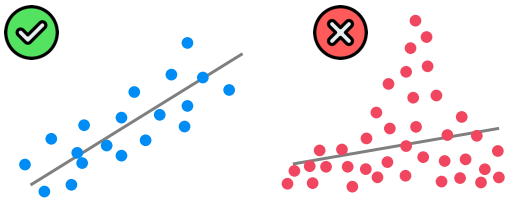
# Hands-on Example (3/3)

```julia
X = [1 0; 1 25; 1 50; 1 75; 1 100]  # Features
y = [14, 38, 54, 76, 95]  # Target

alpha, n, theta = 0.01, 5, [1; .5]  # \alpha, n, \theta = ...

J = []
for k in 1:5
   h_th = X[k, :]' * theta
   println("h_th = $(h_th)")
   cost = (y[k] - h_th)^2
   push!(J, cost);
   theta += theta * (y[k] - h_th) * X[k, :]  # \theta = ...
   println("theta = $(theta)\n")
end
```

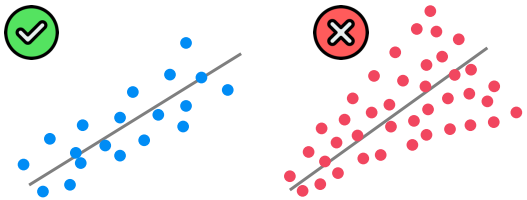
https://github.com/a-mhamdi/journey-into-ML/blob/main/Julia/gradient-descent.jl

# Assumptions of Linear Regression

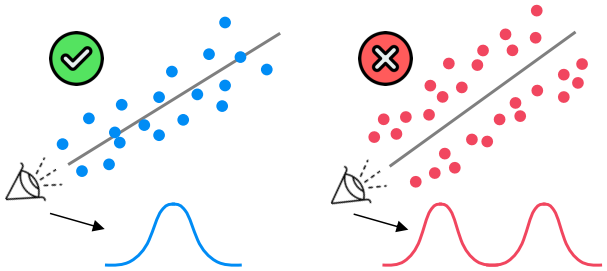
## 1. Linearity
(Linear relationship between Y and each X)



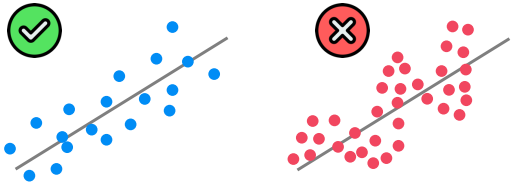
## 2. Homoscedasticity
(Equal variance)



## 3. Multivariate Normality
(Normality of error distribution)



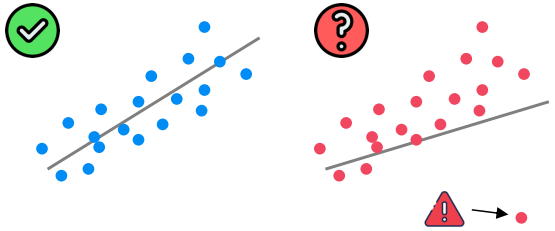
## 4. Independence
(of observations. Includes “no autocorrelation”)



## 5. Lack of Multicollinearity
(Predictors are not correlated with each other)

$X_1 \nsim X_2$ $X_1 \sim X_2$

## 6. The Outlier Check
(This is not an assumption, but an “extra”)



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Source

# Evaluation metrics (1/2)

**Mean Absolute Error (MAE)** measures the average difference of absolute values between predicted and actual targets.

$$\textbf{MAE} = \frac{1}{n}\sum_{k=1}^{n}\left|y^{(k)} - \hat{y}^{(k)}\right|$$

**Root Mean Squared Error (RMSE)** measures the root of the average squared difference between predicted and actual values.

$$\textbf{RMSE} = \sqrt{\frac{1}{n}\sum_{k=1}^{n}\left(y^{(k)} - \hat{y}^{(k)}\right)^2}$$

**Mean Absolute Percentage Error (MAPE)** is a measure of the prediction quality. It is equivalent to doing weighted **MAE**.

$$\textbf{MAPE} = \frac{1}{n}\sum_{k=1}^{n}\left|\frac{y^{(k)} - \hat{y}^{(k)}}{y^{(k)}}\right| \times 100\%$$

*A lower error indicates a better fit of the model to the data.*

# Evaluation metrics (2/2)

**R-squared** is a statistical measure that quantifies the proportion of the variance in the dependent variable that is explained by the independent variables in the model.

$$\mathcal{R}^2 = 1 - \frac{SS_{residuals}}{SS_{total}} = 1 - \frac{\sum_{k=1}^{n} \left(y^{(k)} - \hat{y}^{(k)}\right)^2}{\sum_{k=1}^{n} \left(y^{(k)} - \bar{y}\right)^2}$$

*1 indicates that the model explains **ALL** the variance in the dependent variable*

*0 indicates that the model explains **NONE** of the variance in the dependent variable*

**Adjusted R-squared** is a modified version of R-squared that accounts for the number of independent variables in the model.

$$\textbf{Adjusted } \mathcal{R}^2 = 1 - \left(1 - \mathcal{R}^2\right) \frac{n-1}{n-m-1}$$

# Hands-on Example (1/3)

**Task #3**

We consider a univariate regression problem with only two predictors $m = 2$. Compute the error metrics for the given data.

| $y$ | 1 | 1 | -2 | 5 | -3.5 | 1 |
|---|---|---|---|---|---|---|
| $\hat{y}$ | 0.9 | 0.85 | -2.2 | 4.8 | -3.3 | 1.2 |

# Hands-on Example (2/3)

**MAE:** $$\frac{1}{6}\sum_{k=1}^{6}\left|y^{(k)}-\hat{y}^{(k)}\right| \approx 0.175$$

**RMSE:** $$\sqrt{\frac{1}{6}\sum_{k=1}^{6}\left(y^{(k)}-\hat{y}^{(k)}\right)^2} \approx 0.179$$

**MAPE:** $$\frac{1}{6}\sum_{k=1}^{6}\left|\frac{y^{(k)}-\hat{y}^{(k)}}{y^{(k)}}\right| \times 100\% \approx 10.786$$

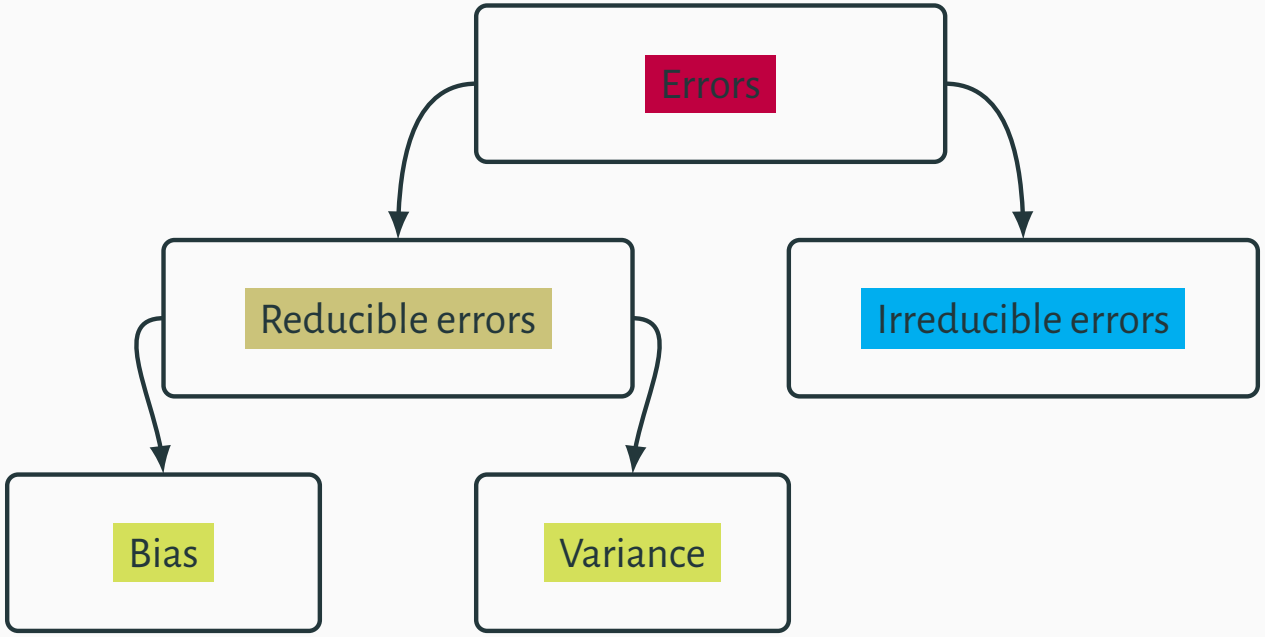
**$\mathcal{R}^2$:** $$1-\frac{\sum_{k=1}^{6}\left(y^{(k)}-\hat{y}^{(k)}\right)^2}{\sum_{k=1}^{6}\left(y^{(k)}-\bar{y}\right)^2} \approx 0.996$$

**Adjusted $\mathcal{R}^2$:** $$1-\left(1-\mathcal{R}^2\right)\frac{6-1}{6-2-1} \approx 0.992$$

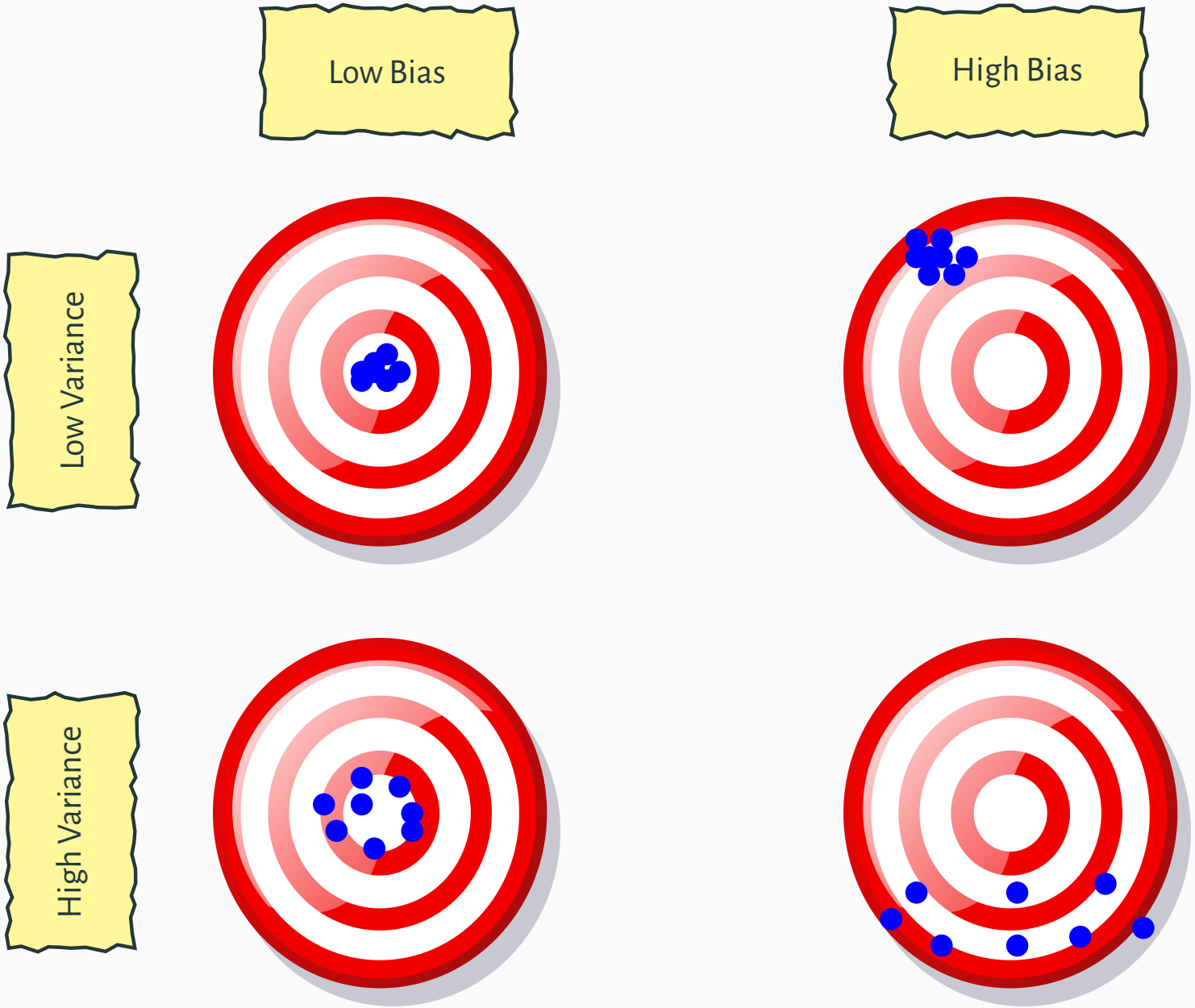
# Hands-on Example (3/3)

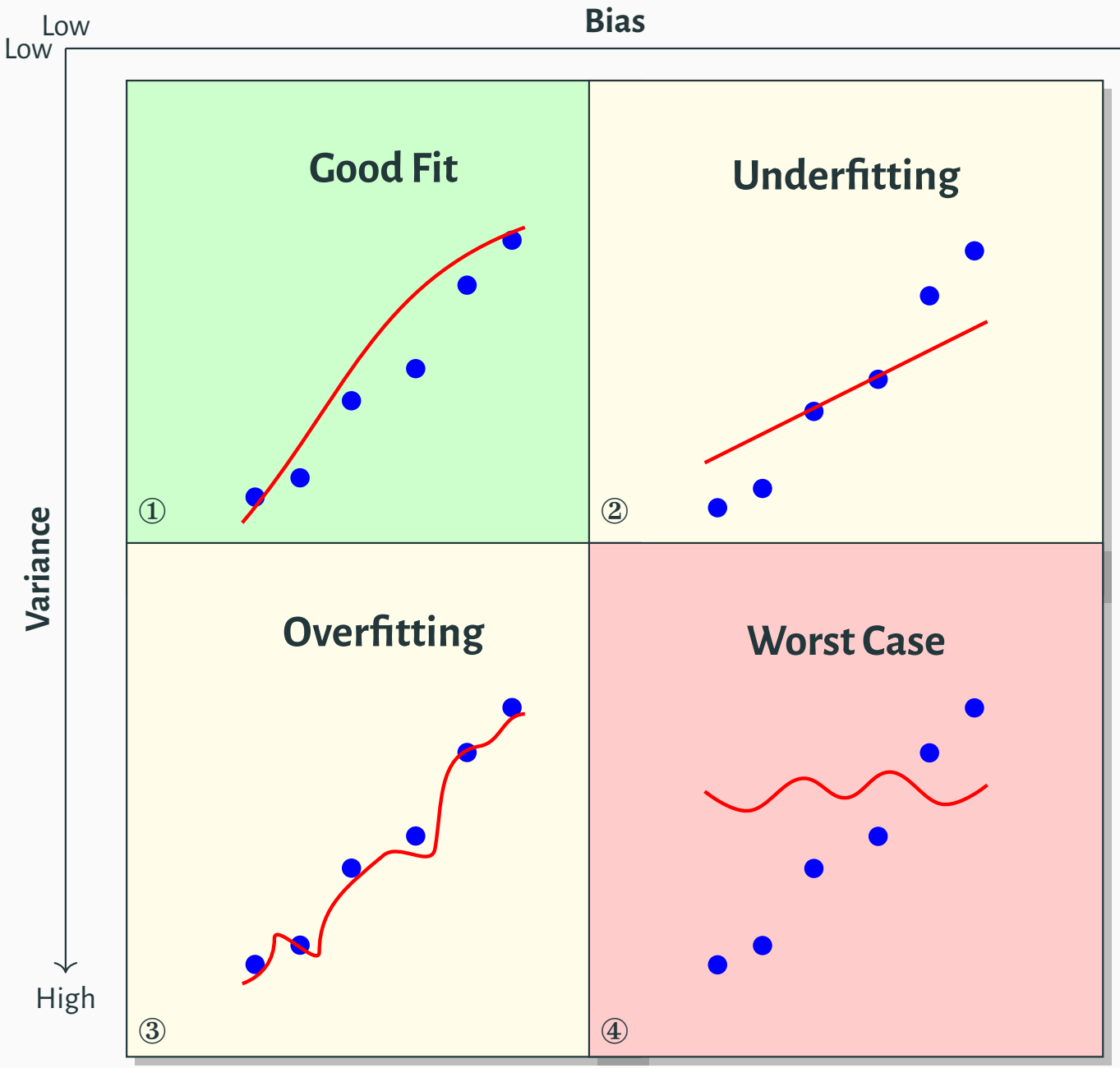
```julia
y = [1 1 -2 5 -3.5 1]
ŷ = [.9 .85 -2.2 4.8 -3.3 1.2]

mae = 1/6 * sum( abs.(y .- ŷ) ) # 0.17500000000000007
rmse = √(1/6 * sum( (y .- ŷ).^2 )) # 0.1791182105017057
mape = 1/6 * sum( abs.((y .- ŷ)./y) ) * 100 # 10.785714285714288

ȳ = 1/6 * sum(y); # 0.41666666666666663 (y\bar = ...)
# using Statistics; ȳ = mean(y) (y\bar = mean(y))
r2 = 1 - sum( (y .- ŷ).^2 )/sum( (y .- ȳ).^2 ) # 0.9955448408871745
adj_r2 = 1 - (1-r2) * (6-1)/(6-2-1) # 0.9925747348119576
```

# Errors in ML


Errors
Reducible errors
Irreducible errors
Bias
Variance

# Bias-Variance Tradeoff






①

- Optimal complexity
- Generalizes well
- Balanced tradeoff

②

- Too simple
- High training error
- Misses patterns

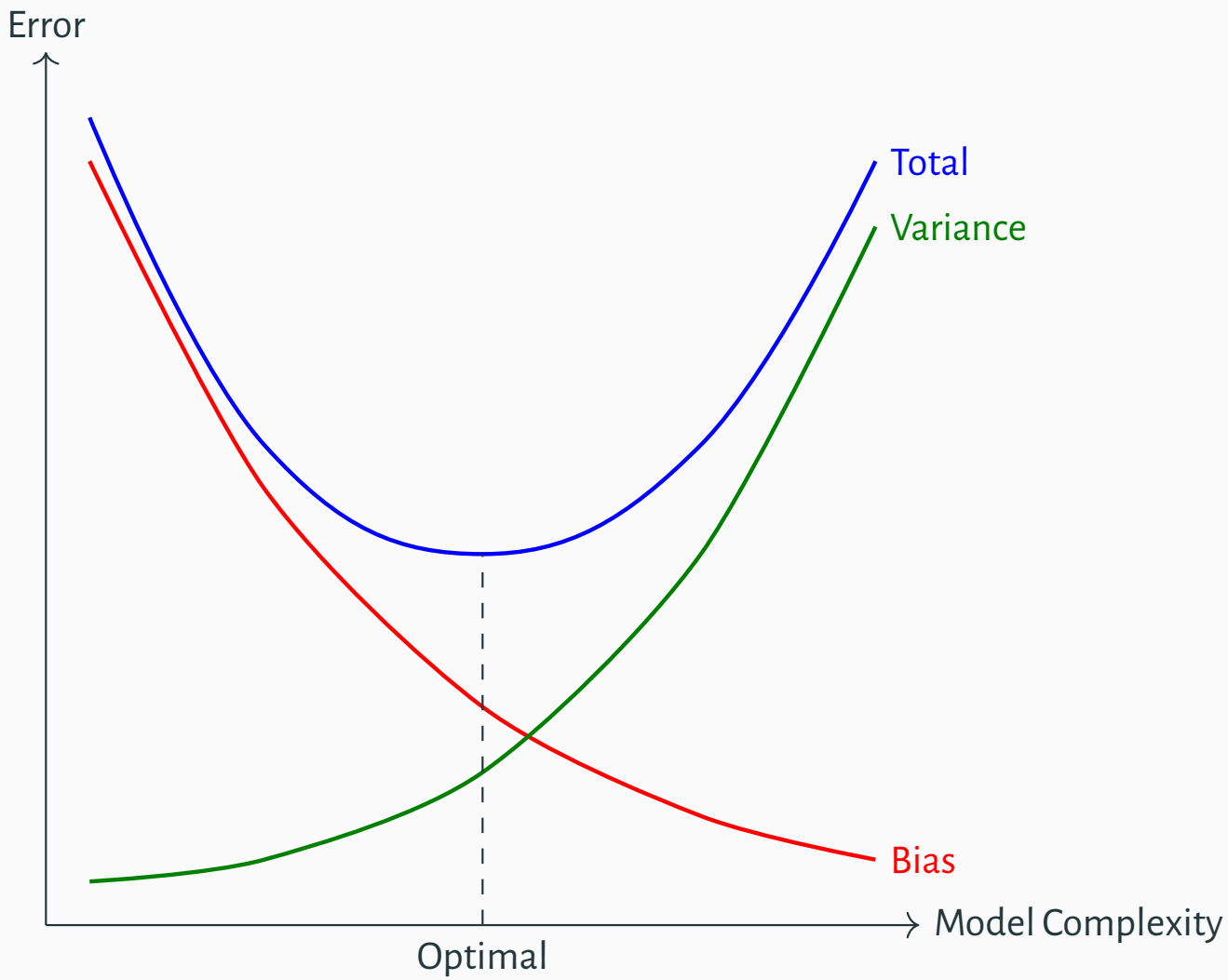
③

- Too complex
- Fits noise
- Poor generalization

④

- Wrong assumptions
- Inconsistent predictions
- Unreliable model

# Error Curve vs. Model Complexity



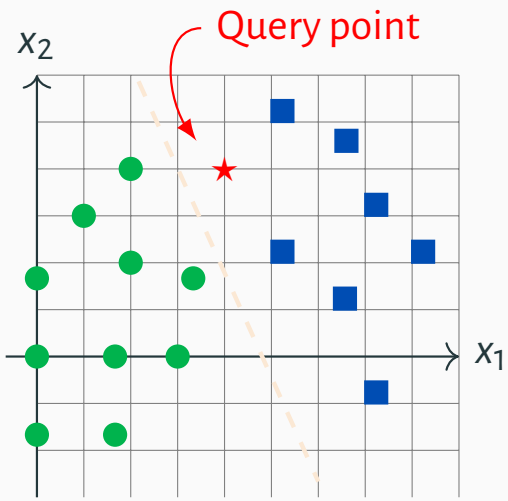
# Bias-Variance Tradeoff: Understanding & Resolving

| Underfitting & High Bias | Overfitting & High Variance |
|---|---|
| Model is too simple to capture data patterns | Model memorizes noise/outliers in training data |
| ▸ High training error<br>▸ Poor performance on both seen/unseen data | ▸ Low training error but high test error<br>▸ Excellent on training data, poor on new data |
| ▸ Oversimplified model architecture<br>▸ Insufficient features<br>▸ Excessive regularization | ▸ Overly complex model<br>▸ Too many features<br>▸ Insufficient training data |

| Fix Underfitting (High Bias) | Fix Overfitting (High Variance) |
|---|---|
| ▸ Increase model complexity<br>▸ Add relevant features<br>▸ Reduce regularization<br>▸ Train longer with better optimization | ▸ Get more training data<br>▸ Use regularization (L1/L2)<br>▸ Implement early stopping<br>▸ Use cross-validation |

# Introduction

Classification is a type of supervised machine learning algorithm. A model is trained on a set of *labeled data*, where each data point is associated with a known class or category. The goal of the algorithm is to learn the relationship between the *input features* $x$ and the corresponding *output classes* $y$, so that it can accurately predict the class of new, unseen query points.



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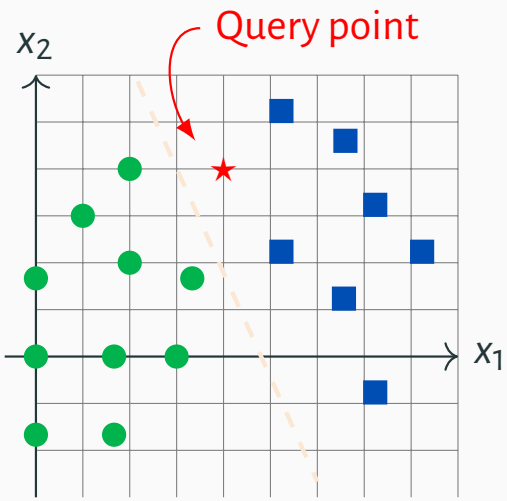


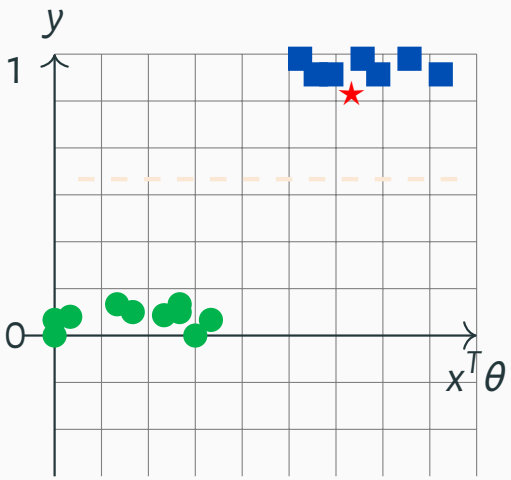

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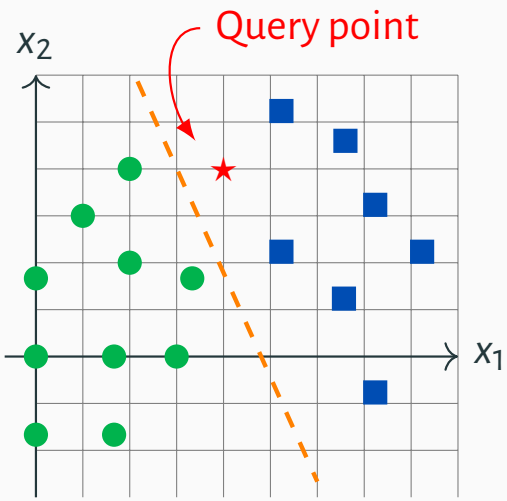


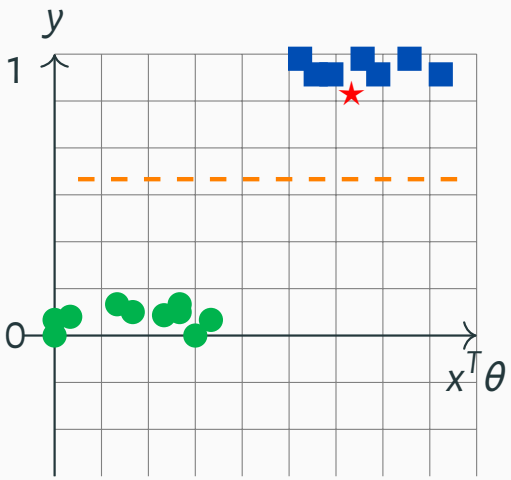

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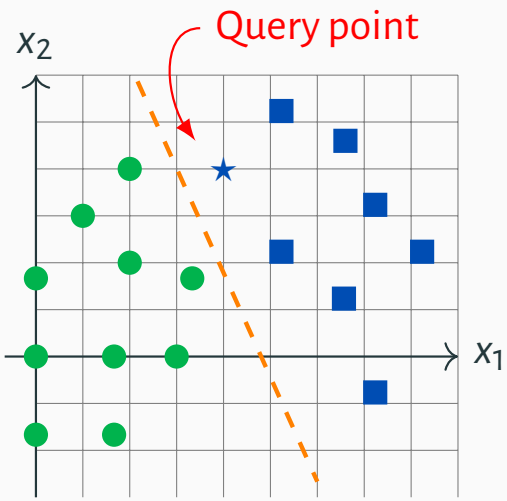
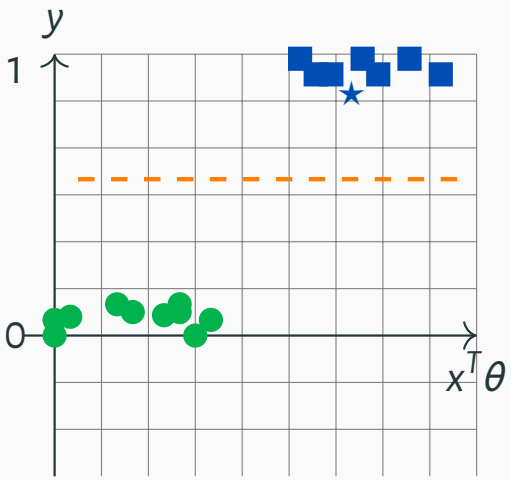
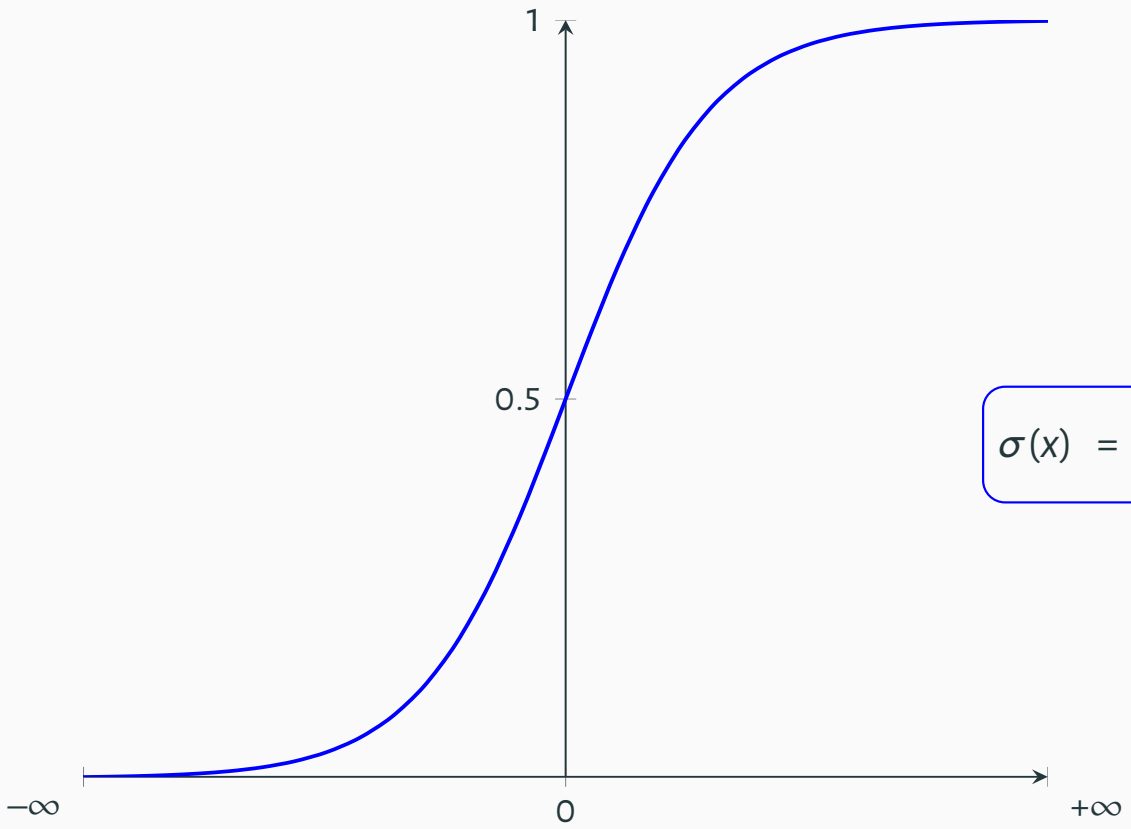


# Logistic or S-shaped function $\sigma$




$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

- $\sigma$ squashes range of distance from $]-\infty, +\infty[$ to $[0, 1]$
- $\sigma$ is differentiable and easy to compute: $\dot{\sigma} = \sigma \times (1 - \sigma)$

# Decision boundary

$$y = \sigma\left(\theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_m x_m\right)$$

$$y = \frac{1}{1 + e^{-\mathbf{x}^T \boldsymbol{\theta}}}$$

**Hypothesis**

$$h_{\boldsymbol{\theta}}(\mathbf{x}) = P\left(y = 1 | \mathbf{x}; \boldsymbol{\theta}\right) = \frac{1}{1 + e^{-\mathbf{x}^T \boldsymbol{\theta}}}$$

For some given $x^{(k)}$

$$h_{\boldsymbol{\theta}}\left(\mathbf{x}^{(k)}\right) = P\left(y = 1 | \mathbf{x}^{(k)}; \boldsymbol{\theta}\right) = \frac{1}{1 + e^{-\mathbf{x}^{(k)\,T} \boldsymbol{\theta}}}$$

**Cost function**

$$\mathcal{J} = \begin{cases} -\ln\left(h_{\boldsymbol{\theta}}(\mathbf{x})\right) & \text{if} \quad y = 1 \\ -\ln\left(1 - h_{\boldsymbol{\theta}}(\mathbf{x})\right) & \text{if} \quad y = 0 \end{cases}$$

$$\boxed{\mathcal{J} = -y \ln\left(h_{\boldsymbol{\theta}}(\mathbf{x})\right) - (1 - y) \ln\left(1 - h_{\boldsymbol{\theta}}(\mathbf{x})\right)}$$

# Gradient Descent

$$\boxed{\theta_i \triangleq \theta_i - \underbrace{\alpha}_{\text{Learning Rate}} \frac{\partial \mathcal{J}}{\partial \theta_i}}$$

Generalizing $\mathcal{J}$ yields:

$$\mathcal{J} = -\frac{1}{n} \sum_{k=1}^{n} \left( y^{(k)} \ln \left( h_{\boldsymbol{\theta}} \left( \mathbf{x}^{(k)} \right) \right) + (1 - y_k) \ln \left( 1 - h_{\boldsymbol{\theta}} \left( \mathbf{x}^{(k)} \right) \right) \right)$$

$$\Longrightarrow \quad \frac{\partial \mathcal{J}}{\partial \theta_i} = -\frac{1}{n} \sum_{k=1}^{n} \left( y_k - h_{\boldsymbol{\theta}} \left( \mathbf{x}^{(k)} \right) \right) x_i^{(k)}$$

$$\boxed{\theta_1 \triangleq \theta_1 + \alpha \frac{1}{n} \sum_{k=1}^{n} \left( y^{(k)} - h_{\boldsymbol{\theta}} \left( \mathbf{x}^{(k)} \right) \right) x_1^{(k)}} \quad \boxed{\theta_2 \triangleq \theta_2 + \alpha \frac{1}{n} \sum_{k=1}^{n} \left( y^{(k)} - h_{\boldsymbol{\theta}} \left( \mathbf{x}^{(k)} \right) \right) x_2^{(k)}}$$

$$\vdots$$

$$\boxed{\theta_m \triangleq \theta_m + \alpha \frac{1}{n} \sum_{k=1}^{n} \left( y^{(k)} - h_{\boldsymbol{\theta}} \left( \mathbf{x}^{(k)} \right) \right) x_m^{(k)}}$$

# Gradient Descent

$$\boxed{\theta_i \triangleq \theta_i - \underbrace{\alpha}_{\text{Learning Rate}} \frac{\partial \mathcal{J}}{\partial \theta_i}}$$

Generalizing $\mathcal{J}$ with $\mathbf{L}_2$ regularization term yields:

$$\mathcal{J} = -\frac{1}{n}\sum_{k=1}^{n}\left(y^{(k)} \ln\left(h_{\boldsymbol{\theta}}\left(\mathbf{x}^{(k)}\right)\right) + (1-y_k)\ln\left(1-h_{\boldsymbol{\theta}}\left(\mathbf{x}^{(k)}\right)\right)\right) + \frac{\lambda}{2n}\sum_{i=2}^{m}\theta_i^2$$

$$\Longrightarrow \quad \frac{\partial \mathcal{J}}{\partial \theta_i} = -\frac{1}{n}\sum_{k=1}^{n}\left(y_k - h_{\boldsymbol{\theta}}\left(\mathbf{x}^{(k)}\right)\right)x_i^{(k)} + \frac{\lambda}{n}\theta_i \quad \text{iff } i \neq 1$$

$$\boxed{\theta_1 \triangleq \left(1 - \cancel{\alpha\frac{\lambda}{n}}\right)\theta_1 + \alpha\frac{1}{n}\sum_{k=1}^{n}\left(y^{(k)} - h_{\boldsymbol{\theta}}\left(\mathbf{x}^{(k)}\right)\right)x_1^{(k)}}$$

$$\boxed{\theta_2 \triangleq \left(1 - \alpha\frac{\lambda}{n}\right)\theta_2 + \alpha\frac{1}{n}\sum_{k=1}^{n}\left(y^{(k)} - h_{\boldsymbol{\theta}}\left(\mathbf{x}^{(k)}\right)\right)x_2^{(k)}}$$

$$\vdots$$

$$\boxed{\theta_m \triangleq \left(1 - \alpha\frac{\lambda}{n}\right)\theta_m + \alpha\frac{1}{n}\sum_{k=1}^{n}\left(y^{(k)} - h_{\boldsymbol{\theta}}\left(\mathbf{x}^{(k)}\right)\right)x_m^{(k)}}$$

# Practical Implementation

The code is available @ github.com/a-mhamdi/jlai → *Codes* → *Julia* → *Part-2*

→ *Pluto* → *logistic-regression.jl*

→ *Jupyter* → *logistic-regression.ipynb*

# Confusion Matrix

| | | Actual | |
|---|---|---|---|
| | | Positive | Negative |
| Predicted | Positive | **TP** | **FP** |
| | Negative | **FN** | **TN** |

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$f1 - score = \frac{2}{\frac{1}{Precision} + \frac{1}{Recall}}$$

| | |
|---|---|
| 183 | 141 |
| 13 | 663 |

$Accuracy = 0.846$

$Precision = 0.565$

$Recall = 0.934$

$f1 - score = 0.704$

| | |
|---|---|
| 320 | 20 |
| 43 | 538 |

$Accuracy = 0.932$

$Precision = 0.941$

$Recall = 0.882$

$f1 - score = 0.910$

# Confusion Matrix

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$$Precision = 0.941$$
$$Recall = 0.882$$
$$f1 - score = 0.910$$

# Evaluation metrics

**Accuracy** **Precision** **Recall** ***f*1-score**

*Accuracy* denotes the ratio of how many we got right over all cases:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}}$$

*Precision* designates how many positives do we get right over all positive predictions:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

*Recall* is the ratio of how many positives we got right over all actual positive cases:

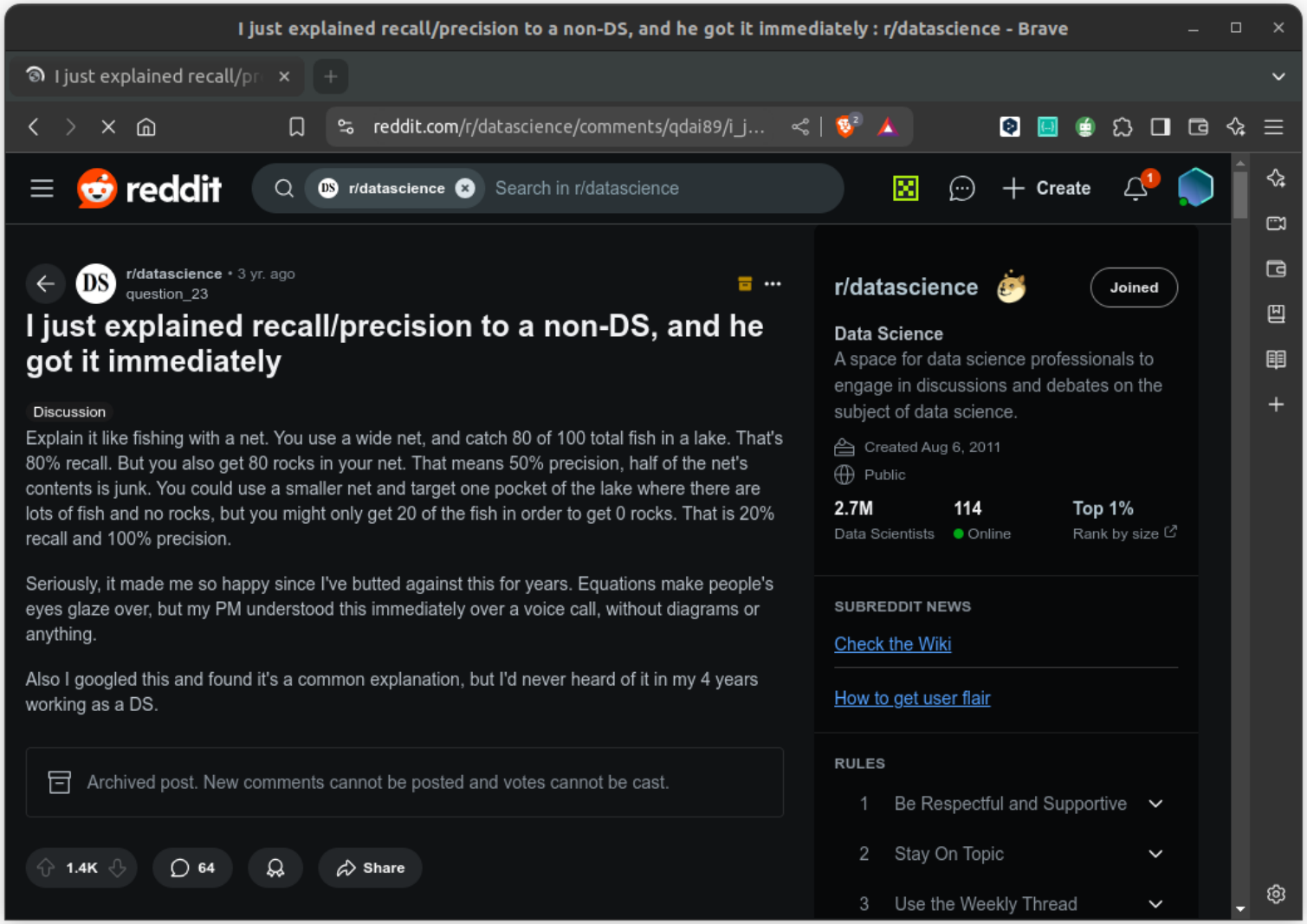
$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

*f*1 – score denotes the Harmonic Mean of *Precision* & *Recall*:

$$\text{f1} - \text{score} = \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}}$$

# Evaluation metrics

## An Analogy




Reddit

# Evaluation Metrics

## Follow up

**$f_\beta$-score**

$$\mathrm{f}_\beta - \mathrm{score} = \frac{1+\beta^2}{\frac{1}{\mathrm{Precision}} + \frac{\beta^2}{\mathrm{Recall}}}$$

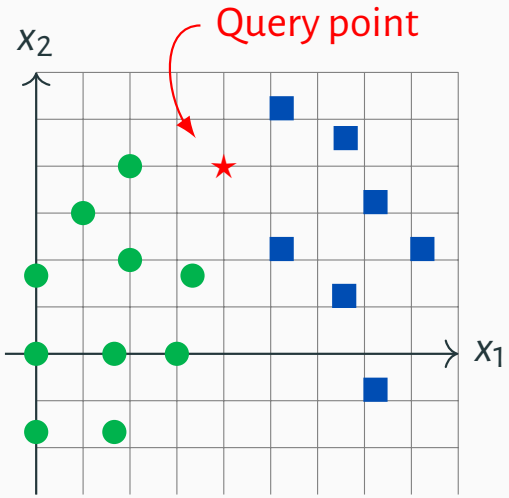
**Case #1:** Prioritize *Precision* over *Recall*, *e.g.*, $\beta = 0.5$

- Mail spam detection
- Predicting appropriate day to launch a satellite

**Case #2:** Prioritize *Recall* over *Precision*, *e.g.*, $\beta = 2$

- Detection of life threatening diseases like cancer
- Fraud detection

# $k$-Nearest Neighbors (1/2)




*Minkowski distance*

$$d(x;\ y) = \left( \sum_{i=1}^{n} |y_i - x_i|^p \right)^{1/p}$$

*Manhattan distance* (p=1)

$$d(x;\ y) = \sum_{i=1}^{n} |y_i - x_i|$$

*Euclidean distance* (p=2)

$$d(x;\ y) = \sqrt{\sum_{i=1}^{n} (y_i - x_i)^2}$$

# $k$-Nearest Neighbors (2/2)

Evelyn Fix and Joseph Hodges, 1951 | Thomas Cover, 1966

**Algorithm 1** Summary Construction

```
procedure How does k-NN work? (Finding Nearest Neighbors)
Input: A query point;
Output: Assign a class label to that point.
    Define how many neighbors will be checked to classify the specific query point;
    Compute the distance d(x; y) of the query point to other data points;
    Count the number of the data points in each category;
    Assign the query point to the class with most frequent neighbors.
end procedure
```

# Hands-on Example (1/2)

**Task #4**
Let be the following coordinate points:

A(1, 6); B(2, 6); C(3, 1); D(4, 2); E(6, 0); F(7, 5); G(7, 3); H(10, 3); I(–4, –1)

Using the Euclidean distance, what are the two closest neighbors of point $P(5,\ 5)$?

$d(A;\ P) = \sqrt{17} \approx 4.12$ $\quad d(B;\ P) = \sqrt{10} \approx 3.16$ $\quad d(C;\ P) = \sqrt{20} \approx 4.47$

$d(D;\ P) = \sqrt{10} \approx 3.16$ $\quad d(E;\ P) = \sqrt{26} \approx 5.1$ $\quad d(F;\ P) = \sqrt{4} = 2$

$d(G;\ P) = \sqrt{8} \approx 2.83$ $\quad d(H;\ P) = \sqrt{29} \approx 5.38$ $\quad d(I;\ P) = \sqrt{117} \approx 10.82$

# Hands-on Example (2/2)

```julia
function dds(a, b) # `a` and `b` are coordinates of some point
    d_squared = (a-5)^2+(b-5)^2
    (d_squared, sqrt(d_squared))
end

dds(1, 6) # Point `A`
dds(2, 6) # Point `B`
```

# Hands-on Example (1/3)

**Task #5**

We try to predict the color of a fruit according to its width ($w$) and height ($h$). The following training data is available:

| Fruit | $F_1$ | $F_2$ | $F_3$ | $F_4$ | $F_5$ | $F_6$ | $F_7$ | $F_8$ |
|---|---|---|---|---|---|---|---|---|
| $w$ | 2 | 5 | 2 | 6 | 1 | 4 | 2 | 6 |
| $h$ | 6 | 6 | 5 | 5 | 2 | 2 | 1 | 1 |
| Color | Red | Yellow | Orange | Purple | Red | Blue | Violet | Green |

The goal here is to study the influence of neighbors on the color property of a fruit. Let $U$ be the new fruit of width $w = 1$ and height $h = 4$

1. What is its color if we consider 1 neighbor?
2. What is its color if we consider 3 neighbors?
3. Rather than majority voting, we would like to consider the vote of neighbors weighted by the distance. Each neighbor votes according to a weight inversely proportional to the square of its distance: $\frac{1}{d^2}$. We take 3 neighbors, what is the color of $U$? Compare your results to those in question 2.

# Hands-on Example (2/3)

$d(U;\ F_1) = \sqrt{5} \approx 2.24$ $\quad d(U;\ F_2) = \sqrt{20} \approx 4.47$ $\quad d(U;\ F_3) = \sqrt{2} \approx 1.41$

$d(U;\ F_4) = \sqrt{26} \approx 5.1$ $\quad d(U;\ F_5) = \sqrt{4} = 2$ $\quad d(U;\ F_6) = \sqrt{13} \approx 3.6$

$d(U;\ F_7) = \sqrt{10} \approx 3.16$ $\quad d(U;\ F_8) = \sqrt{34} \approx 5.83$

1. Color of $U$ is Orange because $d(U;\ F_3)$ is the smallest.
2. Color of $U$ is Red: $F_1$ and $F_5$ *(+2 to Red class)*, $F_3$ *(+1 to Orange class)*
3. Color of $U$ is Orange

$$\boxed{\mathcal{S}(\text{Red}) = \frac{1}{d^2(U;\ F_1)} + \frac{1}{d^2(U;\ F_5)} = 0.45} \quad \boxed{\mathcal{S}(\text{Orange}) = \frac{1}{d^2(U;\ F_3)} = 0.5}$$

# Hands-on Example (3/3)

```julia
function dds(w, h) # `w` and `h` are width and height of some fruit
    d_squared = (w-1)^2+(h-4)^2
    (d_squared, sqrt(d_squared))
end

dds(2, 6) # Fruit `F_1`
dds(5, 6) # Fruit `F_2`
```

# Practical Implementation

The code is available @ github.com/a-mhamdi/jlai ⟶ *Codes* ⟶ *Julia* ⟶ *Part-2*

⟶ *Pluto* ⟶ *knn.jl*

⟶ *Jupyter* ⟶ *knn.ipynb*

# Rule of Thumb to Choose $k$

| | |
|---|---|
| $k$ **is even** | if the number of classes is odd |
| $k$ **is odd** | if the number of classes is even |

$k$ is an important hyperparameter that can affect the performance of the model.

1. Larger values of $k$ will result in a smoother decision boundary, which can lead to a more generalized model.
2. Smaller values of $k$ will result in a more complex decision boundary, which can lead to a model that is more prone to overfitting.
3. The optimal value of $k$ may depend on the specific dataset and the characteristics of the data.

# Support Vector Machines (SVMs)

Support Vector Machines (SVMs) are supervised learning algorithms for classification and regression. They identify the optimal hyperplane to separate data into distinct classes.

**Hyperplane** A decision boundary separating classes in feature space.

**Support Vectors** Closest data points to the hyperplane, determining its position and orientation.

**Margin** Distance between the hyperplane and the nearest data points. SVM maximizes this margin.

For a dataset $\{(\mathbf{x}_k, y_k)\}_{k=1}^{n}$, where $\mathbf{x}_k \in \mathbb{R}^d$ and $y_k \in \{-1, +1\}$:

- Find a hyperplane $\mathbf{x}^\top \mathbf{w} + b = 0$ that maximizes the margin.
- Constrained optimization:

$$\min_{\mathbf{w},\, b} \frac{1}{2} \|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_k(\mathbf{x}_k^\top \mathbf{w} + b) \geq 1 \quad \forall k$$

**Types of SVMs**

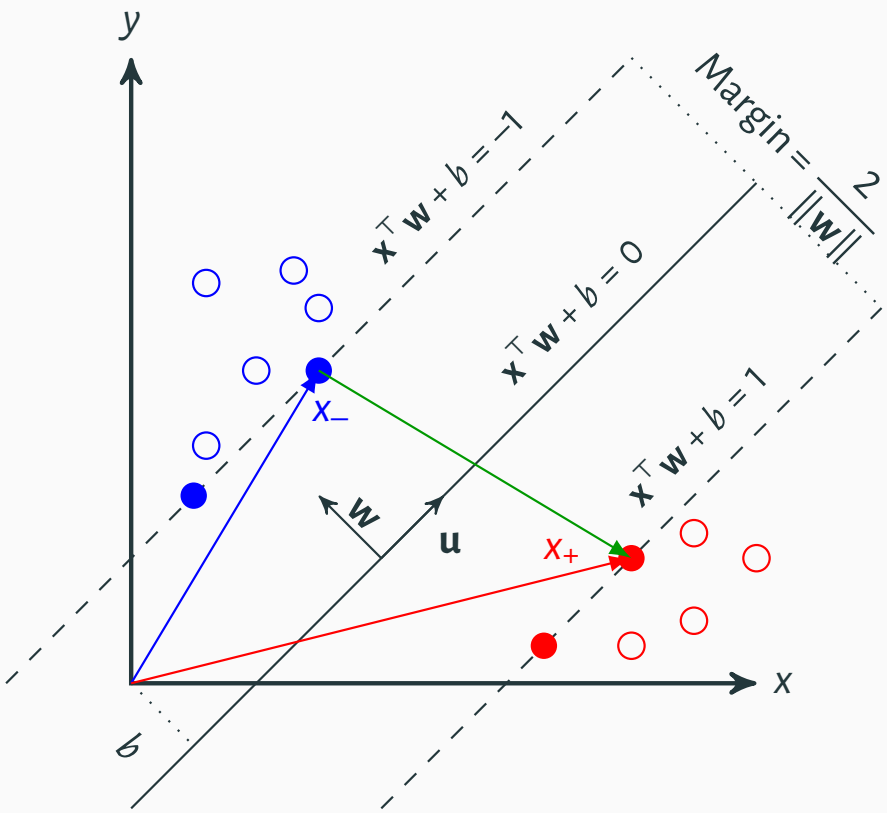
- **Linear SVM:** For linearly separable data.
- **Kernel SVM:** Maps data to higher dimensions for non-linear separation.

# Visualization of SVM

$\mathbf{u}$ is a direction vector of $\Delta$.

$$\Delta : \mathbf{x}^\top \mathbf{w} + b = 0 \quad \Longrightarrow \quad \mathbf{w} \perp \mathbf{u}$$

$$\begin{aligned}
\text{Margin} &= \frac{\left|(\mathbf{x}_+ - \mathbf{x}_-) \cdot \mathbf{w}\right|}{\|\mathbf{w}\|} \\
&= \frac{\left|\mathbf{x}_+^\top \mathbf{w} - \mathbf{x}_-^\top \mathbf{w}\right|}{\|\mathbf{w}\|} \\
&= \frac{\left|(1-b) - (-1-b)\right|}{\|\mathbf{w}\|} \\
&= \frac{2}{\|\mathbf{w}\|}
\end{aligned}$$



# SVM: Loss Function

**Hinge Loss Function:** The hinge loss is used in SVMs to penalize misclassifications and maximize the margin:

$$L(y, h(\mathbf{x})) = \max(0, 1 - yh(\mathbf{x})),$$

where:

- $y \in \{-1, +1\}$ is the true label,
- $h(x) = \mathbf{x}^\top \mathbf{w} + b$ is the predicted score.

The loss is zero if $yh(\mathbf{x}) \geq 1$ (correct classification with margin), and increases linearly otherwise.

**Why SVMs are powerful**

- **Robustness:** Focuses on support vectors, making it less sensitive to outliers.
- **High-Dimensional Data:** Performs well even with a large number of features.
- **Kernel Trick:** Enables non-linear classification by mapping data to higher dimensions.

# Solving the Lagrangian Problem in SVM (1/3)

## Primal Problem *(Hard-Margin SVM)*

The objective is to maximize the margin between two classes. The primal optimization problem is:

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2 \quad \text{subject to} \quad y_k\left(\mathbf{x}_k^\top \mathbf{w} + b\right) \geq 1, \quad \forall k$$

## Construct the Lagrangian

Introduce Lagrange multipliers $\alpha_k \geq 0$ for each constraint and form the Lagrangian:

$$\mathcal{L}(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{1}{2}\|\mathbf{w}\|^2 - \sum_{k=1}^{n} \alpha_k \left(y_k\left(\mathbf{x}_k^\top \mathbf{w} + b\right) - 1\right)$$

## Take Derivatives and Set to Zero

Minimize $\mathcal{L}$ with respect to $\mathbf{w}$ and $b$:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \mathbf{w} - \sum_{k=1}^{n} \alpha_k y_k \mathbf{x}_k = 0 \implies \mathbf{w} = \sum_{k=1}^{n} \alpha_k y_k \mathbf{x}_k$$

# Solving the Lagrangian Problem in SVM (2/3)

$$\frac{\partial \mathcal{L}}{\partial b} = -\sum_{k=1}^{n} \alpha_k y_k = 0 \implies \sum_{k=1}^{n} \alpha_k y_k = 0$$

Substitute Back into the Lagrangian

Plugging **w** and the constraint into $\mathcal{L}$, we obtain the **dual problem**:

$$\mathcal{L}_D(\boldsymbol{\alpha}) = \sum_{k=1}^{n} \alpha_k - \frac{1}{2} \sum_{k,l=1}^{n} \alpha_k \alpha_l y_k y_l \mathbf{x}_k^\top \mathbf{x}_l$$

Solve the Dual Problem

Maximize $\mathcal{L}_D(\boldsymbol{\alpha})$ subject to:

$$\alpha_k \geq 0 \quad \text{and} \quad \sum_{k=1}^{n} \alpha_k y_k = 0$$

# Solving the Lagrangian Problem in SVM (3/3)

## Obtain the Solution

- The optimal $\mathbf{w}^\star$ is a linear combination of support vectors:

$$\mathbf{w}^\star = \sum_{k \in SV} \alpha_k y_k \mathbf{x}_k$$

- The bias $b^\star$ is computed using any support vector $\mathbf{x}_k$:

$$b^\star = y_k - \mathbf{x}_k^\top \mathbf{w}^\star$$

## Final Decision Function

For a new input $\mathbf{x}$, the prediction is:

$$f(\mathbf{x}) = \text{sign}\left(\mathbf{x}^\top \mathbf{w}^\star + b^\star\right)$$

# Hands-on Example (1/5)

**Task #6**
You are given a dataset with two classes: **Class A** ($y = +1$) and **Class B** ($y = -1$). The features ($x_1$, $x_2$) and corresponding labels are:

| **Data Point** | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $x_1$ | 2 | 3 | 1 | 6 | 7 | 8 |
| $x_2$ | 2 | 4 | 1 | 2 | 3 | 2 |
| $y$ | +1 | +1 | +1 | -1 | -1 | -1 |

1. Plot the data points on a 2D plane.
2. Identify the support vectors.
3. Determine the optimal hyperplane separating the two classes using SVM.
4. Calculate the margin width if the weight vector is $\mathbf{w} = \left[-1/2,\ 1/4\right]^\top$.

**Hint!**

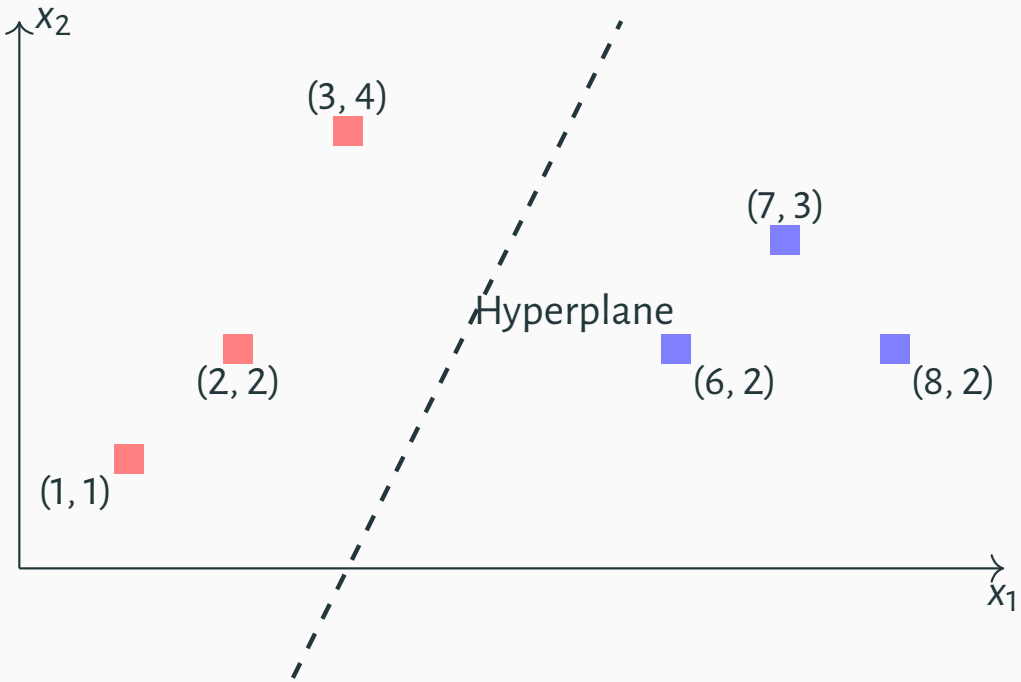
Support vectors are the data points closest to the hyperplane. The margin width is given by:

$$\text{Margin Width} = \frac{2}{\|\mathbf{w}\|}$$

# Hands-on Example (2/5)

**Q. #1: Plot the Data Points**

The data points are visualized on the plane $(x_1, x_2)$:



# Hands-on Example (3/5)

### Q. #2: Support Vectors

The support vectors are the points closest to the hyperplane:

$$\boxed{(2,2) \text{ from Class A}, \quad (3,4) \text{ from Class A}, \quad (6,2) \text{ from Class B.}}$$

### Q. #3: Optimal Hyperplane

$$\begin{aligned}
\mathcal{L}_D(\boldsymbol{\alpha}) &= \sum_{k=1}^{6} \alpha_k - \frac{1}{2} \sum_{k,l=1}^{6} \alpha_k \alpha_l y_k y_l \mathbf{x}_k^\top \mathbf{x}_l \\
&= \alpha_1 + \alpha_2 + \alpha_4 \\
&\quad -\frac{1}{2} \left( \alpha_1 \left[ 8\alpha_1 + 14\alpha_2 - 16\alpha_4 \right] + \alpha_2 \left[ 14\alpha_1 + 25\alpha_2 - 26\alpha_4 \right] - \alpha_4 \left[ 16\alpha_1 + 26\alpha_2 - 40\alpha_4 \right] \right)
\end{aligned}$$

Given

$$\sum_{k=1}^{6} \alpha_k y_k = 0 \therefore \alpha_1 + \alpha_2 - \alpha_4 = 0$$

$$\mathcal{L}_D(\boldsymbol{\alpha}) = 2\alpha_1 + 2\alpha_2 - \frac{1}{2} \left( 16\alpha_1^2 + 13\alpha_2^2 + 24\alpha_1\alpha_2 \right)$$

# Hands-on Example (4/5)

$$\frac{\partial \mathcal{L}_D(\boldsymbol{\alpha})}{\partial \alpha_1} = 0 \quad \therefore \quad 16\alpha_1 + 12\alpha_2 = 2$$

$$\frac{\partial \mathcal{L}_D(\boldsymbol{\alpha})}{\partial \alpha_1} = 0 \quad \therefore \quad 12\alpha_1 + 13\alpha_2 = 2$$

The resulting linear system is:

$$\begin{bmatrix} 16 & 12 \\ 12 & 13 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$\boldsymbol{\alpha}$ is:

$$\boldsymbol{\alpha}^T = \begin{bmatrix} 1/32 & 1/8 & 0 & 5/32 & 0 & 0 \end{bmatrix}$$

- The optimal $\mathbf{w}^\star$ is a linear combination of support vectors:

$$\mathbf{w}^\star = 1/32 \begin{bmatrix} 2 \\ 2 \end{bmatrix} + 1/8 \begin{bmatrix} 3 \\ 4 \end{bmatrix} - 5/32 \begin{bmatrix} 6 \\ 2 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 1/4 \end{bmatrix}$$

# Hands-on Example (5/5)

- The bias $b^\star$ is computed using any support vector $\mathbf{x}_k$:

$$b^\star = y_k - \mathbf{x}_k^T \mathbf{w}^\star = 3/2$$

The separating hyperplane is given by:

$$x_1 w_1 + x_2 w_2 + b = 0 \quad \text{where} \quad \mathbf{w} = \begin{bmatrix} -1/2 \\ 1/4 \end{bmatrix}^T, \; b = 3/2$$

Thus, the equation becomes:

$$-2x_1 + x_2 + 6 = 0$$

**Q. #4: Margin Width**

The margin width is calculated as:

$$\text{Margin Width} = \frac{2}{\|\mathbf{w}\|} = \frac{2}{\sqrt{(1/2)^2 + (1/4)^2}} = \frac{8}{\sqrt{5}}$$

# Hard Margin vs. Soft Margin SVM

- **Data Assumption**

  **Hard:** Strict linear separability
  **Soft:** Tolerates noise/overlap

- **Primal Form**

  **Hard:** $\min \frac{1}{2}\|w\|^2 \qquad \text{s.t. } y_k\left(\mathbf{x}_k^\top \mathbf{w} + b\right) \geq 1$
  **Soft:** $\min \frac{1}{2}\|w\|^2 + C\sum \xi_k \qquad \text{s.t. } y_k\left(\mathbf{x}_k^\top \mathbf{w} + b\right) \geq 1 - \xi_k$

- **Key Features**

  **Hard:** No slack variables ($\xi$), $C \longrightarrow \infty$
  **Soft:** Slack variables ($\xi_i \geq 0$), finite $C$ (trade-off)

- **Dual Form**

  **Hard:** $0 \leq \alpha_i$
  **Soft:** $0 \leq \alpha_i \leq C$
  **Both:** $\sum \alpha_i y_i = 0$

- **Practicality**

  **Hard:** Rarely used (overfits)
  **Soft:** Robust to noise, works with kernels

# Kernel Trick in SVMs

**Dual Form with Kernel**:

$$\mathcal{L}_D(\boldsymbol{\alpha}) = \sum_{k=1}^{n} \alpha_k - \frac{1}{2} \sum_{k,l=1}^{n} \alpha_k \alpha_l y_k y_l \underbrace{\mathbf{x}_k^\top \mathbf{x}_l}_{\mathcal{K}(\mathbf{x}_k, \mathbf{x}_l)}$$

$$\mathcal{K}\left(\mathbf{x}_k, \mathbf{x}_l\right) = \Phi^\top\left(\mathbf{x}_k\right) \Phi\left(\mathbf{x}_l\right)$$

**Kernel Trick**: Replace inner products with kernel functions - no need to compute $\Phi(\mathbf{x})$ explicitly.

**Common Kernels**:

- **Linear**: $\mathcal{K}(\mathbf{x}_k, \mathbf{x}_l) = \mathbf{x}_k^\top \mathbf{x}_l$
- **Polynomial**: $\mathcal{K}\left(\mathbf{x}_k, \mathbf{x}_l\right) = \left(\gamma \mathbf{x}_k^\top \mathbf{x}_l + r\right)^d$
- **RBF (Gaussian)**: $\mathcal{K}\left(\mathbf{x}_k, \mathbf{x}_l\right) = \exp\left(-\gamma \left\|\mathbf{x}_k - \mathbf{x}_l\right\|^2\right)$
- **Sigmoid**: $\mathcal{K}(\mathbf{x}_k, \mathbf{x}_l) = \tanh\left(\gamma \mathbf{x}_k^\top \mathbf{x}_l + r\right)$

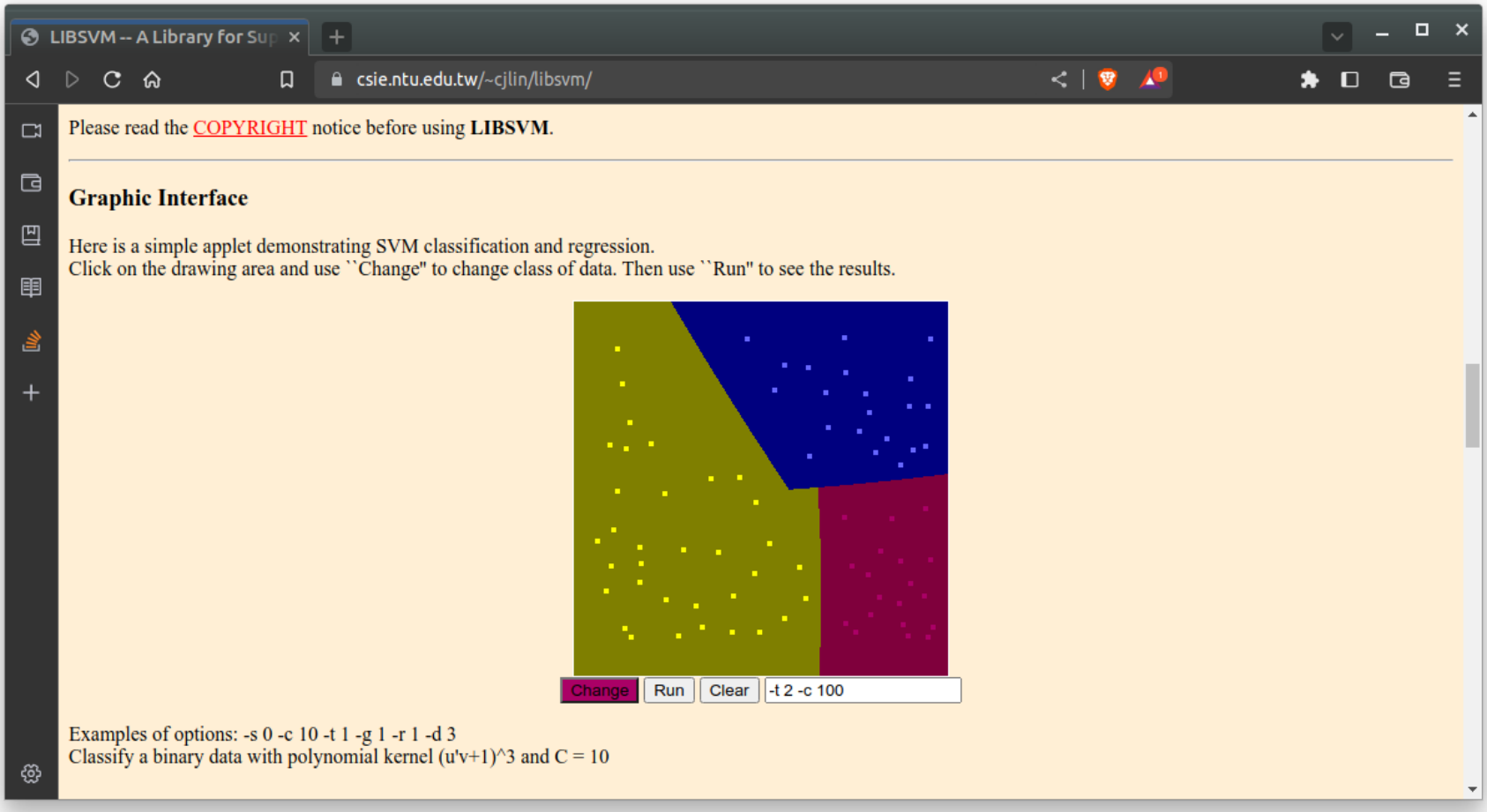
**Parameters**

$d$: Polynomial degree (#non-linear terms)

$\gamma$: Kernel scale (RBF/poly)

$C$: Regularization (controls margin vs. errors)

# LibSVM Library




https://www.csie.ntu.edu.tw/~cjlin/libsvm/

# Practical Implementation

The code is available @ github.com/a-mhamdi/jlai → *Codes* → *Julia* → *Part-2*

→ *Pluto* → *svc.jl*

→ *Jupyter* → *svc.ipynb*

# Naive Bayes

## COVID-19 Testing

**Task #7**
Imagine we have the following information:

- 1% of the population has COVID-19 (prior probability)
- A COVID test has 95% sensitivity (true positive rate) - if you have COVID, there's a 95% chance the test will be positive
- The test has 90% specificity (true negative rate) - if you don't have COVID, there's a 90% chance the test will be negative

Now, if someone receives a positive test result, what's the probability they actually have COVID-19?

# Naive Bayes

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Now, if someone receives a positive test result, what's the probability they actually have COVID-19?

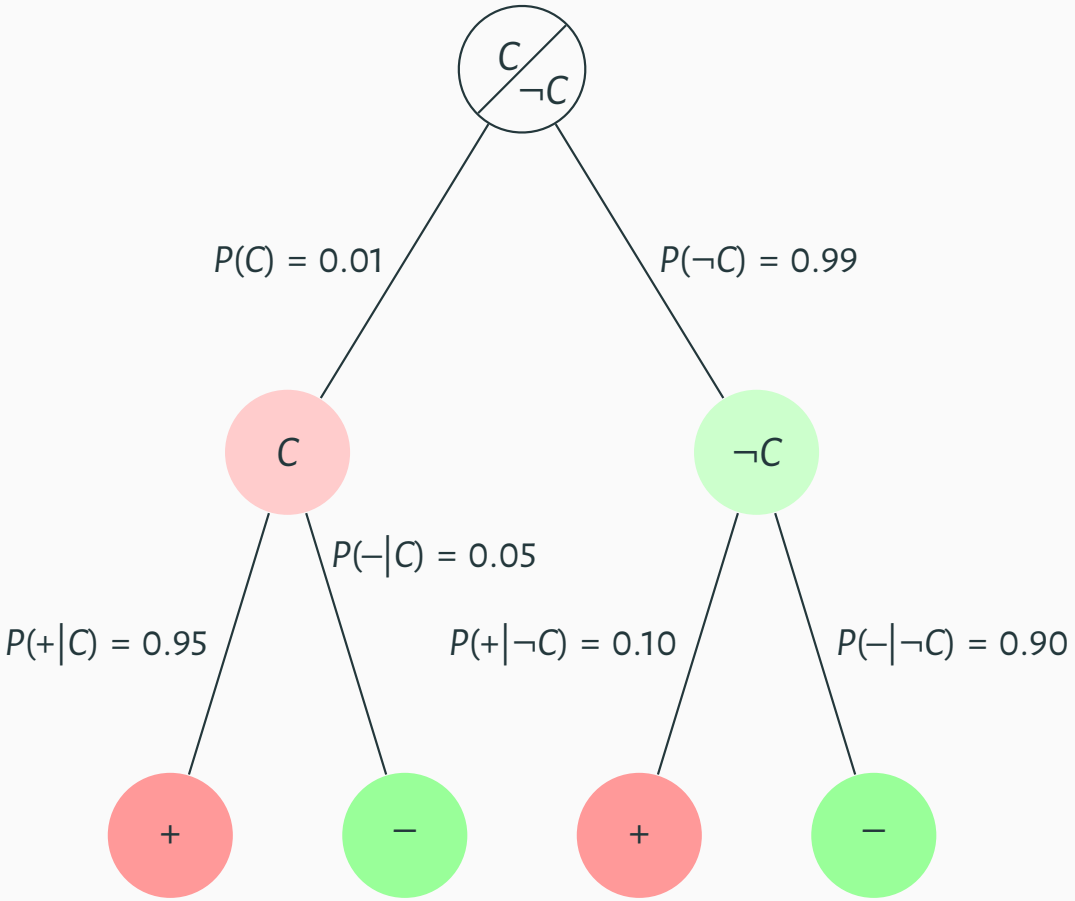
$$P(+) = P(C \cap +) + P(\neg C \cap +) = 0.0095 + 0.099 = 0.1085$$

$$P(C|+) = \frac{P(C \cap +)}{P(+)} = \frac{0.0095}{0.1085} \approx 0.088 \ \textbf{(8.8\%)}$$

$\therefore$ Despite a positive test, there's only an 8.8% chance the person actually has COVID-19.

# Naive Bayes

## COVID-19 Testing




$P(C \cap +) = 0.01 \times 0.95 = 0.0095$

$P(C \cap -) = 0.01 \times 0.05 = 0.0005$

$P(\neg C \cap +) = 0.99 \times 0.10 = 0.099$

$P(\neg C \cap -) = 0.99 \times 0.90 = 0.891$

# Naive Bayes (1/2)

## Definition and Key Concept

Naive Bayes is a probabilistic machine learning algorithm based on Bayes' Theorem. It is commonly used for classification tasks, such as spam detection and text classification.

**Bayes' Theorem:**

$$P(y|X) = \frac{P(X|y) \cdot P(y)}{P(X)}$$

where:

- $P(y|X)$: Posterior probability of class $y$ given features $X$.
- $P(X|y)$: Likelihood of features $X$ given class $y$.
- $P(y)$: Prior probability of class $y$.
- $P(X)$: Marginal probability of features $X$.

$$\hat{y} = \arg\max_{y \in Y} P(y|X) = \arg\max_{y \in Y} P(X|y) \cdot P(y)$$

# Naive Bayes (2/2)

## Definition and Key Concept

**Naive Bayes for Spam Detection**

**Features:** Words in an email (e.g., "free", "offer", "money").

**Classes:** "Spam" ($S$) or "Not Spam" ($\neg S$).

**Steps:**

① Calculate $P(S)$ and $P(\neg S)$ from the training data.

② Calculate $P(\text{Word}|S)$ and $P(\text{Word}|\neg S)$ for each word.

③ Use Bayes' Theorem to predict the class of a new email.

# Hands-on Example (1/4)

**Task #8**
You are given a dataset of emails with the following word frequencies and labels:

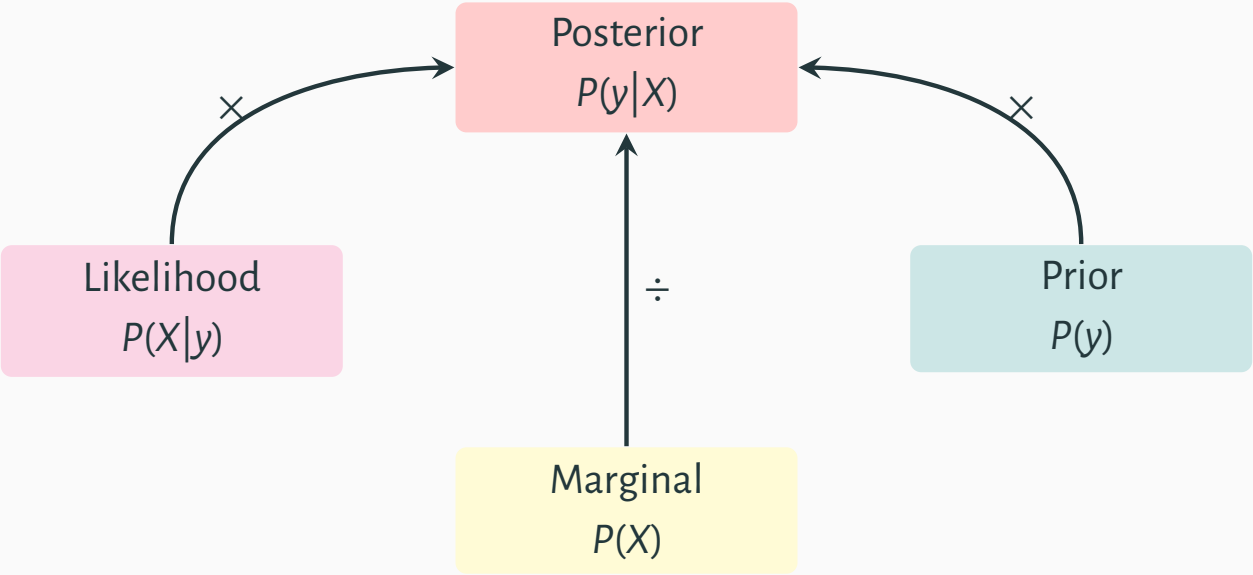
| Email | "free" | "money" | Label |
|---|---|---|---|
| 1 | 1 | 0 | S |
| 2 | 0 | 1 | S |
| 3 | 0 | 0 | ¬S |
| 4 | 0 | 1 | ¬S |

1. Calculate the prior probabilities $P(S)$ and $P(\neg S)$.
2. Calculate the likelihoods $P(\text{"free"}|S)$, $P(\text{"money"}|S)$, $P(\text{"free"}|\neg S)$, and $P(\text{"money"}|\neg S)$.
3. Use Naive Bayes to classify a new email with the words "free" and "money".

**Hint!**

- Use Laplace smoothing (add +1 smoothing) to handle zero probabilities.

# Hands-on Example (2/4)




**Step 1: Calculate Prior Probabilities**

- $P(S) = \dfrac{\text{Number of Spam Emails}}{\text{Total Emails}} = \dfrac{2}{4} = 0.5$
- $P(\neg S) = \dfrac{\text{Number of Not Spam Emails}}{\text{Total Emails}} = \dfrac{2}{4} = 0.5$

# Hands-on Example (3/4)

**Step 2: Calculate Likelihoods with Laplace Smoothing**

$$P\left(w_i|c\right) = \frac{\text{count}\left(w_i, c\right) + k}{\sum_{w \in V} \text{count}(w, c) + k|V|}$$

$k$: Smoothing parameter ($k \geq 0$)

$|V|$: Vocabulary size

**count** $\left(w_i, c\right)$: Frequency of word $w_i$ in class $c$

- $P(\text{“free”}|S) = \dfrac{\text{Count of “free” in Spam} + 1}{\text{Count of Words in Spam} + |\text{Vocabulary}|} = \dfrac{1+1}{2+2} = \dfrac{2}{4} = 0.5$
- $P(\text{“money”}|S) = \dfrac{\text{Count of “money” in Spam} + 1}{\text{Count of Words in Spam} + |\text{Vocabulary}|} = \dfrac{1+1}{2+2} = \dfrac{2}{4} = 0.5$
- $P(\text{“free”}|\neg S) = \dfrac{\text{Count of “free” in Not Spam} + 1}{\text{Count of Words in} \neg \text{Spam} + |\text{Vocabulary}|} = \dfrac{0+1}{1+2} = \dfrac{1}{3} \approx 0.333$
- $P(\text{“money”}|\neg S) = \dfrac{\text{Count of “money” in Not Spam} + 1}{\text{Count of Words in} \neg \text{Spam} + |\text{Vocabulary}|} = \dfrac{1+1}{1+2} = \dfrac{2}{3} \approx 0.666$

# Hands-on Example (4/4)

**Step 3: Classify a New Email**

- For the email with words "free" and "money":

$$P(S \mid \text{“free” and “money”}) \propto P(\text{”free”} \mid S) \cdot P(\text{”money”} \mid S) \cdot P(S)$$
$$= 1/2 \cdot 1/2 \cdot 1/2 = 1/6$$

$$P(\neg S \mid \text{“free” and “money”}) \propto P(\text{“free”} \mid \neg S) \cdot P(\text{“money”} \mid \neg S) \cdot P(\neg S)$$
$$= 1/3 \cdot 2/3 \cdot 1/2 = 1/9$$

- Since $1/6 > 1/9$, the email is classified as **Spam**.

# Naive Bayes: Key Assumptions and Types

## Key Assumptions

**Independence:** Features are assumed to be conditionally independent given the class.

**Equal Importance:** All features contribute equally to the prediction.

## Types

**Gaussian Naive Bayes:** Assumes features follow a normal distribution.

**Multinomial Naive Bayes:** Used for discrete data (e.g., word counts).

**Bernoulli Naive Bayes:** Used for binary features.

- ▲ Simple and fast to train.
- ▲ Works well with high-dimensional data (e.g., text).
- ▲ Requires less training data compared to other algorithms.

- ▼ Assumes feature independence, which is rarely true in real-world data.
- ▼ Struggles with zero probabilities (requires smoothing techniques).
- ▼ Less accurate for complex relationships between features.

# Practical Implementation

The code is available @ github.com/a-mhamdi/jlai ⟶ *Codes* ⟶ *Julia* ⟶ *Part-2*

⟶ *Pluto* ⟶ *naive-bayes.jl*

⟶ *Jupyter* ⟶ *naive-bayes.ipynb*

# Decision Tree: Definition and Key Concepts

A decision tree is a supervised learning algorithm used for classification and regression. It partitions the feature space into regions by applying a series of decision rules.

## Components

**Root Node:** Represents the entire dataset.

**Decision Nodes:** Split the dataset based on a feature and a threshold.

**Leaf Nodes:** Represent the final output (class label or regression value).

## Key Concepts

**Splitting Criteria:** Choose the feature and threshold that maximize information gain or minimize impurity.

**Pruning:** Remove unnecessary branches to prevent overfitting.

**Feature Importance:** Rank features based on their contribution to the model.

# Gini Index vs Entropy

## Gini Index

- Measures the impurity of a node.
- Lower values indicate purer nodes.

$$G = 1 - \sum_{i=1}^{k} p_i^2$$

## Entropy

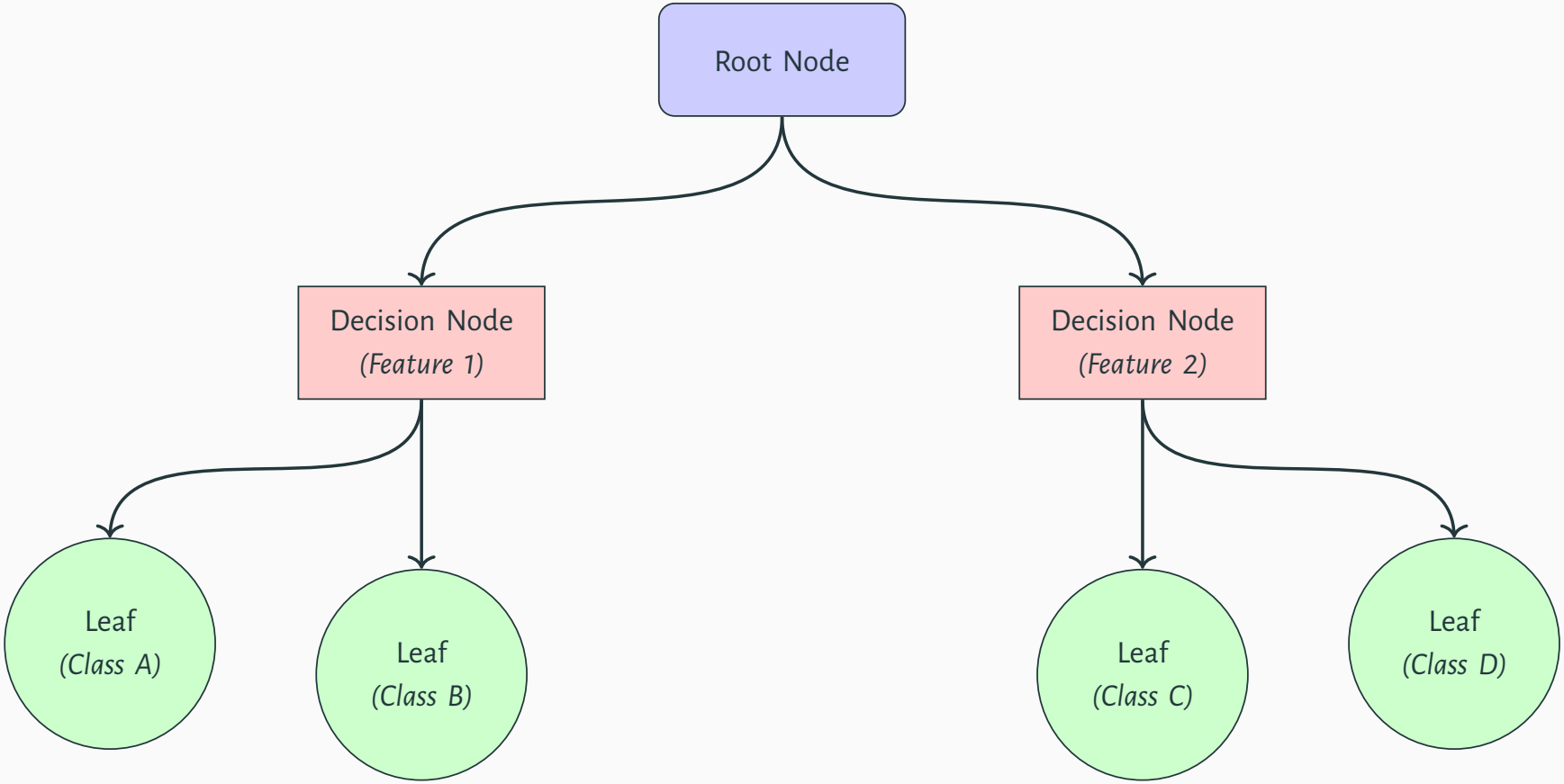
- Measures the uncertainty of a node.
- Lower values indicate more certainty.

$$H = -\sum_{i=1}^{k} p_i \log(p_i)$$

## Comparison

- Both are used for splitting criteria in decision trees;
- Gini Index is faster to compute;
- Entropy provides more precise splits in some cases.

# Decision Tree Visualization



# Hands-on Example (1/9)

**Task #9**
Considering the following restaurant wait time dataset, we need to predict if a reservation will be honored (Yes/No). Construct a **decision tree** using Gini Index:

| Party Size | Day of Week | Special Occasion? | Reservation Honored? |
|---|---|---|---|
| 2 | Weekday | No | Yes |
| 4 | Weekend | Yes | No |
| 6 | Weekend | Yes | No |
| 2 | Weekend | No | Yes |
| 4 | Weekday | No | Yes |
| 8 | Weekend | Yes | No |
| 2 | Weekday | Yes | Yes |
| 4 | Weekend | No | No |
| 6 | Weekday | No | Yes |
| 2 | Weekend | Yes | No |

# Hands-on Example (2/9)

## Step 1: Root Node Gini Impurity

$$G_{Root} = 1 - \left( \left( \frac{5}{10} \right)^2 + \left( \frac{5}{10} \right)^2 \right) = 1 - (0.25 + 0.25) = 0.5$$

## Step 2: Feature Split Analysis

*Split 1: "Party Size"*

$$G_{ps=2} = 1 - \left( \left( \frac{3}{4} \right)^2 + \left( \frac{1}{4} \right)^2 \right) = 0.375$$

$$G_{ps=4} = 1 - \left( \left( \frac{1}{3} \right)^2 + \left( \frac{2}{3} \right)^2 \right) = 0.444$$

$$G_{ps=6} = 1 - \left( \left( \frac{1}{2} \right)^2 + \left( \frac{1}{2} \right)^2 \right) = 0.5$$

$$G_{ps=8} = 1 - \left( \left( \frac{0}{1} \right)^2 + \left( \frac{1}{1} \right)^2 \right) = 0$$

$$G_{split} = \frac{4}{10} \times 0.375 + \frac{3}{10} \times 0.444 + \frac{2}{10} \times 0.5 + \frac{1}{10} \times 0 = 0.383$$

# Hands-on Example (3/9)

*Split 2: "DOTW"'*

$$G_{\text{weekday}} = 1 - \left(\left(\frac{4}{4}\right)^2 + \left(\frac{0}{4}\right)^2\right) = 0$$

$$G_{\text{weekend}} = 1 - \left(\left(\frac{1}{6}\right)^2 + \left(\frac{5}{6}\right)^2\right) = 0.278$$

$$G_{\text{split}} = \frac{4}{10} \times 0 + \frac{6}{10} \times 0.278 = 0.167$$

*Split 3: "Special Occasion?"*

$$G_{\text{special}} = 1 - \left(\left(\frac{1}{5}\right)^2 + \left(\frac{4}{5}\right)^2\right) = 0.32$$

$$G_{\text{no special}} = 1 - \left(\left(\frac{4}{5}\right)^2 + \left(\frac{1}{5}\right)^2\right) \approx 0.32$$

$$G_{\text{split}} = \frac{5}{10} \times 0.32 + \frac{5}{10} \times 0.32 \approx 0.32$$

# Hands-on Example (4/9)

**Step 3: Optimal Split Selection**

- **DOTW** has the lowest Gini impurity ($G = 0.167$)
- **Party Size?** ($G = 0.383$) and **Special Occasion?** ($G \approx 0.32$) are less optimal

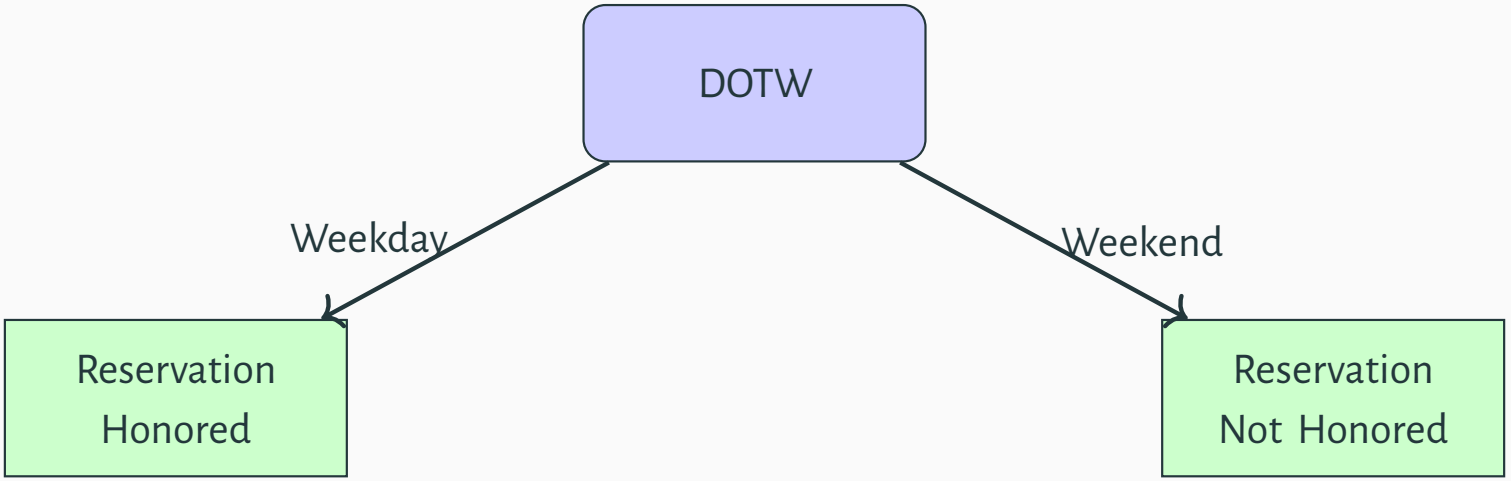
## Resulting Decision Tree

**1st split:** "Weekend?"

**Weekday:** Predict "Reservation Honored" (4/4 samples) *(pure leaf node)*

**Weekend:** Predict "Reservation Not Honored" (5/6 samples)

**Subsequent splits:** Refine with "Party Size" or "Special Occasion?"



# Hands-on Example (5/9)

| Party Size | SO? | RH? |
|---|---|---|
| 2 | No | Yes |
| 4 | No | Yes |
| 2 | Yes | Yes |
| 6 | No | Yes |

**Weekday Node Gini Impurity**

$$G_{\text{Weekday}} = 1 - \left(\left(\frac{4}{4}\right)^2 + \left(\frac{0}{4}\right)^2\right) = 0$$

| Party Size | SO? | RH? |
|---|---|---|
| 4 | Yes | No |
| 6 | Yes | No |
| 2 | No | Yes |
| 8 | Yes | No |
| 4 | No | No |
| 2 | Yes | No |

**Weekend Node Gini Impurity**

$$G_{\text{Weekend}} = 1 - \left(\left(\frac{1}{6}\right)^2 + \left(\frac{5}{6}\right)^2\right) = 0.278$$

# Hands-on Example (6/9)

*"Party Size?" Split*

$$G_{ps=2} = 1 - \left(\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2\right) = 0.5$$

$$G_{ps=4} = 1 - \left(\left(\frac{0}{2}\right)^2 + \left(\frac{2}{2}\right)^2\right) = 0$$

$$G_{ps=6} = 1 - \left(\left(\frac{0}{1}\right)^2 + \left(\frac{1}{1}\right)^2\right) = 0$$

$$G_{ps=8} = 1 - \left(\left(\frac{0}{1}\right)^2 + \left(\frac{1}{1}\right)^2\right) = 0$$

$$G_{split} = \frac{2}{6} \times 0.5 + \frac{2}{6} \times 0 + \frac{1}{6} \times 0 + \frac{1}{6} \times 0 = 0.167$$

# Hands-on Example (7/9)

*"Special Occasion?" Split*

$$
\begin{aligned}
G_{\text{special}} &= 1 - \left( \left( \frac{0}{4} \right)^2 + \left( \frac{4}{4} \right)^2 \right) &&= 0 \\
G_{\text{no special}} &= 1 - \left( \left( \frac{1}{2} \right)^2 + \left( \frac{1}{2} \right)^2 \right) &&\approx 0.5 \\
G_{\text{split}} &= \frac{4}{6} \times 0 + \frac{2}{6} \times 0.5 &&\approx 0.167
\end{aligned}
$$

# Hands-on Example (8/9)

**Optimal Split Selection**

Both "Party Size" and "Special Occasion?" have the same Gini Index (0.1667) for the "Weekend" branch.

In this case, we can choose either one. Let's use "Special Occasion?" as our second split.

Resulting Decision Tree

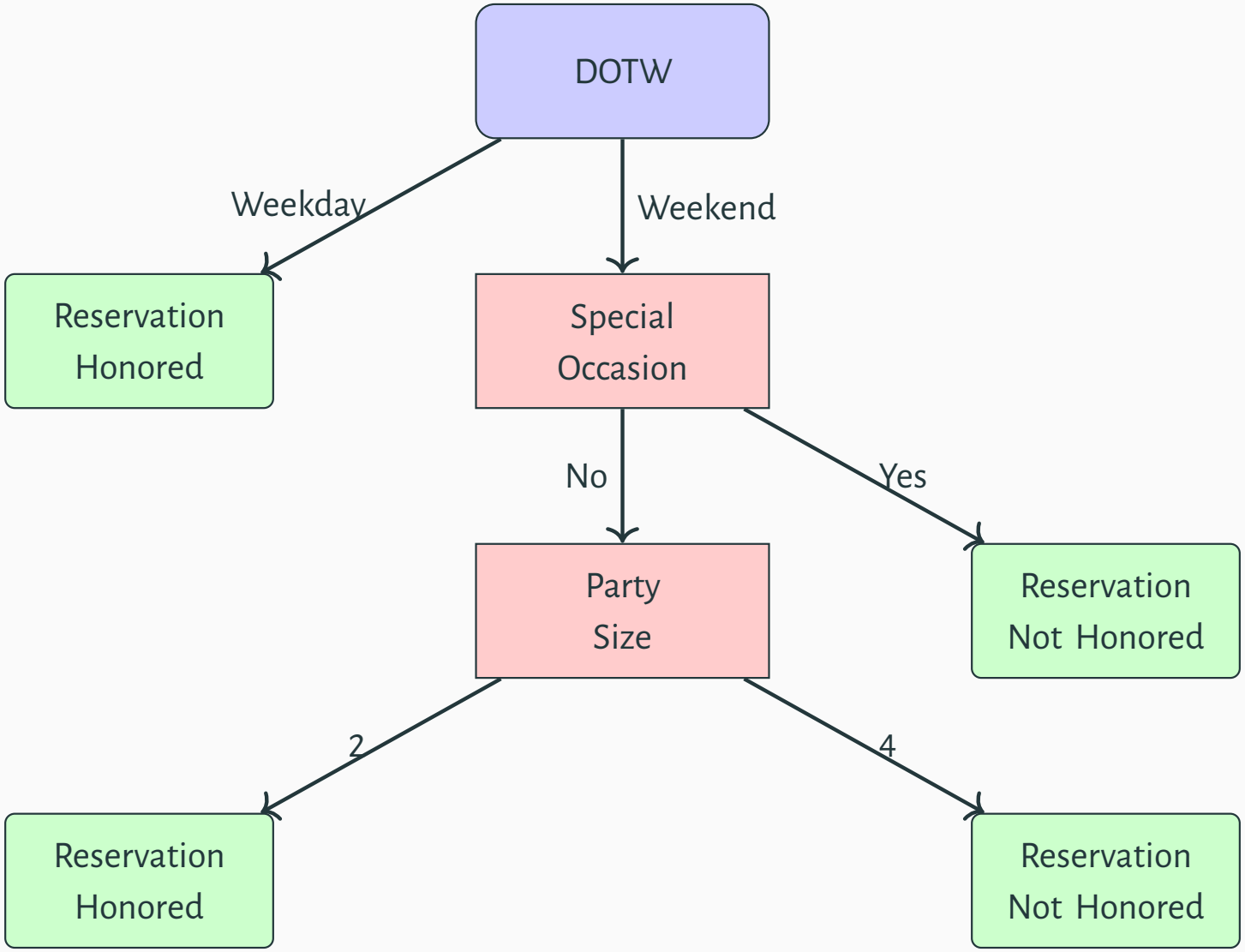
**2nd split:** Special Occasion?

- **Yes:** Predict "Reservation Not Honored" (4 samples)
- **No:** Refine with "Party Size" (2 samples)

**3rd split:** Party Size *(pure leaf)*

- **2:** Predict "Reservation Honored" (1/2 sample)
- **4:** Predict "Reservation Not Honored" (1/2 sample)

# Hands-on Example (9/9)



# Practical Implementation

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⟶ *Pluto* ⟶ *decision-tree-*.jl*

⟶ *Jupyter* ⟶ *decision-tree-*.ipynb*

# Ensemble Approach: Random Forest (1/2)

A random forest is an ensemble method that combines multiple decision trees to improve classification accuracy. It uses bagging (bootstrap aggregation) and random feature selection.

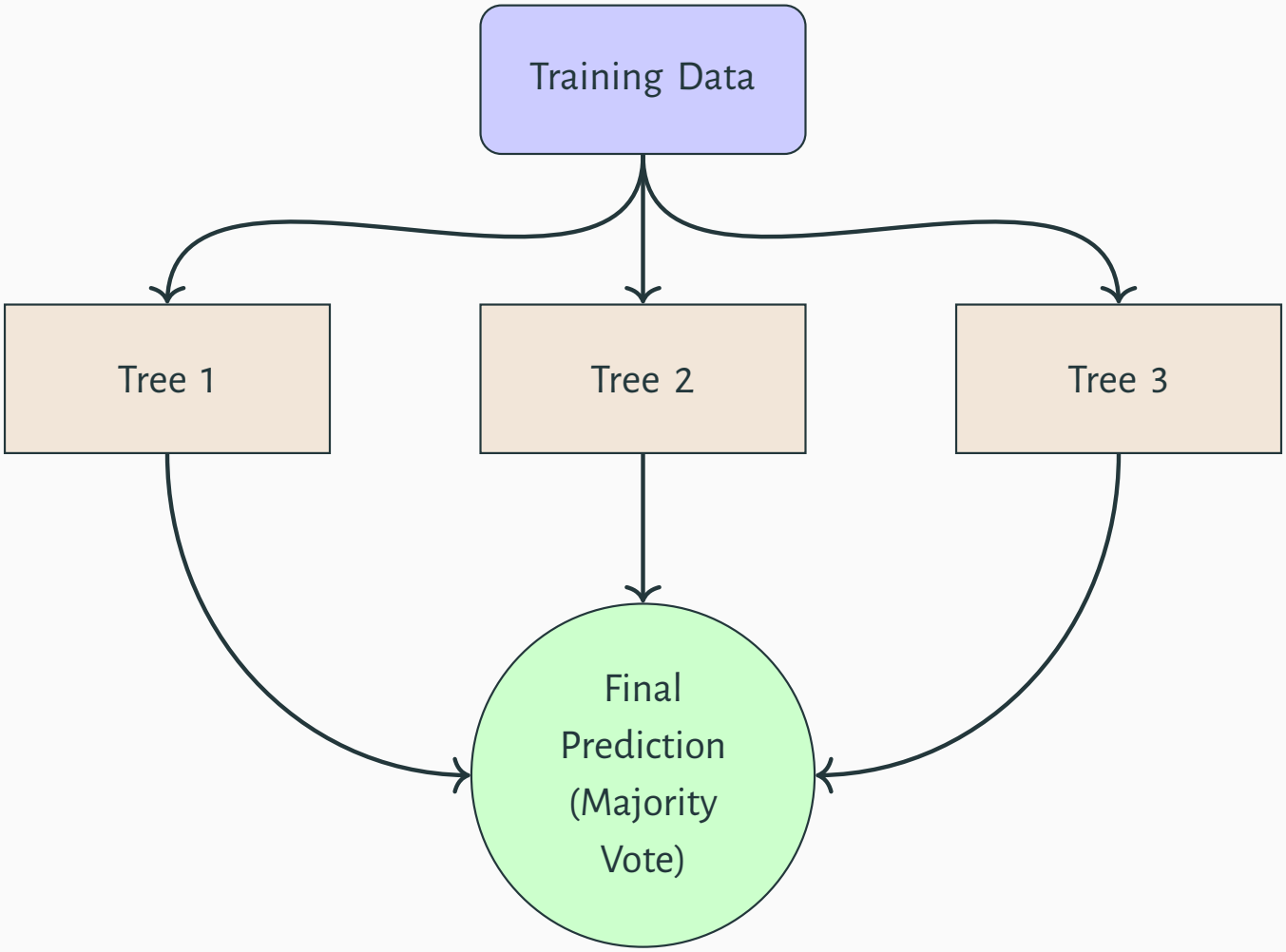
## Key Features

- **Diversity:** Trees are trained on different subsets of data and features.
- **Voting Mechanism:** For classification, the final output is the mode of the tree predictions.

## Loss Function

- Combines the loss functions of individual trees, e.g., Gini Index or Entropy, during splits.

# Ensemble Approach: Random Forest (2/2)



# Practical Implementation

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⟶ *Pluto* ⟶ *random-forest-*.jl*

⟶ *Jupyter* ⟶ *random-forest-*.ipynb*

# Summary

| Method | | Pros | | Cons |
|---|---|---|---|---|
| *Logistic Regression* | ▲ | Probabilistic | ▼ | Almost linearly separable data |
| *k-NN* | ▲ | Fast and efficient | ▼ | # of neighbors $k$ |
| | | | ▼ | Detecting outliers[2] |
| *SVM* | ▲ | Memory efficient | ▼ | Kernel's choice |
| | ▲ | Versatile | ▼ | Large datasets |
| | ▲ | Noise and outliers | ▼ | Overlapping classes |
| | ▲ | High dimension | ▼ | Interpretability |
| *Naive Bayes* | ▲ | Simplicity and efficiency | ▼ | Independence between features |
| | ▲ | High dimension | ▼ | $\exists$ of irrelevant features |
| *Decision Tree* | ▲ | Interpretability | ▼ | Overfitting |
| | ▲ | Numerical and categorical data | ▼ | Unstable |
| | ▲ | Robust to outliers | ▼ | Continuous variables |
| | ▲ | High accuracy | ▼ | # of input features |
| *Random Forest* | ▲ | Less prone to overfitting | ▼ | Computation |
| | ▲ | High dimension | ▼ | Interpretability |

[2] Points that differ significantly from the rest of the data points.

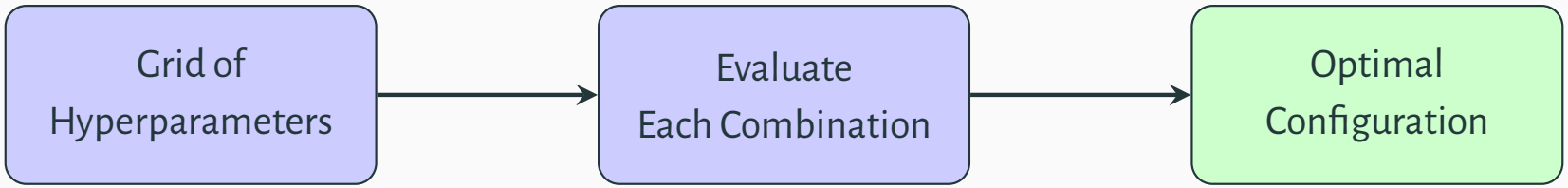
# Grid Search: Optimizing Hyperparameters

Grid Search is a systematic approach to finding the best hyperparameter combination for a machine learning model by evaluating all possible configurations.

## Key Concepts

- Defines a **grid** of hyperparameter values.
- Evaluates all combinations using a performance metric (e.g., accuracy, precision).
- Identifies the **optimal configuration** for the model.

## Process




## Challenges

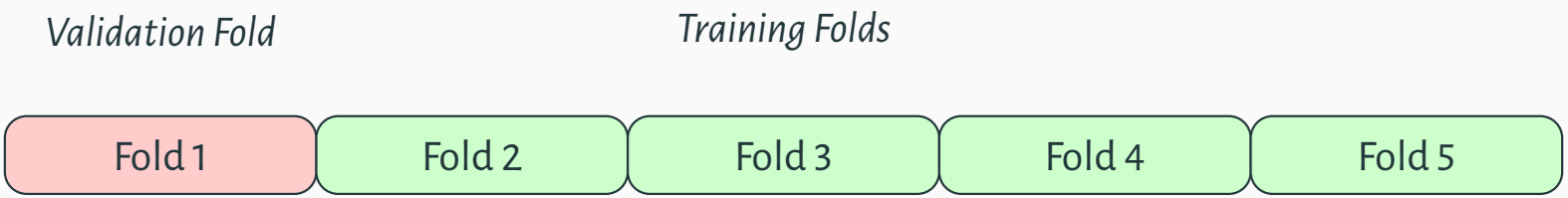
**Computationally expensive:** Large grids require significant resources.

**Alternative:** Randomized Search samples combinations to reduce computation.

# Cross-Validation: Ensuring Robustness

Cross-validation is a resampling technique used to evaluate model performance by dividing the dataset into multiple subsets (folds) and validating the model on each subset.

1. Split the dataset into $k$ subsets (folds).
2. Train the model on $k-1$ folds and validate on the remaining fold.
3. Repeat $k$ times, using a different fold for validation each time.
4. Compute the average performance metric across all folds.




Example: 5-Fold Cross-Validation

**Benefits**

- Reduces overfitting by validating on multiple subsets.
- Provides a reliable estimate of model performance.

# Practical Implementation

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⟶ *Pluto* ⟶ *hp.jl*

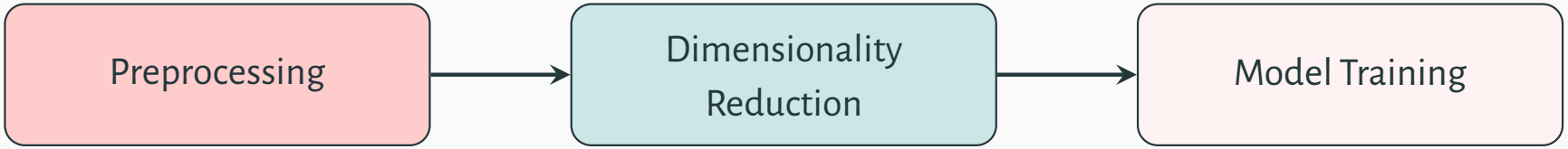
⟶ *Jupyter* ⟶ *hp.ipynb*

# Pipeline Mechanism: Automating Workflow

A pipeline automates the sequential application of preprocessing steps and model training, ensuring a consistent and efficient workflow.

## Steps in a Pipeline

1. Preprocess data (e.g., scaling, normalization).
2. Apply dimensionality reduction or feature selection.
3. Train a machine learning model.




## Advantages

- Ensures consistency across training and testing data.
- Simplifies hyperparameter optimization.

# Integrating Pipelines, Grid Search, and Cross-Validation

## Overview

**Pipeline:** Automates preprocessing and training.

**Grid Search:** Finds the best hyperparameters.

**Cross-Validation:** Ensures robust performance evaluation.

## End-to-End Workflow

1. Define preprocessing steps in a pipeline.
2. Use Grid Search to optimize hyperparameters.
3. Validate results using K-Folds Cross-Validation.

# Overall Methodology (1/2)

1. **Problem Definition**
   - Understand the business or research objective.
   - Define success metrics (e.g., accuracy, F1-score, ROI).
2. **Data Collection**
   - Gather raw data from databases, APIs, or files.
   - Ensure data quality, relevance, and representativeness.
3. **Data Preprocessing & Exploratory Analysis (EDA)**
   - Handle missing values, duplicates, and outliers.
   - Perform statistical and visual analysis.
4. **Feature Engineering & Selection**
   - Transform/create meaningful features (e.g., scaling, encoding).
   - Select optimal features (e.g., using PCA, feature importance).
5. **Model Selection & Training**
   - Split data into training, validation, and test sets.
   - Train multiple algorithms (e.g., regression, neural networks).
   - Apply cross-validation techniques.

# Overall Methodology (2/2)

6. **Model Evaluation**
   - Test on unseen data (validation/test sets).
   - Compare performance metrics (e.g., precision, RMSE, AUC-ROC).
7. **Hyperparameter Tuning & Optimization**
   - Fine-tune models using GridSearch, RandomSearch, or Bayesian methods.
8. **Deployment & Monitoring**
   - Deploy the model (e.g., as an API, cloud service, or embedded system).
   - Continuously monitor performance and data drift.
9. **Communication & Maintenance**
   - Document methodology, results, and limitations.
   - Schedule retraining and updates as needed.

# Unsupervised Learning

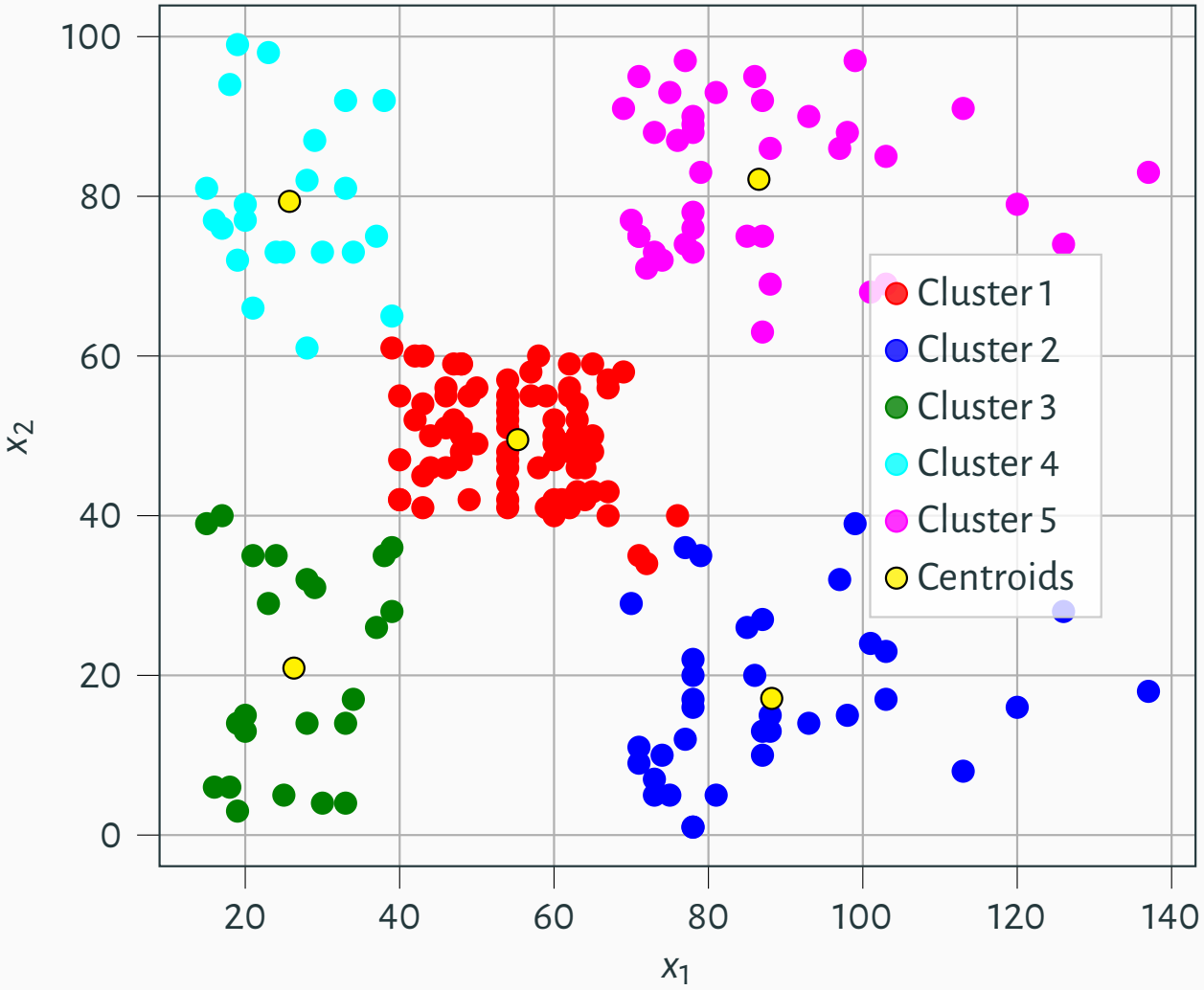
# *K*-Means Clustering (1/3)

The algorithm *K*-**Means** allows to display regularities or patterns in unlabeled data.

- The term 'means' refers to averaging the data when computing each centroid;
- A centroid is the arithmetic mean of all the data points belonging to a particular cluster.

This technique identifies a certain number of centroids within a data set. The algorithm then allocates every data point to the nearest cluster as it attempts to keep the clusters as small as possible. At the same time, *K*-Means attempts to keep the other clusters as different as possible.

# $K$-Means Clustering (2/3)



# *K*-Means Clustering (3/3)

**Algorithm 2** Summary Construction

```
procedure How does K-Means work? (Discovering similarities)
Input: Unlabeled data sets;
Output: Grouping into clusters.
    Define how many clusters will be used to group the data sets;
    Initialize all the coordinates of the k cluster centers
    repeat
        Assign each point to its nearest cluster;
        Update the centroids coordinates;
    until No changes to the centers of the clusters
    Assign new cases to one of the clusters
end procedure
```

**Task #10**[3]

Of the following examples, which would you address using an unsupervised learning algorithms? *(Check all that apply.)*

1. Given email labeled as spam/not spam, learn a spam filter
2. Given a set of news articles found on the web, group them into set of articles about the same story
3. Given a database of customer data, automatically discover market segments and group customers into different market segments
4. Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not.

[3] From 'Machine Learning' course on 'Coursera'

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**Task #11**[4]

Use *K*-Means algorithm to cluster the following eight points intro three clusters:

$$A(2,\ 10);\ B(2,\ 5);\ C(8,\ 4);\ D(5,\ 8);\ E(7,\ 5);\ F(6,\ 4);\ G(1,\ 2) \text{ and } H(4,\ 9).$$

- Initial cluster centers are: $\alpha(2,\ 10)$; $\beta(5,\ 8)$ and $\gamma(1,\ 2)$
- The distance between two points: $M(x_m,\ y_m)$ and $N(x_n,\ y_n)$ is defined as

$$d(M;\ N) \ = \ |x_m - x_n| + |y_m - y_n|$$

[4]Credit: Shokoufeh Mirzaei, PhD

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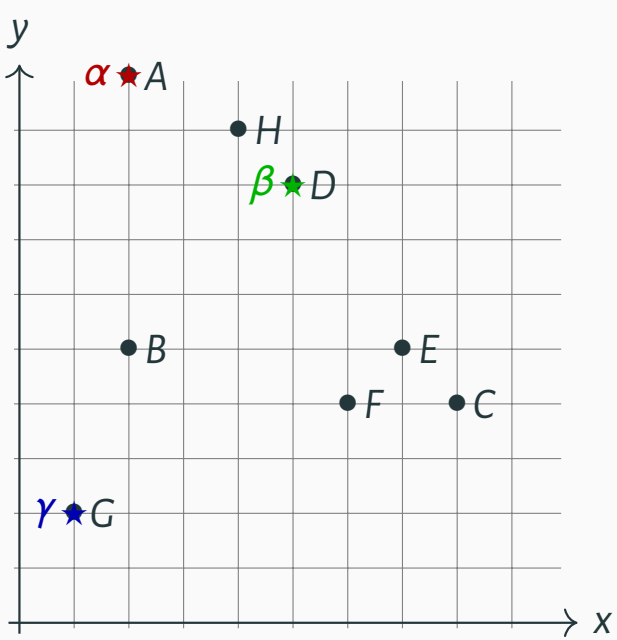


[4]Credit: Shokoufeh Mirzaei, PhD

## Task #11[4]

Use $K$-Means algorithm to cluster the following eight points intro three clusters:

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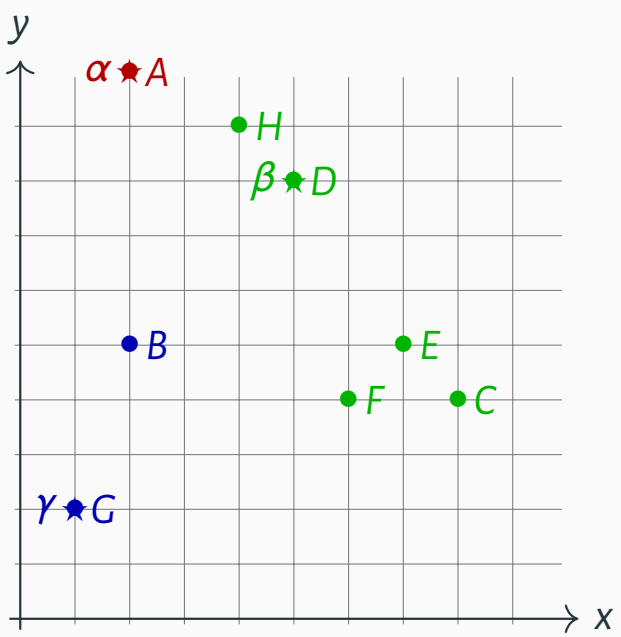
| Point | $\alpha(2,\ 10)$ | $\beta(5,\ 8)$ | $\gamma(1,\ 2)$ | # |
|---|---|---|---|---|
| $A(2,\ 10)$ | 0 | 5 | 9 | 1 |
| $B(2,\ 5)$ | 5 | 6 | 4 | 3 |
| $C(8,\ 4)$ | 12 | 7 | 9 | 2 |
| $D(5,\ 8)$ | 5 | 0 | 10 | 2 |
| $E(7,\ 5)$ | 10 | 5 | 9 | 2 |
| $F(6,\ 4)$ | 10 | 5 | 7 | 2 |
| $G(1,\ 2)$ | 9 | 10 | 0 | 3 |
| $H(4,\ 9)$ | 3 | 2 | 10 | 2 |

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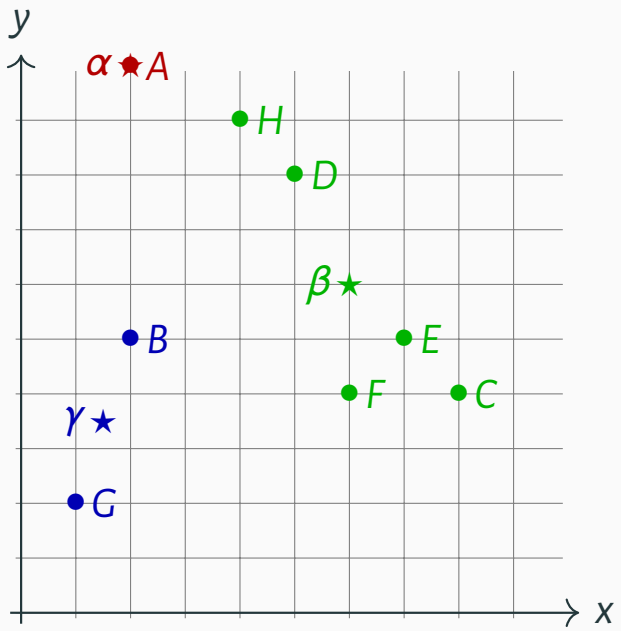
$\alpha(2,\ 10)$ $\beta(6,\ 6)$ $\gamma(1.5,\ 3.5)$

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| Point | $\alpha(2,\ 10)$ | $\beta(6,\ 6)$ | $\gamma(1.5,\ 3.5)$ | # |
|---|---|---|---|---|
| $A(2,\ 10)$ | 0 | 8 | 7 | 1 |
| $B(2,\ 5)$ | 5 | 5 | 2 | 3 |
| $C(8,\ 4)$ | 12 | 4 | 7 | 2 |
| $D(5,\ 8)$ | 5 | 3 | 8 | 2 |
| $E(7,\ 5)$ | 10 | 2 | 7 | 2 |
| $F(6,\ 4)$ | 10 | 2 | 5 | 2 |
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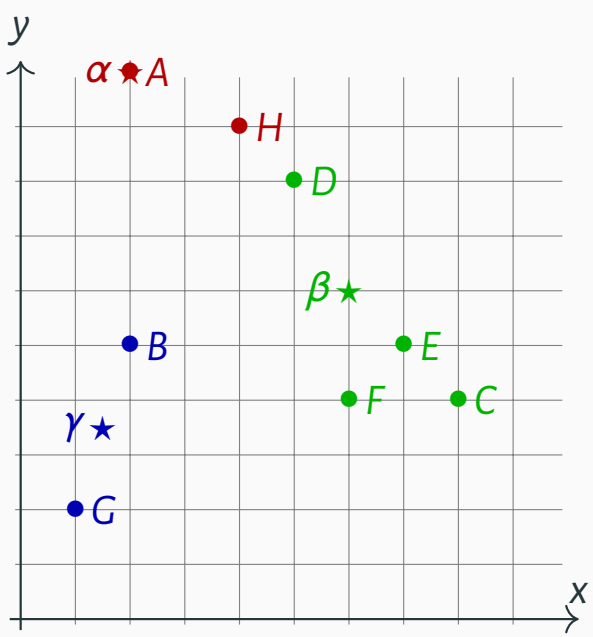


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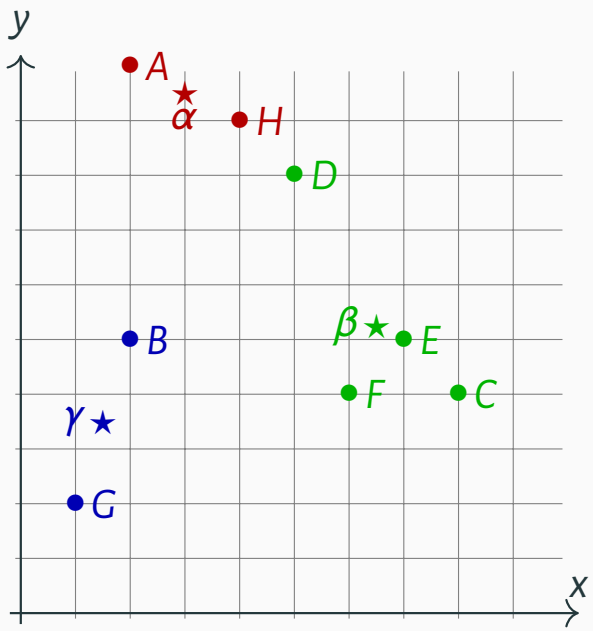
$\alpha(3,\ 9.5)$ $\beta(6.5,\ 5.25)$ $\gamma(1.5,\ 3.5)$

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| Point | $\alpha(3,\ 9.5)$ | $\beta(6.5,\ 5.25)$ | $\gamma(1.5,\ 3.5)$ | # |
|---|---|---|---|---|
| $A(2,\ 10)$ | 1.5 | 9.25 | 7 | 1 |
| $B(2,\ 5)$ | 5.5 | 4.75 | 2 | 3 |
| $C(8,\ 4)$ | 10.5 | 2.75 | 7 | 2 |
| $D(5,\ 8)$ | 3.5 | 4.25 | 8 | 1 |
| $E(7,\ 5)$ | 8.5 | 0.75 | 7 | 2 |
| $F(6,\ 4)$ | 8.5 | 1.75 | 5 | 2 |
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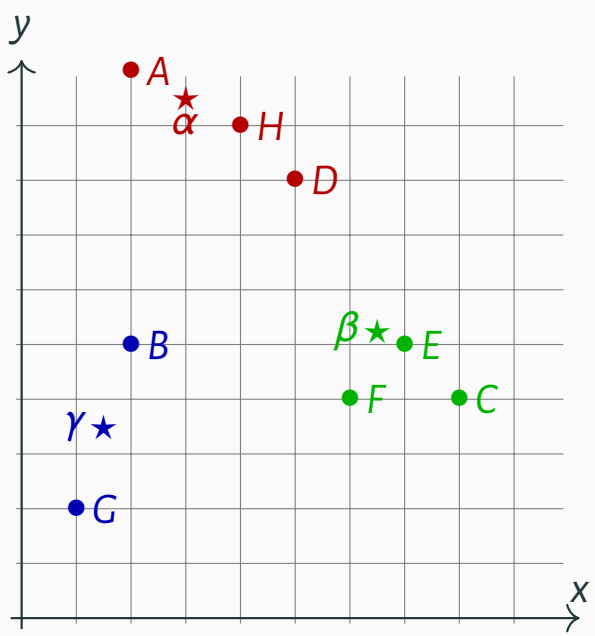


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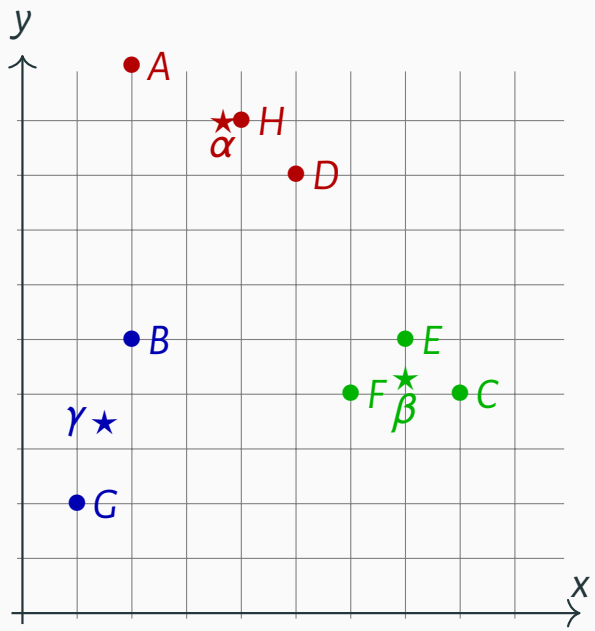
$\alpha(3.67,\ 9)$ $\beta(7,\ 4.3)$ $\gamma(1.5,\ 3.5)$

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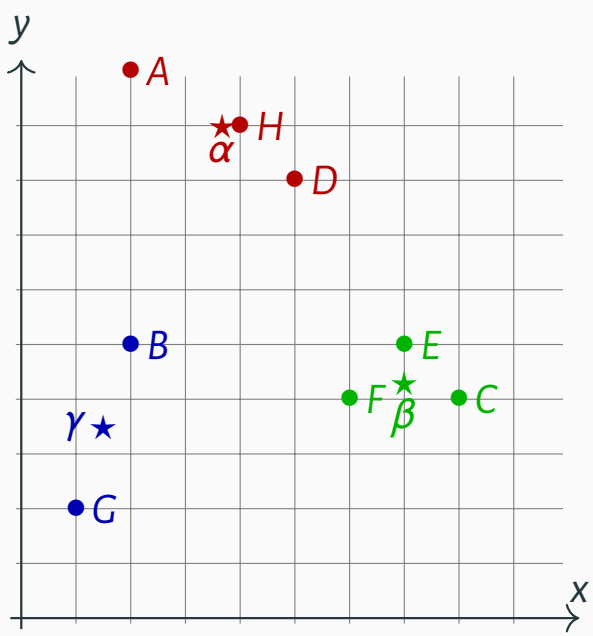
| Point | $\alpha(3.67,\ 9)$ | $\beta(7,\ 4.3)$ | $\gamma(1.5,\ 3.5)$ | # |
|---|---|---|---|---|
| $A(2,\ 10)$ | 2.67 | 10.7 | 7 | 1 |
| $B(2,\ 5)$ | 5.67 | 5.7 | 2 | 3 |
| $C(8,\ 4)$ | 9.33 | 1.3 | 7 | 2 |
| $D(5,\ 8)$ | 2.33 | 5.7 | 8 | 1 |
| $E(7,\ 5)$ | 7.33 | 0.7 | 7 | 2 |
| $F(6,\ 4)$ | 7.33 | 1.3 | 5 | 2 |
| $G(1,\ 2)$ | 9.67 | 8.3 | 2 | 3 |
| $H(4,\ 9)$ | 0.33 | 7.7 | 8 | 1 |

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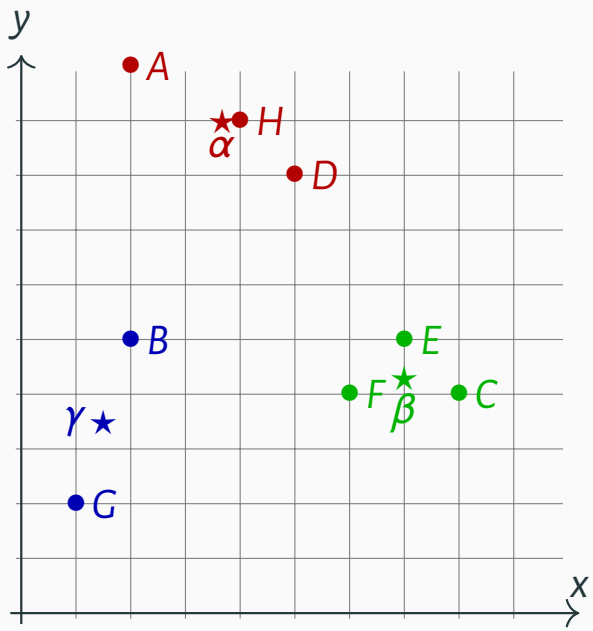


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# Practical Implementation

The code is available @ github.com/a-mhamdi/jlai ⟶ *Codes* ⟶ *Julia* ⟶ *Part-2*

⟶ *Pluto* ⟶ *kmeans.jl*

⟶ *Jupyter* ⟶ *kmeans.ipynb*

# DBSCAN: Overview

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is a density-based clustering algorithm that groups together points closely packed in space while marking points in sparse regions as outliers.

**Epsilon ($\epsilon$):** Maximum distance between two points to consider them as neighbors.

**`MinPts`:** Minimum number of points required to form a dense region.

**Core Point:** A point with at least `MinPts` neighbors within $\epsilon$-distance.

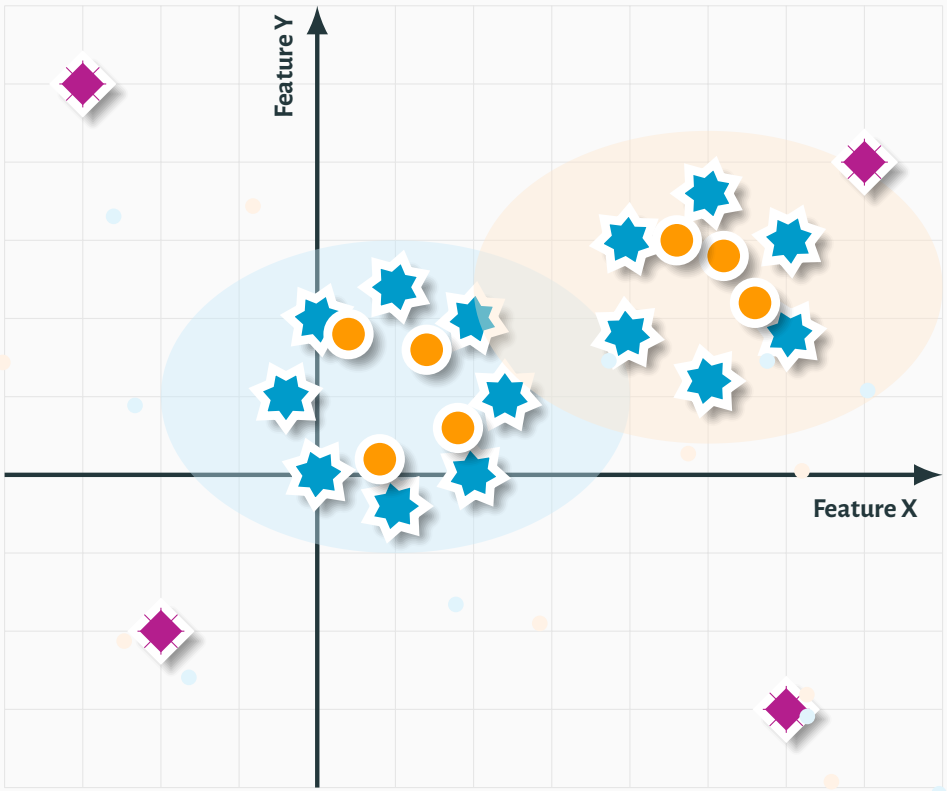
**Border Point:** A point within $\epsilon$-distance of a core point but with fewer than `MinPts` neighbors.

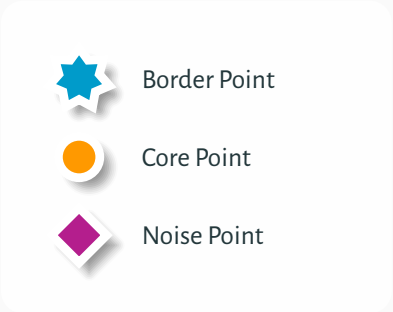
**Noise Point:** A point that is neither a core point nor a border point.

## Steps of DBSCAN

1. Select an unvisited point and check its $\epsilon$-neighborhood.
2. Mark it as a core, border, or noise point based on `MinPts`.
3. Expand clusters iteratively by connecting core points.
4. Continue until all points are visited.

# DBSCAN: Visual Representation




- ▲ Detects clusters of arbitrary shapes.
- ▲ Robust to noise and outliers.

- ▼ Sensitive to the choice of $\epsilon$ and `MinPts`.
- ▼ Struggles with clusters of varying densities.
- ▼ Performance degrades with high-dimensional data.

# Hands-on Example (1/3)

**Task #12**
Consider the following synthetic 2D dataset with 5 points:

| X | Y |
|---|---|
| 1.0 | 1.0 |
| 1.1 | 1.1 |
| 4.0 | 4.0 |
| 4.1 | 4.1 |
| 10.0 | 10.0 |

```
                    Y
                    |
                    |                                (10, 10)
                    |
                    |              (4.1, 4.1)
                    |             (4.0, 4.0)
                    |
                    |     (1.1, 1.1)
                    |    (1.0, 1.0)
                0   +-----+-----+-----+-----+-----+---> X
                    0     2     4     6     8    10
```

Apply DBSCAN with $\epsilon = 1.5$ and `MinPts` = 2.

# Hands-on Example (2/3)

## Compute Pairwise Distances

| | (1, 1) | (1.1, 1.1) | (4, 4) | (4.1, 4.1) | (10, 10) |
|---|---|---|---|---|---|
| (1, 1) | 0.00 | 0.141 | 4.243 | 4.384 | 12.728 |
| (1.1, 1.1) | 0.141 | 0.00 | 4.101 | 4.243 | 12.586 |
| (4, 4) | 4.243 | 4.101 | 0.00 | 0.141 | 8.485 |
| (4.1, 4.1) | 4.384 | 4.243 | 0.141 | 0.00 | 8.344 |
| (10, 10) | 12.728 | 12.586 | 8.485 | 8.344 | 0.00 |

## Identify Core Points

A point is a **core point** if $\geq$ `MinPts` neighbors exist within $\epsilon$:

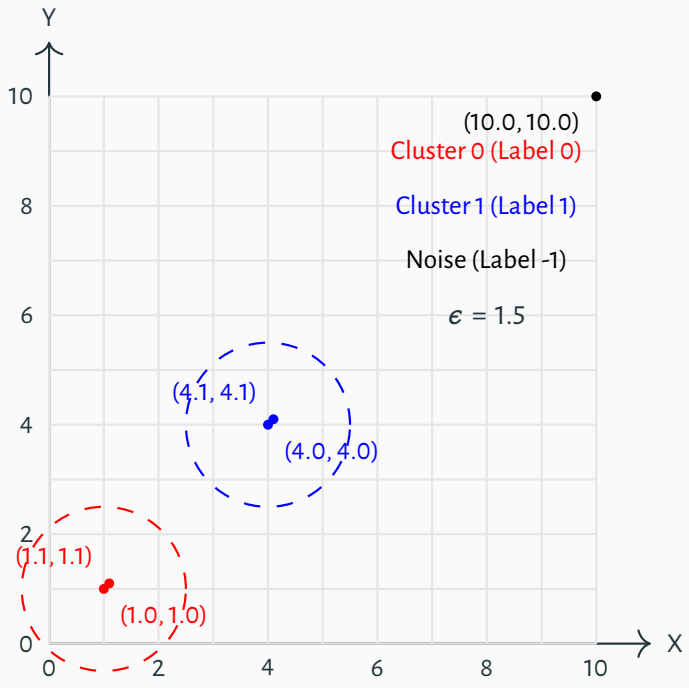
- **(1.0, 1.0)**: Neighbors = $\{(1.1, 1.1)\} \longrightarrow 1$ (<2) $\rightarrow$ *Border*
- **(1.1, 1.1)**: Neighbors = $\{(1.0, 1.0)\} \longrightarrow 1$ (<2) $\rightarrow$ *Border*
- **(4.0, 4.0)**: Neighbors = $\{(4.1, 4.1)\} \longrightarrow 1$ (<2) $\rightarrow$ *Border*
- **(4.1, 4.1)**: Neighbors = $\{(4.0, 4.0)\} \longrightarrow 1$ (<2) $\rightarrow$ *Border*
- **(10.0, 10.0)**: No neighbors within $\epsilon \rightarrow$ *Noise*

# Hands-on Example (3/3)

## Cluster Assignment

- **Cluster 1**: {(1.0,1.0), (1.1,1.1)} (mutually reachable via $\epsilon$)
- **Cluster 2**: {(4.0,4.0), (4.1,4.1)} (mutually reachable via $\epsilon$)
- **Noise**: (10.0,10.0) (no nearby points)

| Point (X, Y) | Label | Type |
|---|---|---|
| (1.0, 1.0) | 0 | Border |
| (1.1, 1.1) | 0 | Border |
| (4.0, 4.0) | 1 | Border |
| (4.1, 4.1) | 1 | Border |
| (10.0, 10.0) | -1 | Noise |



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# Principal Component Analysis (PCA): Overview

PCA is a dimensionality reduction technique that transforms a high-dimensional dataset into a lower-dimensional space by identifying the directions of maximum variance.

- Reduce the number of features while retaining most of the dataset's variability.
- Identify patterns in data by capturing principal components.
- Remove redundant or irrelevant information.

## How PCA Works

1. Standardize the dataset (mean = 0, variance = 1).
2. Compute the covariance matrix.
3. Calculate eigenvalues and eigenvectors of the covariance matrix.
4. Project the data onto the eigenvectors with the largest eigenvalues (principal components).

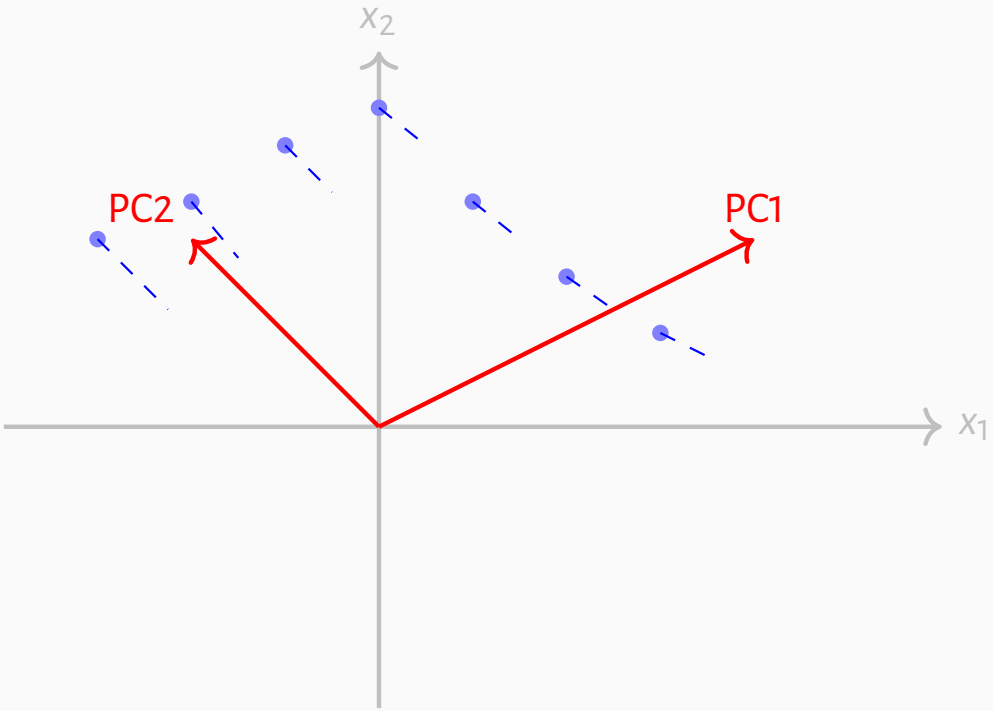
## Applications

- Image compression.
- Data visualization in 2D or 3D.
- Noise reduction.

# Visualizing PCA

## Understanding PCA Through a 2D Example:

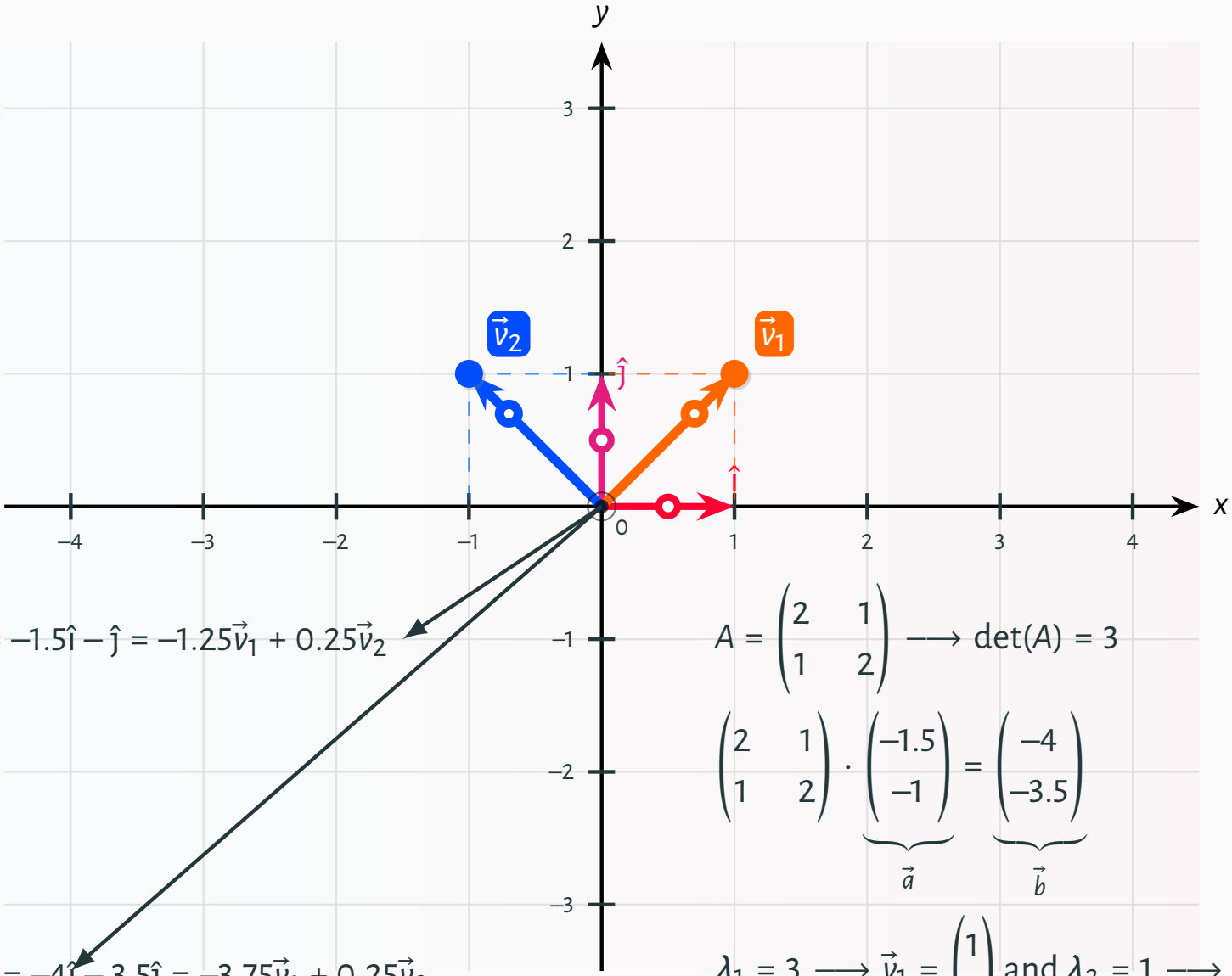
- Dataset with two features $(x_1, x_2)$.
- PCA identifies the principal axes of variance (principal components).
- Data is projected onto these axes.




① PC1: Captures the most variance in the data.

② PC2: Captures the remaining variance, orthogonal to PC1.

# Reminder




$\vec{a} = -1.5\hat{\imath} - \hat{\jmath} = -1.25\vec{v}_1 + 0.25\vec{v}_2$

$\vec{b} = -4\hat{\imath} - 3.5\hat{\jmath} = -3.75\vec{v}_1 + 0.25\vec{v}_2$

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \longrightarrow \det(A) = 3$$

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \cdot \underbrace{\begin{pmatrix} -1.5 \\ -1 \end{pmatrix}}_{\vec{a}} = \underbrace{\begin{pmatrix} -4 \\ -3.5 \end{pmatrix}}_{\vec{b}}$$

$$\lambda_1 = 3 \longrightarrow \vec{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ and } \lambda_2 = 1 \longrightarrow \vec{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

# Hands-on Example (1/4)

## Step-by-Step PCA Example

$$\mathbf{X} = \begin{bmatrix} 2.5 & 2.4 \\ 0.5 & 0.7 \\ 2.2 & 2.9 \\ 1.9 & 2.2 \\ 3.1 & 3.0 \\ 2.3 & 2.7 \\ 2.0 & 1.6 \\ 1.0 & 1.1 \\ 1.5 & 1.6 \\ 1.1 & 0.9 \end{bmatrix}$$

# Hands-on Example (2/4)

## Step-by-Step PCA Example

### Mean Normalization

$$\bar{X} = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} = \begin{bmatrix} 1.81 \\ 1.91 \end{bmatrix}, \quad \mathbf{X}_{\text{normalized}} = \mathbf{X} - \bar{X}$$

### Covariance Matrix

$$\mathbf{C} = \frac{1}{n-1} \mathbf{X}_{\text{normalized}}^{\top} \mathbf{X}_{\text{normalized}} \approx \begin{bmatrix} 0.6165 & 0.6152 \\ 0.6152 & 0.7165 \end{bmatrix}$$

### Eigenvalues and Eigenvectors

$$\text{Eigenvalues: } \lambda_1 \approx 1.284, \quad \lambda_2 \approx 0.049$$

$$\text{Eigenvectors: } \mathbf{v}_1 \approx \begin{bmatrix} 0.678 \\ 0.735 \end{bmatrix}, \quad \mathbf{v}_2 \approx \begin{bmatrix} -0.735 \\ 0.677 \end{bmatrix}$$

# Hands-on Example (3/4)

## Step-by-Step PCA Example

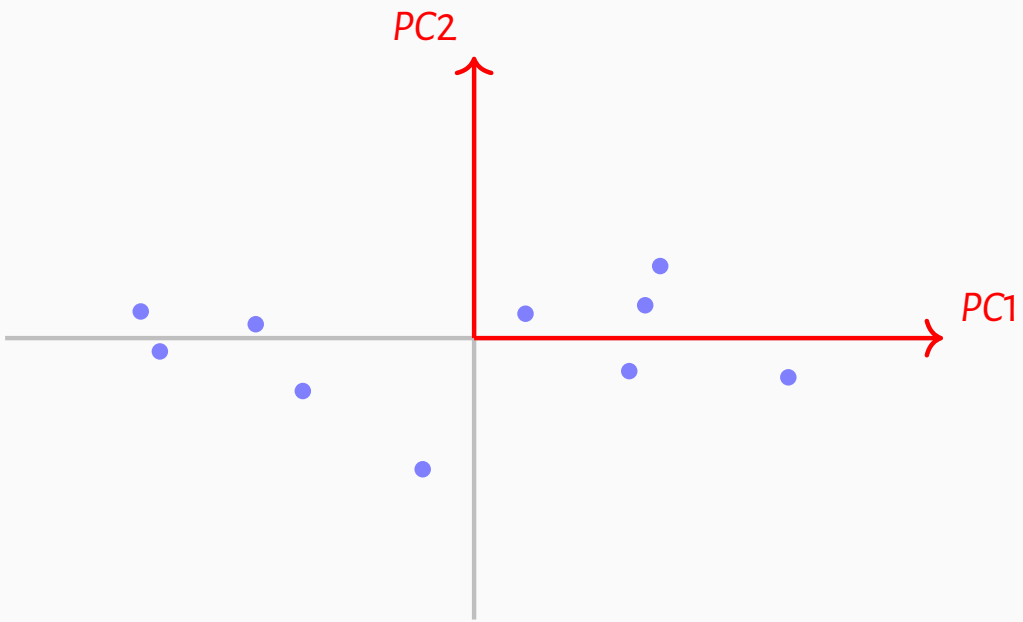
Project Data onto Principal Components

$$\mathbf{Z} = \mathbf{X}_{\text{normalized}}\mathbf{V}, \quad \therefore \quad \mathbf{Z} \approx \begin{bmatrix} 0.827 & -0.175 \\ -1.777 & 0.142 \\ 0.992 & 0.384 \\ 0.274 & 0.130 \\ 1.675 & -0.209 \\ 0.912 & 0.175 \\ -0.099 & -0.35 \\ -1.144 & 0.046 \\ -0.438 & 0.018 \\ -1.224 & -0.163 \end{bmatrix}$$

# Hands-on Example (4/4)

## Step-by-Step PCA Example

Visualization of Principal Components




**Key Observations**

- Most variance is along $PC1$ (horizontal axis).
- Data projected onto $PC2$ (vertical axis) shows minimal variance.

# Practical Implementation

The code is available @ github.com/a-mhamdi/jlai ⟶ *Codes* ⟶ *Julia* ⟶ *Part-2*

⟶ *Pluto* ⟶ *pca.jl*

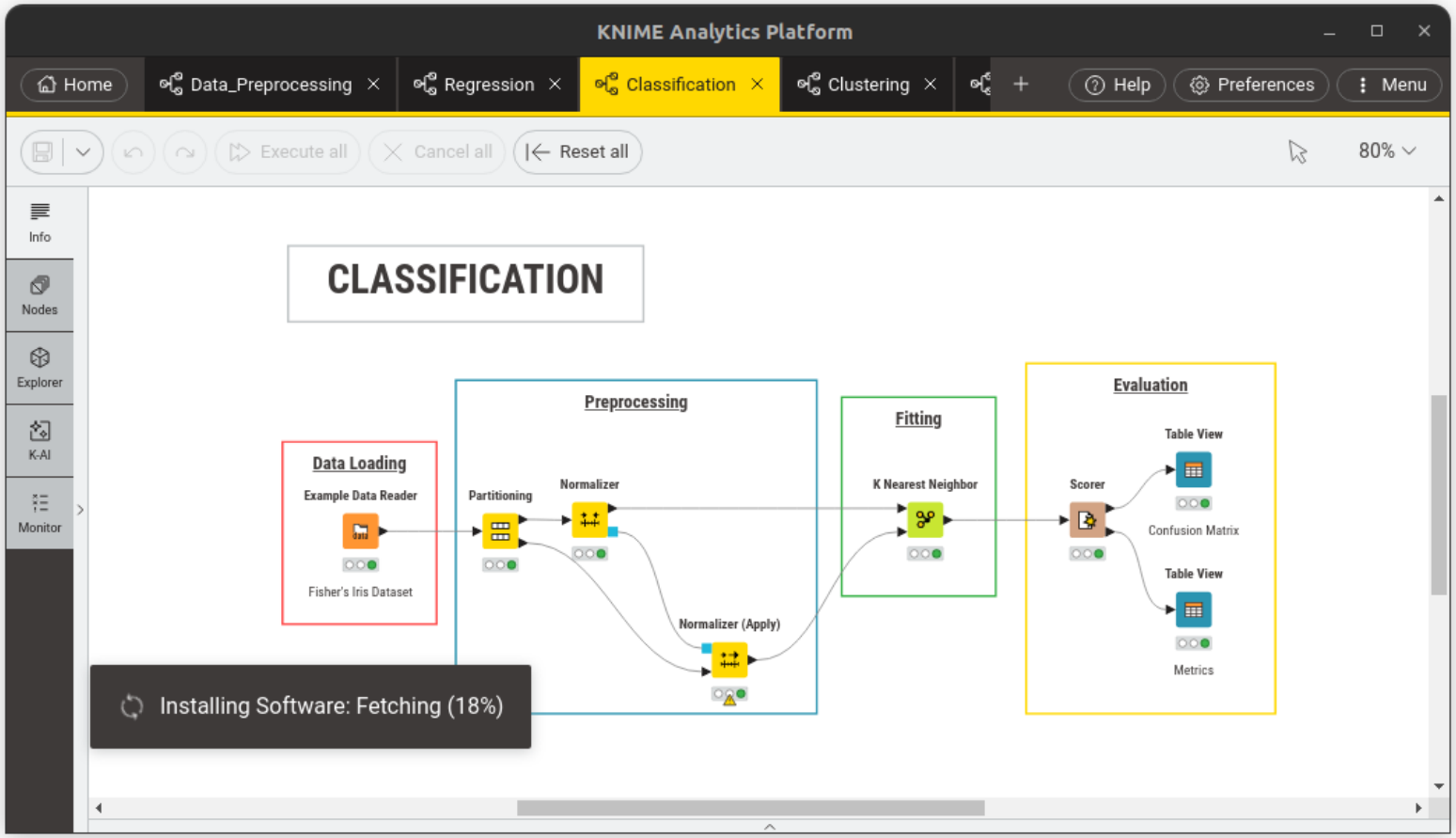
⟶ *Jupyter* ⟶ *pca.ipynb*

# Complementary Lab. Project

On the day of assignment, you will be informed about the **dataset to use** *(source: database)*, **specific features to retain**, and the **machine learning model to implement**. You will be asked to:

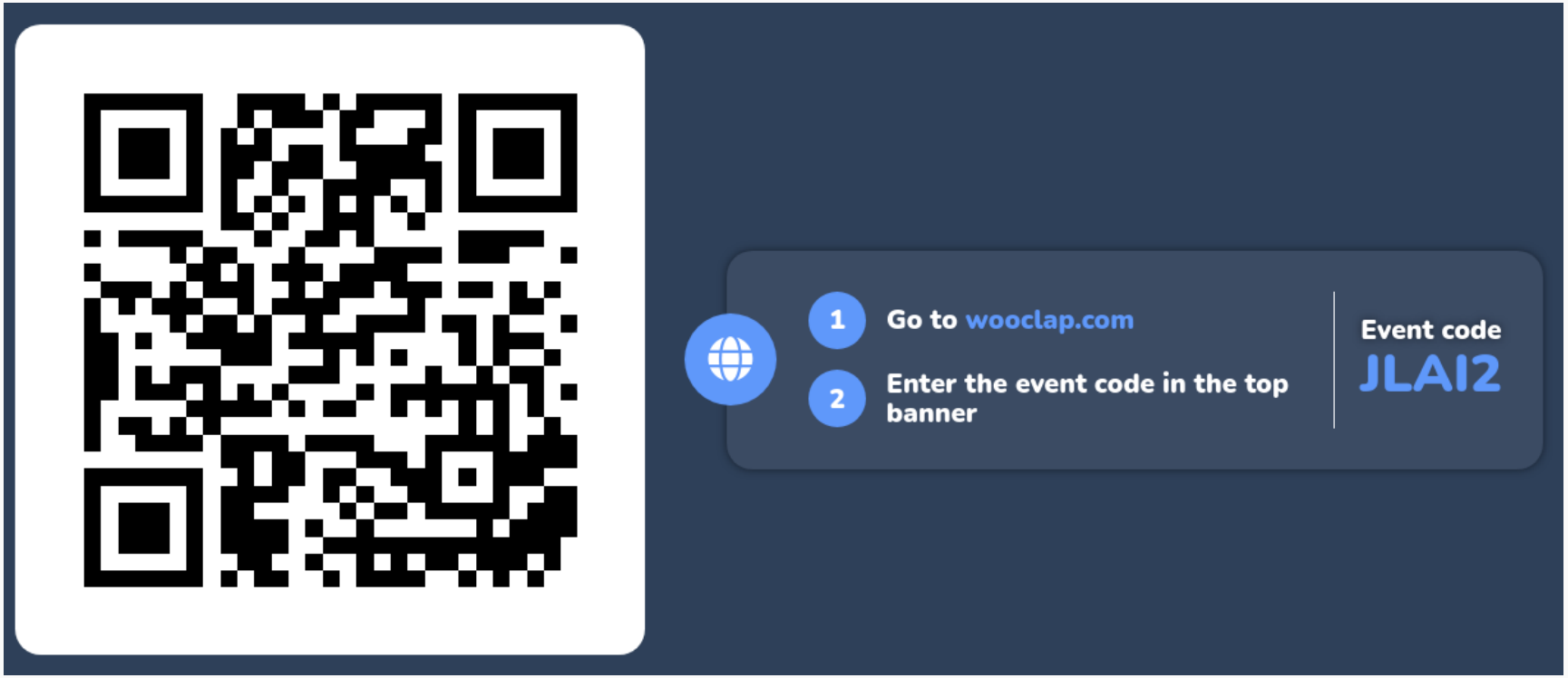
① ETL*(Extract, Transform, Load)* data from a database;

② Build and evaluate the **ML** model *(pipeline, featurization, split, etc.)*.



# ML Landscape through Quizzes

# Knowledge Check




https://app.wooclap.com/JLAI2

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# Further Reading (2/2)


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